

Supporting Materials for “Automatic Differentiation to Improve Parameter Estimation for Partially Observed Markov Processes”

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S1 Proof of Theorem 1 (DMOP- α Functional Forms)

Theorem 1 follows immediately as a consequence of the following results, Lemmas S1 and S2. Here, an A superscript is used to denote ancestral trajectories, so that $x_{1:n,j}^{A,P,\theta}$ contains the history of the prediction particle $x_{n,j}^{P,\theta}$ at observation times $1:n$. Similarly, $g_{i,j}^{A,P,\theta}$ is the ancestral weight of $x_{n,j}^{P,\theta}$ at observation time i .

Lemma S1. Write $\nabla_{\theta} \hat{\ell}^{\alpha}(\theta)$ for the DMOP- α gradient estimate (or, equivalently, the gradient of MOP- α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 1$,

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:N,j}^{A,F,\theta}; \theta), \quad (\text{S1})$$

with the same undiscounted path-space score target as the bootstrap filter estimators of Poyiadjis et al. (2011) and Ścibior and Wood (2021).

Proof. Consider MOP- α with $\alpha = 1$. Hold ϕ , the simulator draws, and the reference-filter re-sampling indices fixed when differentiating, then evaluate at $\theta = \phi$. Write $k_{n,j}$ for k_j at time n . Algorithm 1 gives

$$w_{n,j}^{P,\theta} = w_{n-1,j}^{F,\theta}, \quad w_{n,j}^{F,\theta} = w_{n,k_{n,j}}^{P,\theta} \frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}} = w_{n-1,k_{n,j}}^{F,\theta} \frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}}. \quad (\text{S2})$$

The after-resampling likelihood estimate therefore telescopes:

$$\begin{aligned} \hat{\mathcal{L}}^1(\theta) &= \prod_{n=1}^N L_n^{A,\theta,1} = \prod_{n=1}^N \left(L_n^{\phi} \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \right) \\ &= \left(\prod_{n=1}^N L_n^{\phi} \right) \prod_{n=1}^N \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n-1,j}^{F,\theta}} = \left(\frac{1}{J} \sum_{j=1}^J w_{N,j}^{F,\theta} \right) \prod_{n=1}^N L_n^{\phi}. \end{aligned} \quad (\text{S3})$$

Here each intermediate weight sum cancels, and the initial sum is J because $w_{0,j}^{F,\theta} = 1$. The log-derivative identity gives

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{\nabla_{\theta} \hat{\mathcal{L}}^1(\theta)}{\hat{\mathcal{L}}^1(\theta)} = \frac{J^{-1} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta} \prod_{n=1}^N L_n^{\phi}}{J^{-1} \sum_{j=1}^J w_{N,j}^{F,\theta} \prod_{n=1}^N L_n^{\phi}} = \frac{\sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta}}{\sum_{j=1}^J w_{N,j}^{F,\theta}}. \quad (\text{S4})$$

We next expand the filtered weight along its ancestry. Using the existing A convention, let $g_{i,j}^{A,F,\theta}$ denote the measurement density at time i along the filtered trajectory $x_{1:n,j}^{A,F,\theta}$; the endpoint n is implicit. At time n , this density is $g_{n,k_{n,j}}^{\theta}$. At earlier times, the trajectory agrees with that of the selected parent $x_{n-1,k_{n,j}}^{F,\theta}$. Starting from $w_{0,j}^{F,\theta} = 1$ and repeatedly substituting the filtered-weight recursion therefore gives

$$w_{n,j}^{F,\theta} = w_{n-1,k_{n,j}}^{F,\theta} \frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}} = \left(\prod_{i=1}^{n-1} \frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}} \right) \frac{g_{n,j}^{A,F,\theta}}{g_{n,j}^{A,F,\phi}} = \prod_{i=1}^n \frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}. \quad (\text{S5})$$

The product through $n - 1$ is empty when $n = 1$; for $n > 1$, it is the product already accumulated by the selected parent. The reference factors and ancestor indices are fixed under differentiation, so

$$\begin{aligned} \frac{\nabla_{\theta} w_{n,j}^{F,\theta}}{w_{n,j}^{F,\theta}} &= \nabla_{\theta} \log w_{n,j}^{F,\theta} = \nabla_{\theta} \log \left(\prod_{i=1}^n \frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}} \right) \\ &= \nabla_{\theta} \sum_{i=1}^n \left(\log g_{i,j}^{A,F,\theta} - \log g_{i,j}^{A,F,\phi} \right) = \sum_{i=1}^n \nabla_{\theta} \log g_{i,j}^{A,F,\theta}. \end{aligned} \quad (\text{S6})$$

For the terminal filtered trajectory at time N , conditional independence of the observations gives

$$\begin{aligned} \sum_{n=1}^N \nabla_{\theta} \log g_{n,j}^{A,F,\theta} &= \nabla_{\theta} \log \left(\prod_{n=1}^N g_{n,j}^{A,F,\theta} \right) \\ &= \nabla_{\theta} \log \left(\prod_{n=1}^N f_{Y_n|X_n}(y_n^* | x_{n,j}^{A,F,\theta}; \theta) \right) \\ &= \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:N,j}^{A,F,\theta}; \theta). \end{aligned} \quad (\text{S7})$$

The derivative includes the dependence of the simulated particle locations on θ through the chain rule. Multiplying the log-weight derivative by $w_{N,j}^{F,\theta}$ yields

$$\nabla_{\theta} w_{N,j}^{F,\theta} = w_{N,j}^{F,\theta} \sum_{n=1}^N \nabla_{\theta} \log g_{n,j}^{A,F,\theta} = w_{N,j}^{F,\theta} \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:N,j}^{A,F,\theta}; \theta). \quad (\text{S8})$$

Finally, at $\theta = \phi$, every ratio in Equation (S5) equals one, so $w_{N,j}^{F,\theta} = 1$ and $\sum_j w_{N,j}^{F,\theta} = J$. Substituting into Equation (S4) after differentiating gives

$$\begin{aligned} \nabla_{\theta} \hat{\ell}^1(\theta) \Big|_{\theta=\phi} &= \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta} \Big|_{\theta=\phi} \\ &= \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:N,j}^{A,F,\theta}; \theta) \Big|_{\theta=\phi}, \end{aligned} \quad (\text{S9})$$

as claimed. This has the same undiscounted path-space score target as the estimators of Poyiadjis et al. (2011); Ścibior and Wood (2021) for the bootstrap filter. The reparameterization score contribution need not equal their complete-data likelihood-ratio score contribution particle by particle. $\square$

For $\theta \neq \phi$, the same calculation instead gives the normalized weighted path score

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{\sum_{j=1}^J w_{N,j}^{F,\theta} \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:N,j}^{A,F,\theta}; \theta)}{\sum_{j=1}^J w_{N,j}^{F,\theta}}. \quad (\text{S10})$$

Growth of the unnormalized weight products alone does not establish a variance rate for this normalized expression. When $\theta = \phi$, a generic normalized path-functional inequality gives the coarse upper bound $O(N^4/J)$ after rescaling to the full unnormalized score. Section S7 gives an exact constant-weight calculation with order N^2/J at $\alpha = 1$ and explains why this example does not by itself provide a general finite- J upper bound.

Lemma S2. Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the DMOP- α gradient estimate (or, equivalently, the gradient of MOP- α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 0$,

$$\nabla_\theta \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_\theta \log \left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right), \quad (\text{S11})$$

yielding the estimate of Naesseth et al. (2018) when applied to the bootstrap filter.

Proof. First, write

$$s_{n,j} = \frac{g_{n,k_{n,j}}^\theta}{g_{n,k_{n,j}}^\phi} \quad (\text{S12})$$

as shorthand for the selected measurement density ratios, with the reference parameter and resampling indices held fixed. Observe that, when $\alpha = 0$, the likelihood estimate becomes

$$\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N \left(L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \right) \quad (\text{S13})$$

$$= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) \quad (\text{S14})$$

$$= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J \frac{f_{Y_n|X_n}(y_n^*|x_{n,k_{n,j}}^{P,\theta}; \theta)}{f_{Y_n|X_n}(y_n^*|x_{n,k_{n,j}}^{P,\phi}; \phi)} \right). \quad (\text{S15})$$

We lose the nice telescoping property observed in the MOP-1 case, but this expression still yields something useful. This is because its gradient when $\theta = \phi$ is therefore

$$\nabla_\theta \hat{\ell}^0(\theta) := \sum_{n=1}^N \nabla_\theta \log \left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) \quad (\text{S16})$$

$$= \sum_{n=1}^N \frac{\nabla_\theta \left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)}{\left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)} \quad (\text{S17})$$

$$= \sum_{n=1}^N \frac{\sum_{j=1}^J \nabla_\theta s_{n,j}}{\sum_{j=1}^J s_{n,j}} \quad (\text{S18})$$

$$= \sum_{n=1}^N \frac{1}{J} \sum_{j=1}^J \frac{\nabla_\theta f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta)}{f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\phi}; \phi)} \quad (\text{S19})$$

$$= \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left(f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta) \right), \quad (\text{S20})$$

where we use the log-derivative trick in the second equality, observe that $\sum_{j=1}^J s_{n,j} = J$ when $\theta = \phi$ in the fourth equality, and use the log-derivative trick again while noting that $\theta = \phi$ in the fifth equality. This yields the desired result. $\square$

S2 Proof of Theorem 2 (Optimization Convergence Analysis)

All derivatives in this section are evaluated at the common filtering and evaluation parameter $\theta = \phi$. Let Z_n contain the complete ancestral path through time n , the parameter-independent simulator variables, and the first- and second-order gradients propagated by automatic differentiation. Set

$$G_n^{\theta}(Z_n) = f_{Y_n|X_n}(y_n^* | X_n^{\theta}; \theta), \quad W_{\theta}(Z_N) = \prod_{n=1}^N G_n^{\theta}(Z_n), \quad (\text{S21})$$

$$\psi_{\theta}(Z_N) = \nabla_{\theta} \log W_{\theta}(Z_N), \quad \Xi_{\theta}(Z_N) = \nabla_{\theta}^2 \log W_{\theta}(Z_N). \quad (\text{S22})$$

Differentiate through X_n^{θ} in $G_n^{\theta}(Z_n)$, holding the simulator draws and resampling indices fixed.

Write $\rho_N^J = J^{-1} \sum_{j=1}^J \delta_{Z_{N,j}}$ for the empirical measure of the after-resampling genealogical particles, and define the normalized Feynman–Kac path measure by

$$\rho_N(F) = \frac{\mathbb{E}\{W_{\theta}(Z_N)F(Z_N)\}}{\mathbb{E}\{W_{\theta}(Z_N)\}}. \quad (\text{S23})$$

Assumptions (A3) and (A5) give $|\partial_{\theta_i} W_{\theta}| \leq N \bar{G}^N G'(\theta)$, so dominated convergence and the log-derivative identity yield

$$\nabla_{\theta} \ell(\theta) = \rho_N(\psi_{\theta}), \quad \nabla_{\theta} \hat{\ell}_J^{A,1}(\theta) = \rho_N^J(\psi_{\theta}). \quad (\text{S24})$$

The second equality is Lemma S1; importantly, it is one terminal empirical average of the complete path score, not a sum of empirical filtering expectations at the individual observation times.

For completeness, multinomial resampling gives the required genetic Feynman–Kac particle system explicitly. For semigroup indexing, let Z_0 be the augmented prediction path at observation 1; for $q = 1, \dots, N-1$, let Z_q be the prediction path at observation $q+1$; and let Z_N be the terminal path after the final resampling. For $0 \leq q \leq N$, write $\rho_q^J = J^{-1} \sum_{j=1}^J \delta_{Z_{q,j}}$, where the $Z_{0,j}$ are independent draws from the unweighted prediction law of Z_0 and $Z_{N,j} = Z_{N,j}$. Write $V_q^{\theta}(Z_q) = G_{q+1}^{\theta}(Z_q)$ for $q = 0, \dots, N-1$. Let the locally indexed augmented kernel $\mathcal{M}_{q+1}^{\theta}$ append the next independent simulator innovation and its gradients for $0 \leq q \leq N-2$, and put $\mathcal{M}_N^{\theta} = \text{Id}$ for the final selection. The one-step population map is

$$\Phi_{q+1}^{\theta}(\mu)(dz') = \frac{\int \mu(dz) V_q^{\theta}(z) \mathcal{M}_{q+1}^{\theta}(z, dz')}{\mu(V_q^{\theta})}, \quad 0 \leq q < N. \quad (\text{S25})$$

Conditionally on the current cloud, the J children are independent with common law $\Phi_{q+1}^{\theta}(\rho_q^J)$. The final observation is handled by the $q = N-1$ transition, namely multinomial selection with potential $V_{N-1}^{\theta} = G_N^{\theta}$ followed by the identity mutation. Thus ρ_N^J is exactly the terminal empirical

measure of this augmented historical system. Systematic resampling does not have this conditional-independence property and is not covered by the calculation.

We next write out the particle concentration constant used below. For $0 \leq q \leq N$, let

$$Q_{q,N}^\theta F(z_q) = \mathbb{E} \left[F(Z_N) \prod_{r=q}^{N-1} V_r^\theta(Z_r) \middle| Z_q = z_q \right], \quad (\text{S26})$$

$$P_{q,N}^\theta F = \frac{Q_{q,N}^\theta F}{Q_{q,N}^\theta 1}, \quad g_{q,N} = \sup_{z,z'} \frac{Q_{q,N}^\theta 1(z)}{Q_{q,N}^\theta 1(z')}, \quad b_{q,N} = \sup_{\text{osc}(F) \leq 1} \text{osc}(P_{q,N}^\theta F), \quad (\text{S27})$$

where $Q_{N,N}^\theta F = F$ and $\text{osc}(F) = \sup_{z,z'} |F(z) - F(z')|$. Thus $0 \leq b_{q,N} \leq 1$. With

$$R_G = \frac{\bar{G}}{\underline{G}}, \quad S_N = \sum_{q=0}^N g_{q,N}^3 b_{q,N}, \quad K_N = 32 S_N, \quad K_N^{\text{pf}} = 32 \sum_{r=0}^N R_G^{3r}, \quad (\text{S28})$$

the potential bounds give, term by term,

$$\underline{G}^{N-q} \leq Q_{q,N}^\theta 1(z_q) \leq \bar{G}^{N-q}, \quad g_{q,N} \leq R_G^{N-q}, \quad K_N \leq K_N^{\text{pf}}. \quad (\text{S29})$$

In particular, the constant used in the theorem is the closed-form quantity

$$K_N^{\text{pf}} = \begin{cases} 32(N+1), & R_G = 1, \\ 32 \frac{R_G^{3(N+1)} - 1}{R_G^3 - 1}, & R_G > 1. \end{cases} \quad (\text{S30})$$

Here is the constant calculation behind the tail bound used below. Let r_N^{DM} and β_N^{DM} denote the second- and first-order coefficients in Theorem 1.2 of Del Moral and Rio (2011). The Feynman–Kac bounds in their Appendix A.3 give

$$r_N^{\text{DM}} \leq 4S_N, \quad (\beta_N^{\text{DM}})^2 \leq 4 \sum_{q=0}^N g_{q,N}^2 b_{q,N}^2 \leq 4S_N, \quad (\text{S31})$$

where the last inequality uses $g_{q,N} \geq 1$ and $0 \leq b_{q,N} \leq 1$. If $S_N = 0$, the particle error is identically zero. Hence suppose $S_N > 0$. For $u \in (0, 1/2]$, set $x = Ju^2/(32S_N)$. If $x \leq \log 6$, then $6e^{-x} \geq 1$ and the desired probability bound is trivial. If $x > \log 6$, use $h^{-1}(x) \leq 2x + 2\sqrt{x}$ for $h(t) = \{t - \log(1+t)\}/2$ and $1 + 2x + 2\sqrt{x} \leq 8x$ to obtain

$$\begin{aligned} \frac{r_N^{\text{DM}}}{J} \{1 + h^{-1}(x)\} &\leq \frac{u^2}{8x} \{1 + 2x + 2\sqrt{x}\} \leq u^2 \leq \frac{u}{2}, \\ \beta_N^{\text{DM}} \sqrt{\frac{2x}{J}} &\leq 2\sqrt{S_N} \sqrt{\frac{2x}{J}} = \frac{u}{2}. \end{aligned}$$

Theorem 1.2 applied to both F and $-F$, after centering and scaling F to oscillation one, therefore gives the explicit Feynman–Kac corollary

$$\Pr(|(\rho_N^J - \rho_N)(F)| > \text{osc}(F)u) \leq 6 \exp\left(-\frac{Ju^2}{K_N}\right) \leq 6 \exp\left(-\frac{Ju^2}{K_N^{\text{pf}}}\right). \quad (\text{S32})$$

for every F with $\text{osc}(F) > 0$ and every $u \in (0, 1/2]$. If F is constant, $(\rho_N^J - \rho_N)(F) = 0$ identically. Equations S28–S32 replace all unnamed model-dependent concentration constants.

Lemma S3 (Explicit concentration of the DMOP-1 gradient). *Assume multinomial resampling and let*

$$A_N(\theta) = \max_{1 \leq i \leq p} \text{osc}(\psi_{\theta,i}), \quad u_g(J, \delta) = \sqrt{\frac{K_N^{\text{pf}}}{J} \log \frac{6p}{\delta}}. \quad (\text{S33})$$

If $J \geq 4K_N^{\text{pf}} \log(6p/\delta)$, then, with probability at least $1 - \delta$,

$$\|\nabla_{\theta} \hat{\ell}_J^{A,1}(\theta) - \nabla_{\theta} \ell(\theta)\|_2 \leq A_N(\theta) \sqrt{p} u_g(J, \delta). \quad (\text{S34})$$

Moreover, Assumption (A3) gives $A_N(\theta) \leq 2NG'(\theta)$. Consequently, the explicit condition

$$J \geq K_N^{\text{pf}} \log \frac{6p}{\delta} \max \left\{ 4, \frac{4pN^2 G'(\theta)^2}{\epsilon^2} \right\} \quad (\text{S35})$$

implies that the left side of equation S34 is at most ϵ with probability at least $1 - \delta$.

Proof. The lower bound on J makes $u_g(J, \delta) \leq 1/2$. Apply equation S32 to each normalized coordinate $\psi_{\theta,i}/A_N(\theta)$; a constant coordinate has zero error and needs no normalization. A union bound over the p coordinates gives

$$\max_{1 \leq i \leq p} |(\rho_N^J - \rho_N)(\psi_{\theta,i})| \leq A_N(\theta) u_g(J, \delta)$$

with probability at least $1 - \delta$. Now use $\|v\|_2 \leq \sqrt{p}\|v\|_{\infty}$ and equation S24. Finally, $|\psi_{\theta,i}| \leq NG'(\theta)$, so $A_N(\theta) \leq 2NG'(\theta)$; substituting this inequality and solving $2NG'(\theta)\sqrt{p}u_g(J, \delta) \leq \epsilon$ gives equation S35. $\square$

The next result treats the actual particle Hessian. It retains the outer-product and empirical-mean terms created by differentiating the logarithm of the terminal empirical average.

Lemma S4 (Explicit concentration of the DMOP-1 particle Hessian). *Suppose in addition that Assumption (A3) gives the entrywise bound $|\Xi_{\theta,ab}(Z_N)| \leq NG''(\theta)$ for $1 \leq a, b \leq p$. Define*

$$d_p = \frac{p(p+3)}{2}, \quad H_N(\theta) = 2p\{NG''(\theta) + 3N^2 G'(\theta)^2\}, \quad u_H(J, \delta) = \sqrt{\frac{K_N^{\text{pf}}}{J} \log \frac{6d_p}{\delta}}. \quad (\text{S36})$$

If $J \geq 4K_N^{\text{pf}} \log(6d_p/\delta)$, then, with probability at least $1 - \delta$,

$$\|\nabla_{\theta}^2 \hat{\ell}_J^{A,1}(\theta) - \nabla_{\theta}^2 \ell(\theta)\|_{\text{op}} \leq H_N(\theta) u_H(J, \delta). \quad (\text{S37})$$

Fix $0 < \kappa < \gamma$. If

$$\gamma I_p \preceq -\nabla_{\theta}^2 \ell(\theta) \preceq \Gamma I_p \quad \text{and} \quad J \geq K_N^{\text{pf}} \log \frac{6d_p}{\delta} \max \left\{ 4, \frac{H_N(\theta)^2}{\kappa^2} \right\}, \quad (\text{S38})$$

then, with probability at least $1 - \delta$,

$$(\gamma - \kappa) I_p \preceq -\nabla_{\theta}^2 \hat{\ell}_J^{A,1}(\theta) \preceq (\Gamma + \kappa) I_p. \quad (\text{S39})$$

In particular, the raw particle curvature is then positive definite and invertible.

Proof. For fixed reference-filter random numbers and resampling indices, the telescoping identity in the proof of Lemma S1 gives

$$\hat{\ell}_J^{A,1}(\theta) = C_\phi + \log \left\{ \frac{1}{J} \sum_{j=1}^J \exp(a_j(\theta)) \right\}, \quad a_j(\theta) = \log w_{N,j}^{F,\theta}, \quad a_j(\phi) = 0,$$

where C_ϕ is constant in θ . Twice differentiating this log-sum-exp at $\theta = \phi$ gives the exact finite-particle identity

$$\nabla_\theta^2 \hat{\ell}_J^{A,1} = \rho_N^J(\Xi_\theta + \psi_\theta \psi_\theta^T) - \rho_N^J(\psi_\theta) \rho_N^J(\psi_\theta)^T. \quad (\text{S40})$$

Assumption (A3) also gives the entrywise dominating bound

$$|\partial_{\theta_a \theta_b}^2 W_\theta| = W_\theta |\Xi_{\theta,ab} + \psi_{\theta,a} \psi_{\theta,b}| \leq \bar{G}^N \{NG''(\theta) + N^2 G'(\theta)^2\}.$$

Thus differentiation through second order may pass under the expectation, and differentiating $\log \mathbb{E} W_\theta$ gives the population identity

$$\nabla_\theta^2 \ell = \rho_N(\Xi_\theta + \psi_\theta \psi_\theta^T) - \rho_N(\psi_\theta) \rho_N(\psi_\theta)^T. \quad (\text{S41})$$

Let $A_1 = 2NG'(\theta)$ and $A_2 = 2\{NG''(\theta) + N^2 G'(\theta)^2\}$. The p scalar functions ψ_a have oscillation at most A_1 , and the symmetric entries of $\Xi + \psi\psi^T$ have oscillation at most A_2 . Apply equation S32 jointly to these functions, using $d_p = p + p(p+1)/2$ scalar events. On the resulting event,

$$\begin{aligned} \|(\rho_N^J - \rho_N)(\psi)\|_2 &\leq A_1 \sqrt{p} u_H(J, \delta), \\ \|(\rho_N^J - \rho_N)(\Xi + \psi\psi^T)\|_{\text{op}} &\leq p A_2 u_H(J, \delta). \end{aligned}$$

Both $\|\rho_N^J(\psi)\|_2$ and $\|\rho_N(\psi)\|_2$ are at most $\sqrt{p} NG'(\theta)$. Therefore

$$\begin{aligned} \|\rho_N^J(\psi) \rho_N^J(\psi)^T - \rho_N(\psi) \rho_N(\psi)^T\|_{\text{op}} \\ \leq 2\sqrt{p} NG'(\theta) \|(\rho_N^J - \rho_N)(\psi)\|_2 \leq 4p N^2 G'(\theta)^2 u_H(J, \delta). \end{aligned}$$

Combining the last three displays with equation S40 and equation S41 proves equation S37. Condition equation S38 makes its right side at most κ ; Weyl's eigenvalue inequality then gives equation S39. $\square$

For the optimization theorem, no stochastic Hessian claim is needed if the curvature is regularized deterministically. If $\tilde{B} = Q \text{diag}(\lambda_1, \dots, \lambda_p) Q^T$ is an orthogonal eigendecomposition of any symmetric raw estimate, define

$$B = Q \text{diag}([\lambda_1]_c^C, \dots, [\lambda_p]_c^C) Q^T, \quad [x]_c^C = \min\{C, \max\{c, x\}\}. \quad (\text{S42})$$

For every $v \in \mathbb{R}^p$, direct expansion in the eigenbasis gives

$$c\|v\|_2^2 \leq v^T B v \leq C\|v\|_2^2, \quad C^{-1}\|v\|_2^2 \leq v^T B^{-1} v \leq c^{-1}\|v\|_2^2. \quad (\text{S43})$$

Thus $cI_p \preceq B \preceq CI_p$ and $C^{-1}I_p \preceq B^{-1} \preceq c^{-1}I_p$. Lemma S4 shows that the unclipped particle Hessian has these properties with $c = \gamma - \kappa$ and $C = \Gamma + \kappa$ under equation S38; equation S42 guarantees the chosen c, C bounds for every realization.

Proof of Theorem 2. Let $\mathcal{D}$ be the convex region in Theorem 2, and set

$$G'_{\mathcal{D}} = \sup_{\theta \in \mathcal{D}} G'(\theta), \quad J_0 = \left\lceil K_N^{\text{pf}} \log \left(\frac{6pM}{\delta} \right) \max \left\{ 4, \frac{4pN^2(G'_{\mathcal{D}})^2}{\epsilon^2} \right\} \right\rceil. \quad (\text{S44})$$

Thus it is enough to take $J \geq J_0$. With $g_m = -\nabla_{\theta} \hat{\ell}_J^{A,1}(\theta_m)$, define

$$\tau = \min \{m \in \{0, \dots, M-1\} : \|g_m\|_2 \leq \sigma\epsilon\}, \quad \min \emptyset = M, \quad (\text{S45})$$

and

$$\mathcal{E}_M = \bigcap_{m=0}^{\min\{\tau, M-1\}} \{ \|g_m - \nabla f(\theta_m)\|_2 \leq \epsilon \}. \quad (\text{S46})$$

If $\mathcal{F}_m$ is the history before the m th particle-filter call, then θ_m and $\{m \leq \min(\tau, M-1)\}$ are $\mathcal{F}_m$ -measurable. Lemma S3, with failure probability δ/M , gives

$$\begin{aligned} \Pr(\mathcal{E}_M^c \mid \theta_0) &\leq \sum_{m=0}^{M-1} \mathbb{E} [\mathbf{1}_{\{m \leq \min(\tau, M-1)\}} \Pr(\|g_m - \nabla f(\theta_m)\|_2 > \epsilon \mid \mathcal{F}_m) \mid \theta_0] \\ &\leq \sum_{m=0}^{M-1} \frac{\delta}{M} = \delta. \end{aligned} \quad (\text{S47})$$

This calculation is valid at the adaptive iterates because each call uses a fresh particle-randomness block.

Work on $\mathcal{E}_M$ and write $p_m = -B_m^{-1}g_m$. If $\|g_m\|_2 \leq \sigma\epsilon$, then

$$\|\nabla f(\theta_m)\|_2 \leq \|g_m\|_2 + \epsilon \leq (\sigma + 1)\epsilon. \quad (\text{S48})$$

Strong convexity implies $f(\theta) - f(\theta^*) \leq \|\nabla f(\theta)\|_2^2 / (2\gamma)$; therefore

$$f(\theta_m) - f(\theta^*) \leq \frac{(\sigma + 1)^2 \epsilon^2}{2\gamma}. \quad (\text{S49})$$

Suppose instead that $\|g_m\|_2 > \sigma\epsilon$. Since $g_m = -B_m p_m$ and $cI_p \preceq B_m \preceq CI_p$,

$$c\|p_m\|_2 \leq \|g_m\|_2 \leq C\|p_m\|_2, \quad \epsilon\|p_m\|_2 < \frac{C}{\sigma}\|p_m\|_2^2. \quad (\text{S50})$$

The smoothness inequality and the gradient-error event give

$$\begin{aligned} f(\theta_m + \eta p_m) - f(\theta_m) &\leq \eta \nabla f(\theta_m)^T p_m + \frac{\Gamma \eta^2}{2} \|p_m\|_2^2 \\ &\leq -\eta p_m^T B_m p_m + \eta \epsilon \|p_m\|_2 + \frac{\Gamma \eta^2}{2} \|p_m\|_2^2. \end{aligned} \quad (\text{S51})$$

The choices of σ and η imply

$$\frac{C}{\sigma} \|p_m\|_2^2 \leq \frac{c(1-\beta)}{2} \|p_m\|_2^2, \quad \frac{\Gamma \eta}{2} \|p_m\|_2^2 \leq \frac{c(1-\beta)}{2} \|p_m\|_2^2.$$

Using $p_m^T B_m p_m \geq c \|p_m\|_2^2$ in equation S51 now gives

$$f(\theta_{m+1}) - f(\theta_m) \leq -\eta \beta p_m^T B_m p_m. \quad (\text{S52})$$

On a nonstopping iteration,

$$\|\nabla f(\theta_m)\|_2 \leq \|g_m\|_2 + \epsilon < \frac{\sigma + 1}{\sigma} \|g_m\|_2. \quad (\text{S53})$$

Consequently, equation S43 and strong convexity give

$$\begin{aligned} p_m^T B_m p_m &= g_m^T B_m^{-1} g_m \\ &\geq \frac{1}{C} \left(\frac{\sigma}{\sigma + 1} \right)^2 \|\nabla f(\theta_m)\|_2^2 \\ &\geq \frac{2\gamma}{C} \left(\frac{\sigma}{\sigma + 1} \right)^2 \{f(\theta_m) - f(\theta^*)\}. \end{aligned} \quad (\text{S54})$$

Combining equation S52 and equation S54 proves

$$f(\theta_{m+1}) - f(\theta^*) \leq q \{f(\theta_m) - f(\theta^*)\}, \quad q = 1 - \frac{2\eta\beta\gamma}{C} \left(\frac{\sigma}{\sigma + 1} \right)^2. \quad (\text{S55})$$

The step-size bound also verifies the claimed range of the contraction:

$$0 < 1 - q < 2\beta(1 - \beta) \frac{c\gamma}{C\Gamma} \leq \frac{1}{2}, \quad \text{so} \quad \frac{1}{2} < q < 1.$$

Iteration gives the q^m conclusion in the theorem, while equation S49 gives its stopping conclusion after substituting $f = -\ell$. $\square$

The result is deliberately restricted to DMOP-1. For $\alpha < 1$, the population limit is a discounted gradient. A theorem for that case must add a finite-sample bound for the discount bias relative to the exact score; choosing α close to one does not by itself provide such a bound.

S3 Feynman-Kac Models and Monte Carlo Approximations

In this section, we introduce the Feynman-Kac convention of Del Moral (2004) that has since become commonplace (Karjalainen et al., 2023) for the analysis of the particle filter. The mathematical formalization and notation introduced here will be adopted in the remainder of the analysis, in order to prove Theorems 3, 4, and 5. Let $(\eta_n)_{n=1}^N$ and $(\pi_n)_{n=1}^N$ be the prediction and filtering probability measures on the current state space $\mathcal{X}$. The posterior measure ρ_n is on simulator histories through time n , including the base variables, as defined in Equation (S153). Its state-path marginal is $f_{X_{0:n}|Y_{1:n}}$. These are the measures we seek to approximate with the particle filter. For any measurable bounded functional h , we adopt the following functional-analytic notation, borrowed from Del Moral (2004); Chopin and Papaspiliopoulos (2020); Karjalainen et al. (2023). We choose our specific choice of notation and definitions to be in line with that of Karjalainen et al. (2023).

Markov kernels and the process model: A Markov kernel M with source $\mathcal{X}_1$ and target $\mathcal{X}_2$ is a map $M : \mathcal{X}_1 \times \mathcal{B}(\mathcal{X}_2) \rightarrow [0, 1]$ such that for every set $A \in \mathcal{B}(\mathcal{X}_2)$ and every point $x \in \mathcal{X}_1$, the map $x \mapsto M(x, A)$ is a measurable function of x , and the map $A \mapsto M(x, A)$ is a probability measure on $\mathcal{X}_2$. The quantity $M(x, A)$ can be thought of as the probability of transitioning to the set A given that we are at the point x . If this yields a density, this then corresponds to the process density $f_{X_2|X_1}$ conditional on x and integrated over A .

Markov kernels and measures: For any measure η , any Markov kernel M on $\mathcal{X}$, any point $x \in \mathcal{X}$ and any measurable subset $A \subseteq \mathcal{X}$, let

$$\eta(h) = \int h d\eta = \int h(x)\eta(dx), \quad (\text{S56})$$

$$(\eta M)(A) = \int \eta(dx)M(x, A), \quad (\text{S57})$$

$$(Mh)(x) = \int M(x, dy)h(y). \quad (\text{S58})$$

Compositions of Markov kernels: The composition of a Markov kernel M_1 with another Markov kernel M_2 is another Markov kernel, given by

$$(M_1 M_2)(x, A) = \int M_1(x, dy)M_2(y, A). \quad (\text{S59})$$

Total variation distance: The total variation distance between two measures μ and ν on $\mathcal{X}$ is

$$\|\mu - \nu\|_{\text{TV}} = \sup_{\|h\|_\infty \leq 1/2} |\mu(h) - \nu(h)| = \sup_{\text{osc}(h) \leq 1} |\mu(h) - \nu(h)|. \quad (\text{S60})$$

Dobrushin contraction: The Dobrushin contraction coefficient β_{TV} of a Markov kernel M is given by

$$\beta_{\text{TV}}(M) = \sup_{x, y \in \mathcal{X}} \|M(x, \cdot) - M(y, \cdot)\|_{\text{TV}} = \sup_{\mu, \nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\text{TV}}}{\|\mu - \nu\|_{\text{TV}}}. \quad (\text{S61})$$

Potential functions and the measurement model: A potential function $G : \mathcal{X} \rightarrow [0, \infty)$ is a non-negative function of an element of the state space $x \in \mathcal{X}$. In our case, this corresponds to the measurement model, and in our previous notation is written as $g_{n,j} = f_{Y_n|X_n}(y_n^*|x_{n,j}^P; \theta) = G_n(x_{n,j}^P)$, where we suppress the dependence on θ for notational simplicity. Note that $G_n(\cdot) = f_{Y_n|X_n}(y_n^*|\cdot)$ is the conditional density of the observed measurement at time n , evaluated at a state in $\mathcal{X}$. At the filtered particle, $G_n(x_{n,j}^F) = g_{n,k_{n,j}}$.

Feynman-Kac models: A Feynman-Kac model on $\mathcal{X}$ is a tuple $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$ of an initial probability measure on the state space π_0 , a sequence of transition kernels $(M_n)_{n=1}^N$, and a sequence of potential functions $(G_n)_{n=1}^N$. In the notation used in the main text, this corresponds to the starting distribution f_{X_0} , the sequence of transition densities $f_{X_n|X_{n-1}}$, and the measurement densities $f_{Y_n|X_n}$. This induces a set of mappings from the set of probability measures on $\mathcal{X}$ to itself, $\mathcal{P}(\mathcal{X}) \rightarrow \mathcal{P}(\mathcal{X})$, as follows:

- The update from the prediction to the filtering distributions is given by

$$\pi_n(dx) = \Psi_n(\eta_n)(dx) = \frac{G_n(x) \cdot \eta_n(dx)}{\eta_n(G_n)}. \quad (\text{S62})$$

- The map from the prediction distribution at timestep n to timestep $n + 1$ is given by

$$\Phi_{n+1}(\eta_n) = \Psi_n(\eta_n)M_{n+1}. \quad (\text{S63})$$

- The composition of maps between prediction distributions yields the map from the prediction distribution at time k to the prediction distribution at time n where $k \leq n$,

$$\Phi_{k,n} = \Phi_n \circ \dots \circ \Phi_{k+1}. \quad (\text{S64})$$

The particle filter: The particle filter then yields a Monte Carlo approximation to the above Feynman-Kac model, via a sequence of mixture Dirac measures. When one resamples at every timestep, the prediction measure at timestep n is then given by

$$\eta_n^J = \frac{1}{J} \sum_{j=1}^J \delta_{x_{n,j}^P}, \quad (\text{S65})$$

and the filtering measure at timestep n is given by

$$\pi_n^J = \frac{\sum_{j=1}^J g_{n,j} \delta_{x_{n,j}^P}}{\sum_{j=1}^J g_{n,j}} \approx \frac{1}{J} \sum_{j=1}^J \delta_{x_{n,j}^F}. \quad (\text{S66})$$

In a slight abuse of notation, we will identify $x_{n,1:J}^P \equiv \eta_n^J$, and $x_{n,1:J}^F \equiv \pi_n^J$. As in Karjalainen et al. (2023), one can view this as an inhomogenous Markov process evolving on $\mathcal{X}^J$. For $\theta = \phi$ and multinomial resampling, the corresponding Markov transition kernel is

$$\mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = (\Phi_n(\eta_{n-1}^J))^{\otimes J} = \left(\Phi_n \left(\frac{1}{J} \sum_{j=1}^J \delta_{x_{n-1,j}^P} \right) \right)^{\otimes J}, \quad (\text{S67})$$

and the composition of Markov kernels on particles from timestep n to timestep $n + k$ is written

$$\mathbf{M}_{n,n+k} = \mathbf{M}_{n+1} \cdots \mathbf{M}_{n+k}. \quad (\text{S68})$$

The kernel acts on the full prediction cloud. Conditional on the preceding prediction cloud, its law is

$$\text{Law}(x_{n,1:J}^P \mid x_{n-1,1:J}^P) = \mathbf{M}_n(x_{n-1,1:J}^P, \cdot). \quad (\text{S69})$$

Forgetting of the particle filter: The above formalization yields a result from Karjalainen et al. (2023) on the forgetting of the particle filter that we require for our analysis of the bias, variance, and error of MOP- α . That is, Karjalainen et al. (2023) show that

$$\beta_{\text{TV}}(\mathbf{M}_{n,n+k}) \leq (1 - \epsilon)^{\lfloor k/(O(\log J)) \rfloor}, \quad (\text{S70})$$

for some ϵ dependent on $\bar{G}, \underline{G}, \bar{M}, \underline{M}$ in Assumptions (A3) and (A2). As a result, the mixing time of the particle filter is only on the order of $O(\log(J))$ timesteps.

Equipped with the above formalisms and results, we are now in a position to provide guarantees on the performance of MOP- α itself.

S4 Proof of Theorem 3 via a Strong Law of Large Numbers for Triangular Arrays of Particles With Off-Parameter Resampling

Here, we prove Theorem 3 by deriving more general result, from which it will be clear that MOP- α targets the filtering distribution and is strongly consistent for the likelihood. We will prove a strong law of large numbers for triangular arrays of particles with off-parameter resampling, meaning that we resample the particles according to an arbitrary resampling rule that is not necessarily in proportion to the target distribution of interest. Using weights that encode the cumulative discrepancy between the resampling distribution and the target distribution (instead of resampling to equal weights, as in the basic particle filter) provides a sufficient correction to ensure almost sure convergence. We now introduce the precise definition of an off-parameter resampled particle filter.

Definition S1 (Off-Parameter Resampled Particle Filters). *An off-parameter resampled particle filter is a Monte Carlo approximation to a Feynman-Kac model $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$, where we inductively define given some $x_{n-1,j}^F, w_{n-1,j}^F$ comprising the filtering measure approximation at time $n-1$:*

1. *Conditional on the filtered particles and weights at time $n-1$, draw independently $x_{n,j}^P \sim M_n(x_{n-1,j}^F, \cdot)$ using fresh simulator draws. Retain the corresponding parent weight: $w_{n,j}^P = w_{n-1,j}^F$.*
2. *The prediction measure at timestep n is given by*

$$\eta_n^J = \frac{\sum_{j=1}^J w_{n,j}^P \delta_{x_{n,j}^P}}{\sum_{j=1}^J w_{n,j}^P}. \quad (\text{S71})$$

3. *The particles are resampled at every timestep n , yielding indices k_j according to some arbitrary probabilities $(p_{n,j})_{j=1}^J$.*
4. *The filtering measure at timestep n is given by*

$$\pi_n^J = \frac{\sum_{j=1}^J \delta_{x_{n,j}^P} w_{n,j}^P g_{n,j}}{\sum_{j=1}^J w_{n,j}^P g_{n,j}} \approx \frac{\sum_{j=1}^J \delta_{x_{n,k_j}^P} w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}}{\sum_{j=1}^J w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}} = \frac{\sum_{j=1}^J w_{n,j}^F \delta_{x_{n,j}^F}}{\sum_{j=1}^J w_{n,j}^F}. \quad (\text{S72})$$

5. *The prediction weights at the next timestep are given by $w_{n+1,j}^P = w_{n,j}^F = w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}$.*

To prove that the weight correction is sufficient for almost sure convergence, we first introduce what it means for a triangular array of particles to *target* a given target distribution.

Definition S2 (Targeting). *A pair of random vectors (X, W) drawn from some measure g **targets** another measure π if, for any measurable and bounded functional h ,*

$$E_g[h(X) \cdot W] = E_\pi[h(X)]. \quad (\text{S73})$$

*A particle approximation is a triangular array of pairs of random vectors $(X_j^J, W_j^J), j = 1, 2, \dots, J$. The particle approximation **targets** π if, for any measurable and bounded functional h ,*

$$\frac{\sum_{j=1}^J h(X_j^J) W_j^J}{\sum_{j=1}^J W_j^J} \xrightarrow{a.s.} E_\pi[h(X)] \quad (\text{S74})$$

as $J \rightarrow \infty$.

Chopin (2004) asserted without proof that common particle filter algorithms target the filtering distribution in this sense, while Chopin and Papaspiliopoulos (2020) proved a related result assuming bounded densities. We follow a similar approach to Chopin and Papaspiliopoulos (2020), based on showing strong laws of large numbers for triangular arrays, noting that triangular array strong laws do not generally hold without an additional regularity condition such as boundedness.

In order to prove the consistency of our variation on the particle filter, we now present three helper lemmas. The first follows from standard importance sampling arguments, the second from integrating out the marginal, and the third from Bayes' theorem. We state Lemma S5 assuming multinomial resampling, which is convenient for the proof though other resampling strategies may be preferable in practice.

Lemma S5 (Change of Weight Measure). *Suppose that $\{(\tilde{X}_j^J, U_j^J), j = 1, \dots, J\}$ targets f_X . Now, let $\{(Y_j^J, V_j^J), j = 1, \dots, J\}$ be a multinomial sample with indices k_j drawn from $\{(\tilde{X}_j^J, U_j^J)\}$ where $(\tilde{X}_j^J, U_j^J)$ is represented, on average, proportional to $\pi_j^J J$ times. Write*

$$(Y_j^J, V_j^J) = (\tilde{X}_{k_j}^J, U_{k_j}^J / \pi_{k_j}^J). \quad (\text{S75})$$

If the importance sampling weights U_j / π_j are bounded, then $\{(Y_j^J, V_j^J), j = 1, \dots, J\}$ targets f_X .

Proof. Note that as the Y_j^J are a subsample from X_j^J , h can be a function of Y as well as it is one for X . We then expand

$$\frac{\sum_j h(Y_j^J) V_j^J}{\sum_j V_j^J} = \frac{\sum_j h(\tilde{X}_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}}. \quad (\text{S76})$$

By hypothesis,

$$\frac{\sum_j h(\tilde{X}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X} [f(X)]. \quad (\text{S77})$$

We want to show

$$\frac{\sum_j h(X_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}} - \frac{\sum_j h(\tilde{X}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} 0. \quad (\text{S78})$$

For this, it is sufficient to show that

$$\sum_j h(\tilde{X}_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J) \frac{U_j^J}{\pi_j^J} \xrightarrow{a.s.} 0 \quad (\text{S79})$$

since an application of this result with $h(x) = 1$ provides almost sure convergence of the denominator in (S78). Write $g(\tilde{X}_j^J) = h(\tilde{X}_j^J) \frac{U_j^J}{\pi_j^J}$. We therefore need to show that

$$\sum_j Z_j^J := \sum_j \left(g(\tilde{X}_{k_j}^J) - g(\tilde{X}_j^J) \pi_j^J \right) \xrightarrow{a.s.} 0. \quad (\text{S80})$$

Because the functional h and importance sampling weights $u_{k_j}^J/\pi_{k_j}^J$ are bounded, we have that $\mathbb{E}[(Z_j^J)^4] < \infty$. We can then follow the argument of Chopin and Papaspiliopoulos (2020) from this point on, where noting that the Z_j^J are conditionally independent given the $\tilde{X}_j^J$ and U_j^J ,

$$\begin{aligned} \mathbb{E} \left[\left(\sum_j Z_j^J \right)^4 \middle| (\tilde{X}_j^J, U_j^J)_{j=1}^J \right] &= J \mathbb{E} \left[(Z_1^J)^4 | (\tilde{X}_j^J, U_j^J)_{j=1}^J \right] \\ &\quad + 3J(J-1) \left(\mathbb{E}[(Z_j^J)^2 | (\tilde{X}_j^J, U_j^J)_{j=1}^J] \right) \end{aligned} \quad (\text{S81})$$

$$\leq CJ^2, \quad (\text{S82})$$

for some $C > 0$. Taking expectations on both sides yields

$$\mathbb{E} \left[\left(\sum_j Z_j^J \right)^4 \right] \leq CJ^2 \quad (\text{S83})$$

by the tower property. Now by Markov's inequality,

$$\mathbb{P} \left(\left| \frac{1}{J} \sum_{j=1}^J Z_j^J \right| > \epsilon \right) \leq \frac{1}{\epsilon^4 J^4} \mathbb{E} \left[\left(\sum_{j=1}^J Z_j^J \right)^4 \right] \leq \frac{C}{\epsilon^4 J^2}, \quad (\text{S84})$$

and as these terms are summable we can apply Borel-Cantelli to conclude that these deviations happen only finitely often for every $\epsilon > 0$, giving us the almost-sure convergence for

$$\sum_j h(X_{k_j}^J) \frac{u_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(X_j^J) \frac{u_j^J}{\pi_j^J} \xrightarrow{a.s.} 0. \quad (\text{S85})$$

Similarly, we also have that

$$\sum_j \frac{u_j^J}{\pi_j^J} \pi_j^J - \sum_j \frac{u_{k_j}^J}{\pi_{k_j}^J} \pi_{k_j}^J \xrightarrow{a.s.} 0, \quad (\text{S86})$$

and the result is proved. $\square$

Remark: Note that Lemma S5 permits $\pi_{1:J}$ to depend on $\{(X_j, u_j)\}$ as long as the resampling is carried out independently of $\{(X_j, u_j)\}$, conditional on $\pi_{1:J}$.

Lemma S6 (Particle Marginals). *Suppose that $\{(\tilde{X}_j^J, U_j^J), j = 1, \dots, J\}$ targets f_X . Also suppose that $\tilde{Z}_j^J \sim f_{Z|X}(\cdot | \tilde{X}_j^J)$ where $f_{Z|X}$ is a conditional probability density function corresponding to a joint density $f_{X,Z}$ with marginal densities f_X and f_Z . Then, if the U_j^J are bounded, $\{(\tilde{Z}_j^J, U_j^J)\}$ targets f_Z .*

Proof. We want to show that, for any measurable bounded h ,

$$\frac{\sum_j h(\tilde{Z}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_Z}[h(Z)] = \mathbb{E}_{f_X}[\mathbb{E}_{f_{Z|X}}[h(Z)|X]]. \quad (\text{S87})$$

By assumption, for any measurable and bounded functional g with domain $\mathcal{X}$,

$$\frac{\sum_j g(\tilde{X}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}[g(X)]. \quad (\text{S88})$$

Let $\bar{U}_j^J = \frac{J U_j^J}{\sum_j U_j^J}$. Examine the numerator and denominator of the quantity

$$\frac{J^{-1} \sum_j h(\tilde{Z}_j^J) U_j^J}{J^{-1} \sum_j U_j^J} = \frac{J^{-1} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J}{J^{-1} \sum_j \bar{U}_j^J}. \quad (\text{S89})$$

The denominator converges to 1 almost surely. The numerator, on the other hand, is

$$\frac{1}{J} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J, \quad (\text{S90})$$

and by the same fourth moment argument to the above lemma, it converges almost surely to the limit of its expectation,

$$\lim_{J \rightarrow \infty} \mathbb{E} \left[\frac{1}{J} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J \right] = \lim_{J \rightarrow \infty} \mathbb{E} \left[\frac{1}{J} \sum_j \mathbb{E} [h(\tilde{Z}_j^J) \bar{U}_j^J | \tilde{X}_j^J, \bar{U}_j^J] \right] \quad (\text{S91})$$

$$= \lim_{J \rightarrow \infty} \mathbb{E} \left[\frac{1}{J} \sum_j \mathbb{E} [h(Z) | X = \tilde{X}_j^J] \bar{U}_j^J \right]. \quad (\text{S92})$$

Applying (S88) with $g(x) = \mathbb{E}[h(Z)|X = x]$, the average on the right hand side converges almost surely to $\mathbb{E}\{\mathbb{E}[h(Z)|X]\} = \mathbb{E}[h(Z)]$. It remains to swap the limit and expectations. We can do so with the bounded convergence theorem, and therefore obtain

$$\frac{1}{J} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J \xrightarrow{a.s.} \mathbb{E}_{f_Z}[h(Z)]. \quad (\text{S93})$$

□

Lemma S7 (Particle Posteriors). *Suppose that $\{(X_j^J, U_j^J), j = 1, \dots, J\}$ targets f_X . Also suppose that $(X_j'^J, U_j'^J) = (X_j^J, U_j^J f_{Z|X}(z^*|X_j^J))$. Then, if $U_j^J f_{Z|X}(z^*|X_j^J)$ and $U_j'^J f_{Z|X}(z^*|X_j'^J)/f_Z(z^*)$ are bounded, $\{(X_j'^J, U_j'^J)\}$ targets $f_{X|Z}(\cdot|z^*)$.*

Proof. Again, we want to show that

$$\frac{\sum_j h(X_j^J) \cdot U_j^J \cdot f_{Z|X}(z^*|X_j^J)}{\sum_j U_j^J \cdot f_{Z|X}(z^*|X_j^J)} \xrightarrow{a.s.} \mathbb{E}_{f_{X|Z}}[h(X)|z^*]. \quad (\text{S94})$$

We already have that for any measurable bounded g ,

$$\frac{\sum_j g(X_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}[g(X)]. \quad (\text{S95})$$

Consider the following:

$$\frac{J^{-1} \sum_j h(X_j^J) f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \times \left(\frac{J^{-1} \sum_j f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \right)^{-1}. \quad (\text{S96})$$

We will apply Equation (S95) to the numerator and the denominator in the ratio above individually. The numerator converges to

$$\frac{J^{-1} \sum_j h(X_j^J) f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X} [h(X) f_{Z|X}(z^*|X)], \quad (\text{S97})$$

while the reciprocal of the denominator converges to

$$\frac{J^{-1} \sum_j f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X} [f_{Z|X}(z^*|X)] = f_Z(z^*). \quad (\text{S98})$$

Now we take advantage of the identities

$$\frac{\mathbb{E}_{f_X} [h(X) f_{Z|X}(z^*|X)]}{f_Z(z^*)} = \mathbb{E}_{f_X} \left[\frac{h(X) f_{Z|X}(z^*|X)}{f_Z(z^*)} \right] \quad (\text{S99})$$

$$= \mathbb{E}_{f_X} \left[h(X) \frac{f_{X|Z}(X|z^*)}{f_X(X)} \right] = \mathbb{E}_{f_{X|Z}} [h(X)|z^*], \quad (\text{S100})$$

to give the desired result. $\square$

Theorem S1 (Off-Parameter Resampled Particle Filters Target the Filtering Distribution). *The off-parameter resampled particle filter as outlined in Definition S1 targets the filtering distribution.*

Proof. Recursively applying Lemmas S5, S6, and S7, we obtain that the off-parameter resampled particle filter targets the posterior. Specifically, suppose inductively that $\{(X_{n-1,j}^F, w_{n-1,j}^F)\}$ targets π_{n-1} . Then, Lemma S6 tells us that $\{(X_{n,j}^P, w_{n,j}^P)\}$ targets η_n . Lemma S7 tells us that $\{(X_{n,j}^P, w_{n,j}^P g_{n,j}^\theta)\}$ therefore targets π_n . Lemma S5 guarantees that the resampling rule, given by

$$(X_{n,j}^F, w_{n,j}^F) = (X_{n,k_j}^P, w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}), \quad (\text{S101})$$

with resampling probabilities proportional to $p_{n,j}$, therefore also targets π_n . $\square$

Proposition S1 (MOP-1 Targets the Filtering Distribution). *When $\alpha = 1$ or $\phi = \theta$, MOP- α targets the filtering distribution.*

Proof. When $\theta = \phi$, regardless of the value of α , the ratio $\frac{g_{n,j}^\theta}{g_{n,j}^\phi} = 1$, and this reduces to the vanilla particle filter estimate.

When $\alpha = 1$, and $\theta \neq \phi$, the proof is identical to that of S1. Recursively applying Lemmas S5, S6, and S7, we obtain that the MOP-1 filter targets the posterior. Specifically, suppose inductively that $\{(X_{n-1,j}^{F,\theta}, w_{n-1,j}^{F,\theta})\}$ is properly weighted for $f_{X_{n-1}|Y_{1:n-1}}(x_{n-1}|y_{1:n-1}^*; \theta)$. Then, Lemma S6 tells

us that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*; \theta)$. Lemma S7 tells us that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta} g_{n,j}^\theta)\}$ therefore targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*; \theta)$. Lemma S5 guarantees that the resampling rule, given by

$$(X_{n,j}^{F,\theta}, w_{n,j}^{F,\theta}) = (X_{n,k_j}^{P,\theta}, w_{n,k_j}^{P,\theta} g_{n,k_j}^\theta / g_{n,k_j}^\phi), \quad (\text{S102})$$

with resampling weights proportional to $g_{n,j}^\phi$, therefore also targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*; \theta)$. $\square$

This has addressed filtering, but not quite yet the likelihood evaluation. For this we use the following lemma.

Lemma S8 (Likelihood Proper Weighting). *$f_{Y_n|Y_{1:n-1}}(y_n^*|y_{1:n-1}^*; \theta)$ is strongly consistently estimated by either the before-resampling estimate,*

$$L_n^{B,\theta} = \frac{\sum_{j=1}^J g_{n,j}^\theta w_{n,j}^{P,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}}, \quad (\text{S103})$$

or by the after-resampling estimate,

$$L_n^{A,\theta} = L_n^\phi \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}}. \quad (\text{S104})$$

where L_n^ϕ is as defined in the various algorithms.

Here, (S103) is a direct consequence of our earlier result that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*; \theta)$. To see (S104), define the corrected weights before resampling and the reference resampling probabilities by

$$w_{n,j}^{FC,\theta} = w_{n,j}^{P,\theta} \frac{g_{n,j}^\theta}{g_{n,j}^\phi}, \quad p_{n,j} = \frac{g_{n,j}^\phi}{J L_n^\phi}, \quad \sum_{j=1}^J p_{n,j} = 1. \quad (\text{S105})$$

The numerator of (S103) can then be written as

$$\sum_{j=1}^J w_{n,j}^{P,\theta} g_{n,j}^\theta = J L_n^\phi \sum_{j=1}^J p_{n,j} w_{n,j}^{FC,\theta}. \quad (\text{S106})$$

Resampling selects $w_{n,j}^{F,\theta} = w_{n,k_j}^{FC,\theta}$. With the pre-resampling particles, weights, and reference densities fixed, the expectation over multinomial resampling satisfies

$$\begin{aligned} \mathbb{E}_{\text{res}} \left[L_n^\phi \sum_{j=1}^J w_{n,j}^{F,\theta} \right] &= J L_n^\phi \sum_{m=1}^J p_{n,m} w_{n,m}^{FC,\theta} \\ &= \sum_{m=1}^J w_{n,m}^{P,\theta} g_{n,m}^\theta. \end{aligned} \quad (\text{S107})$$

The same conditional triangular-array argument as in Lemma S5 gives consistency of this resampling approximation under its boundedness conditions. Dividing by the unchanged prediction-weight sum gives (S104). Using Lemma S8, we obtain a likelihood estimate,

$$L^{A,\theta} = \prod_{n=1}^N \left(L_n^\phi \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \right). \quad (\text{S108})$$

Since $w_{n,j}^{F,\theta} = w_{n+1,j}^{P,\theta}$, this is a telescoping product. The remaining terms are $\sum_{j=1}^J w_{0,j}^{F,\theta} = J$ on the denominator and $\sum_{j=1}^J w_{N,j}^{F,\theta}$ on the numerator. This derives the MOP-1 likelihood estimates.

$L^{B,\theta}$ should generally be preferred in practice, since there is no reason to include the extra variability from resampling when calculating the conditional log likelihood, but it lacks the nice telescoping product that lets us derive exact expressions for the gradient in Theorem 1.

S5 Proof of Theorem 4 via Consistency of Off-Parameter Resampled Gradient Estimates

We now show that, under sufficient local regularity conditions, one can interchange differentiation and the particle limit. We retain the Arzelà–Ascoli argument, working on a compact neighborhood of the evaluation parameter. Consistency is used for the estimates themselves: differentiating a weighted particle estimate also differentiates its particles, weights, and normalizer, so boundedness alone does not make its derivative an empirical average to which Theorem S1 applies.

Theorem S2 (Off-Parameter Particle Filters Yield Strongly Consistent Estimates of Derivatives of Functionals). *Let K be a closed ball of positive radius centered at ϕ and contained in the interior of $\Theta \subset \mathbb{R}^p$. Let $F_J(\theta)$ be scalar particle estimates, such as $\eta_n^J(h_\theta)$, defined on a common probability space. On one probability-one event, suppose every F_J has two continuous derivatives on a neighborhood of K , with*

$$\sup_{J \geq 1} \sup_{\theta \in K} \|\nabla_\theta F_J(\theta)\|_2 \leq G^*, \quad \sup_{J \geq 1} \sup_{\theta \in K} \|\nabla_\theta^2 F_J(\theta)\|_{\text{op}} \leq H^*$$

for finite constants G^, H^* . Suppose $F_J(\theta) \rightarrow h_\theta^*$ almost surely for each fixed $\theta \in K$, where h^* is deterministic. Then h^* is continuously differentiable in the interior of K , and almost surely*

$$\nabla_\theta F_J(\theta) \longrightarrow \nabla_\theta h_\theta^*$$

uniformly on every compact subset of that interior, in particular at $\theta = \phi$.

Proof. For each fixed pair $x, y \in K$, pointwise almost-sure convergence and the gradient bound give $|h_y^* - h_x^*| \leq G^* \|y - x\|_2$. Thus the deterministic function h^* is Lipschitz, and in particular continuous, on K .

Choose a countable dense subset D of K . Intersect the probability-one event of the derivative bounds with the events of convergence at each $\theta \in D$, and fix ω in this intersection. Suppress ω below. By the gradient bound, the functions F_J are uniformly Lipschitz on the convex set K . Convergence on D , compactness of K , and continuity of h^* therefore imply $F_J \rightarrow h^*$ uniformly on K : for any $\epsilon > 0$, choose a sufficiently fine finite net in D , use convergence at its finitely many points, and use the common Lipschitz bound and uniform continuity of h^* between those points.

The gradients are uniformly bounded and, by the Hessian bound, uniformly equicontinuous. Explicitly, for $\theta, \theta' \in K$,

$$\|\nabla_\theta F_J(\theta) - \nabla_{\theta'} F_J(\theta')\|_2 \leq H^* \|\theta - \theta'\|_2. \quad (\text{S109})$$

Thus, by Arzelà–Ascoli, every subsequence of gradients has a further subsequence converging uniformly on K to a continuous function g . We claim that every such limit is $\nabla_\theta h^*$ in the interior of K . For $x, y \in K$, the fundamental theorem of calculus gives

$$F_J(y) - F_J(x) = \int_0^1 \nabla_\theta F_J(x + t(y - x))^\top (y - x) dt. \quad (\text{S110})$$

Passing to the limit along the convergent subsequence yields

$$h_y^* - h_x^* = \int_0^1 g(x + t(y - x))^\top (y - x) dt. \quad (\text{S111})$$

Taking $y = x + te_i$ for each coordinate direction and letting $t \rightarrow 0$ shows that $\partial_{\theta_i} h_x^* = g_i(x)$ for every interior x . Since g is continuous, h^* is continuously differentiable there. Every subsequential gradient limit is therefore the same on the interior, and hence on K by continuity. Relative compactness and uniqueness of this limit imply convergence of the full gradient sequence. In particular, almost surely, for interior θ ,

$$\lim_{J \rightarrow \infty} \nabla_\theta F_J(\theta) = \nabla_\theta \lim_{J \rightarrow \infty} F_J(\theta) = \nabla_\theta h_\theta^*, \quad (\text{S112})$$

with convergence uniform on every compact subset of the interior, as required. $\square$

For $F_J(\theta) = \eta_n^J(h_\theta)$, the conclusion is convergence to $\nabla_\theta \eta_n(h_\theta)$. No identity with $\eta_n(\nabla_\theta h_\theta)$ is asserted, since the target distribution can itself depend on θ .

Theorem 4 follows by taking $F_J(\theta) = \hat{\ell}_J^1(\theta)$ and $h_\theta^* = \ell(\theta)$. Keep the reference parameter ϕ fixed throughout K . For $\alpha = 1$, the off-parameter consistency result and Lemma S8 give $\hat{\mathcal{L}}_J^1(\theta) \rightarrow \mathcal{L}(\theta) > 0$ almost surely for each fixed $\theta \in K$, not only at $\theta = \phi$. The continuous mapping theorem gives the corresponding convergence of the log likelihoods. The neighborhood form of Assumption (A4) supplies their uniform gradient and Hessian bounds, and Assumption (A5) supplies their pathwise smoothness with the simulator draws and reference-filter resampling indices held fixed. Applying the theorem to these log likelihoods establishes score consistency at ϕ directly.

S6 Proof of Theorem 5 (Bias-Variance Analysis)

In this section, we prove various rates on the bias, variance, and MSE of DMOP- α . The earlier truncation argument is retained below for comparison. The DMOP recursion discounts a time- s derivative contribution by its actual age. We prove the DMOP-0 result below and retain the former direct extension to $\alpha > 0$ for comparison. Section S7 obtains the sharper result by first applying the finite-lag variance calculation and then recovering geometric discounting through an exact mixture identity.

First, we relate cloud contraction to the strong-mixing coefficient used in the covariance bounds below.

Lemma S9. Let $\mathcal{F}_m = \sigma(x_{0,1:J}^P, \dots, x_{m,1:J}^P)$ and $\mathcal{G}_n = \sigma(x_{n,1:J}^P, \dots, x_{N,1:J}^P)$ be the past and future sigma-fields of the Markov particle cloud. For $1 \leq \ell \leq N$, define

$$a(\ell) = \sup_{0 \leq m \leq N-\ell} \sup_{A \in \mathcal{F}_m, B \in \mathcal{G}_{m+\ell}} |\Pr(A \cap B) - \Pr(A) \Pr(B)|. \quad (\text{S113})$$

Then

$$a(\ell) \leq \sup_{0 \leq m \leq N-\ell} \beta_{\text{TV}}(\mathbf{M}_{m,m+\ell}). \quad (\text{S114})$$

The same statement applies to the joint state-derivative cloud under (A6-J), using its transition kernel.

Proof. Fix $A \in \mathcal{F}_m$ and $B \in \mathcal{G}_n$, where $n = m + \ell$. By the Markov property, with $h(x) = \Pr(B \mid x_{n,1:J}^P = x) \in [0, 1]$,

$$\Pr(B \mid \mathcal{F}_m) = (\mathbf{M}_{m,n} h)(x_{m,1:J}^P). \quad (\text{S115})$$

The expectation of this conditional probability is $\Pr(B)$, so

$$\begin{aligned} |\Pr(A \cap B) - \Pr(A) \Pr(B)| &= |\mathbb{E} [\mathbf{1}_A \{(\mathbf{M}_{m,n} h)(x_{m,1:J}^P) - \Pr(B)\}]| \\ &\leq \text{osc}(\mathbf{M}_{m,n} h) \leq \beta_{\text{TV}}(\mathbf{M}_{m,n}) \text{osc}(h) \\ &\leq \beta_{\text{TV}}(\mathbf{M}_{m,n}). \end{aligned} \quad (\text{S116})$$

Taking the suprema proves the claim. The coefficient is defined through sigma-fields, so the argument does not require joint Lebesgue densities. $\square$

The same argument bounds dependence between past and future windows of simulation and resampling updates. A past window ending after resampling at time b is absorbed into the next prediction cloud at $b + 1$; condition there and at the prediction cloud starting the future window, and use the intervening cloud kernel. Fresh update variables are conditionally independent of earlier updates given the incoming cloud. Under (A6), apply this to the physical cloud and the derivative-age windows in Equation (S275); under (A6-J), use the joint cloud. The number of intervening transitions determines the lag in $a(\ell)$.

As a warm-up, we consider a variance bound for DMOP-0. Note that we made Assumptions (A2) and (A3) to leverage the results from Karjalainen et al. (2023) on the forgetting of the particle filter. The $O\{N(1 + \log J)^2/J\}$ variance bound for DMOP-0 additionally uses either (A6) or (A6-J), stated in Section S7. Under (A6-J) the derivative contribution is a bounded current function of the joint state-gradient pair; under (A6) we first expand its simulator history. The former truncation extension to $\alpha > 0$ is retained for comparison. We start by decomposing the MOP- α estimator as follows, for the case $\theta = \phi$,

$$\hat{\mathcal{L}}(\theta) := \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N \left(L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \right) = \prod_{n=1}^N \left(L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J (w_{n-1,j}^{F,\theta})^\alpha} \right). \quad (\text{S117})$$

Now, to illustrate the proof strategy for the general case in an easier context, we first analyze the special case when $\alpha = 0$. We now prove the variance bound for this case, presented below. To our knowledge, this result for the variance of the gradient estimate of Naesseth et al. (2018) is novel.

Theorem S3 (Variance of MOP-0 Gradient Estimate). *Suppose the observations are fixed, $J \geq 2$, $\theta = \phi$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Suppose also that either (A6) or (A6-J) holds. Then the variance of DMOP-0, i.e. the algorithm of Naesseth et al. (2018), is*

$$\text{Var}(\nabla_{\theta} \hat{\ell}^0(\theta)) \lesssim \frac{NpG'(\theta)^2(1 + \log J)^2}{J}. \quad (\text{S118})$$

Proof. First assume (A6-J). In this calculation, a current particle means the pair $(X_n, \partial_{\theta} X_n)$, and the particle-cloud transition is the one induced by its joint kernel. When $\alpha = 0$, we have

$$\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A, \theta, 0} = \prod_{n=1}^N \left(L_n^{\phi} \cdot \frac{\sum_{j=1}^J w_{n,j}^{F, \theta, 0}}{\sum_{j=1}^J w_{n,j}^{P, \theta, 0}} \right) \quad (\text{S119})$$

$$= \prod_{n=1}^N \left(L_n^{\phi} \cdot \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) = \prod_{n=1}^N \left(L_n^{\phi} \cdot \frac{1}{J} \sum_{j=1}^J \frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}} \right). \quad (\text{S120})$$

Similarly to the proof of Lemma S2, we define

$$s_{n,j} = \frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}}, \quad (\text{S121})$$

and from the exact same arguments conclude that its gradient when $\theta = \phi$ is therefore

$$\nabla_{\theta} \hat{\ell}^0(\theta) := \sum_{n=1}^N \nabla_{\theta} \log \left(L_n^{\phi} \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) \quad (\text{S122})$$

$$= \sum_{n=1}^N \frac{\nabla_{\theta} \left(L_n^{\phi} \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)}{\left(L_n^{\phi} \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)} \quad (\text{S123})$$

$$= \sum_{n=1}^N \frac{\sum_{j=1}^J \nabla_{\theta} s_{n,j}}{\sum_{j=1}^J s_{n,j}} \quad (\text{S124})$$

$$= \sum_{n=1}^N \frac{1}{J} \sum_{j=1}^J \frac{\nabla_{\theta} f_{Y_n|X_n}(y_n^*|x_{n,j}^{F, \theta}; \theta)}{f_{Y_n|X_n}(y_n^*|x_{n,j}^{F, \phi}; \phi)} \quad (\text{S125})$$

$$= \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F, \theta}; \theta) \right), \quad (\text{S126})$$

where we use the derivative of the logarithm, observe that $\sum_{j=1}^J s_{n,j} = J$ when $\theta = \phi$, and use the derivative of the logarithm where $\theta = \phi$ again. We do this to establish that the gradient of the log-likelihood estimate is given by the sum of terms over all N and J :

$$\nabla_{\theta} \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F, \theta}; \theta) \right). \quad (\text{S127})$$

Therefore, for a given θ_i in the parameter vector θ ,

$$\text{Var} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}^0(\theta) \right) = \frac{1}{J^2} \text{Var} \left(\sum_{n=1}^N \sum_{j=1}^J \frac{\partial}{\partial \theta_i} \log \left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right) \right) \quad (\text{S128})$$

$$\begin{aligned} &= \frac{1}{J^2} \sum_{n=1}^N \text{Var} \left(\sum_{j=1}^J \frac{\partial}{\partial \theta_i} \log \left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right) \right) \\ &\quad + 2 \sum_{m < n} \text{Cov} \left(\frac{1}{J} \sum_{j=1}^J \frac{\partial}{\partial \theta_i} \log \left(f_{Y_m|X_m}(y_m^*|x_{m,j}^{F,\theta}; \theta) \right), \frac{1}{J} \sum_{j=1}^J \frac{\partial}{\partial \theta_i} \log \left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right) \right) \\ &= \sum_{n=1}^N \text{Var} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right) + 2 \sum_{m < n} \text{Cov} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right). \end{aligned} \quad (\text{S129})$$

Here (A6-J) and (A3) ensure strong mixing of the joint model. We know from Theorem 3.1 of Karjalainen et al. (2023) that when $\mathbf{M}_{n,n+k}$ is the k -step Markov operator from timestep n and $\beta_{\text{TV}}(M) = \sup_{x,y \in E} \|M(x, \cdot) - M(y, \cdot)\|_{\text{TV}} = \sup_{\mu, \nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\text{TV}}}{\|\mu - \nu\|_{\text{TV}}}$ is the Dobrushin contraction coefficient of a Markov operator,

$$\beta_{\text{TV}}(\mathbf{M}_{n,n+k}) \leq (1 - \epsilon)^{\lfloor k/(c \log(J)) \rfloor}, \quad (\text{S130})$$

i.e. the mixing time of this joint particle filter is $O(\log(J))$, where ϵ and c depend on $\bar{M}_D, \underline{M}_D, \bar{G}, \underline{G}$. The physical particle cloud satisfies a bound of the same form under (A2), with its own constants; that version is used under (A6) below.

Under (A6-J), apply Lemma 2.3 of Karjalainen et al. (2023) to the joint prediction cloud. Conditional resampling and the Bayes-ratio calculation give the centered L^4 bound below. Theorem 3.1 and Lemma S9 bound the strong-mixing coefficient $a(k)$ of the score sequence. The shift by one accounts for the endpoint resampling. At $\alpha = 0$ these results give

$$\begin{aligned} \left\| \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) - \mathbb{E} \left[\frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right] \right\|_{L^4} &\leq \frac{CG'(\theta)}{\sqrt{J}}, \\ a(k) &\leq (1 - \epsilon)^{\lfloor (k-1)/[c\{1+\log J\}] \rfloor}. \end{aligned} \quad (\text{S131})$$

Davydov's inequality must be applied to the centered variables. Since covariance is unchanged by centering, Equation (S131) gives

$$\text{Cov} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right) \leq (1 - \epsilon)^{\frac{1}{2} \lfloor \frac{|n-m|-1}{c\{1+\log(J)\}} \rfloor} \frac{G'(\theta)^2}{J}. \quad (\text{S132})$$

The geometric sum is bounded by $C(1 + \log J)$. Putting it all together, we see that

$$\text{Var} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}^0(\theta) \right) = \sum_{n=1}^N \text{Var} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right) + 2 \sum_{m < n} \text{Cov} \left(\frac{\partial}{\partial \theta_i} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right) \quad (\text{S133})$$

$$\leq \frac{NG'(\theta)^2}{J} + 2 \sum_{n=1}^{N-1} (N-n)(1 - \epsilon)^{\frac{1}{2} \lfloor \frac{n-1}{c\{1+\log(J)\}} \rfloor} \frac{G'(\theta)^2}{J} \quad (\text{S134})$$

$$\leq \frac{G'(\theta)^2}{J} \left(N + 2 \sum_{n=1}^{N-1} (N-n)(1-\epsilon)^{\frac{1}{2} \lfloor \frac{n-1}{c\{1+\log(J)\}} \rfloor} \right) \quad (\text{S135})$$

$$\lesssim \frac{G'(\theta)^2 N(1+\log J)}{J}. \quad (\text{S136})$$

It then follows that as $\theta \in \Theta \subseteq \mathbb{R}^p$,

$$\text{Var}(\nabla_{\theta} \hat{\ell}^0(\theta)) \lesssim \frac{NpG'(\theta)^2(1+\log J)}{J}. \quad (\text{S137})$$

Under (A6), apply the age decomposition and covariance calculation in Steps 1a–1d of Lemma S13 with $k = 0$. In particular, Equation (S281) gives

$$\text{Var}\{\nabla_{\theta} \hat{\ell}^0(\theta)\} = \text{Var}\{\nabla_{\theta} \hat{s}_0^1(\theta)\} \lesssim NpG'(\theta)^2(1+\log J)^2/J.$$

That argument uses the physical-cloud mixing bound above and bounded finite-history inputs, not mixing of the full gradient cloud under (A6). $\square$

The factor of N comes from using the unnormalized gradient. Dividing by $\sqrt{N}$ gives a variance bound independent of the horizon. Dividing by N gives a centered L^2 error bound of order $(1+\log J)/\sqrt{NJ}$.

The preceding calculation gives the order $NpG'(\theta)^2(1+\log J)^2/J$ for the DMOP-0 baseline. It is not asserted to be a lower bound for every model or every value of α .

Theorem S4 (MSE and variance bounds for DMOP- α). *Suppose the observations are fixed, $J \geq 2$, $\alpha \in [0, 1]$, $\theta = \phi$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Suppose also that either (A6) or (A6-J), stated in Section S7, holds. Then*

$$\max \left\{ \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2^2, \text{Var} \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\} \right\} \lesssim pG'(\theta)^2 \left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2} \right) \left(\sum_{r=0}^{N-1} \alpha^r \right)^2. \quad (\text{S138})$$

For $0 \leq \alpha < 1$,

$$\begin{aligned} & \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2^2 \\ & \lesssim pG'(\theta)^2 \left[\left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2} \right) \left(\sum_{r=0}^{N-1} \alpha^r \right)^2 \right. \\ & \quad \left. + \frac{(1-\sqrt{2(1-\varrho)})^2(1-\alpha)^2}{(1+\sqrt{2(1-\varrho)})^2} \left\{ N-1 + \sum_{q=1}^{N-2} (N-1-q) \left(\alpha \frac{2\varrho-1}{3-2\varrho} \right)^q \right\}^2 \right] \\ & \lesssim pG'(\theta)^2 \left[\frac{N(1+\log J)^2}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2} \right. \\ & \quad \left. + N^2 \left\{ \frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}} \frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)} \right\}^2 \right]. \end{aligned} \quad (\text{S139})$$

For the same range of α , the population component of the fixed-observation bias satisfies

$$\|\nabla_{\theta}\ell^{\alpha}(\theta) - \nabla_{\theta}\ell(\theta)\|_2 \leq 2(N-1)\sqrt{p}G'(\theta)\frac{1 - \sqrt{2(1-\varrho)}}{1 + \sqrt{2(1-\varrho)}}\frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)}. \quad (\text{S140})$$

Here $\varrho = 1 - \underline{M}/(2\bar{M}) \in [\frac{1}{2}, 1)$ is the same constant as in Equation (S193). The sum beginning at $q = 1$ is empty for $N \leq 2$; at $\varrho = \frac{1}{2}$ the displayed population terms are zero. The constants suppressed by $\lesssim$ depend only on the fixed model bounds and not on N, J , or α . The constants may additionally depend on (A6) or (A6-J). The endpoints satisfy Equations (S333) and (S334); in particular,

$$\mathbb{E}\|\nabla_{\theta}\hat{\ell}^1(\theta) - \nabla_{\theta}\ell(\theta)\|_2^2 \lesssim pG'(\theta)^2 \min\{N^2, N^3/J\}.$$

The population inequality itself needs neither additional condition. Section S7 gives the complete proof. The particle argument first bounds each finite lag truncation and then uses the exact geometric-mixture identity; it does not replace DMOP- α by a hard-cutoff estimator.

Proof. Lemma S13 proves Equation (S138). Theorem S5 proves Equation (S139), and Lemma S14 proves Equation (S140). $\square$

S7 Sharper Bias-Variance Analysis

Write $\nabla_{\theta}\ell_n^{\alpha}(\theta)$ for the n th conditional score of the population recursion. At $\theta = \phi$ and $\alpha = 1$, it equals $\nabla_{\theta}\ell_n(\theta)$. Along each selected lineage,

$$\nabla_{\theta} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} = \alpha \nabla_{\theta} \log w_{n-1,k_j}^{F,\theta} \Big|_{\theta=\phi} + \nabla_{\theta} \log g_{n,k_j}^{\theta} \Big|_{\theta=\phi}. \quad (\text{S141})$$

The last term is the measurement gradient h_n defined in Equation (S155). This follows by differentiating

$$\begin{aligned} \log w_{n,j}^{P,\theta} &= \alpha \log w_{n-1,j}^{F,\theta}, \\ \log w_{n,j}^{F,\theta} &= \alpha \log w_{n-1,k_j}^{F,\theta} + \log g_{n,k_j}^{\theta} - \log g_{n,k_j}^{\phi}. \end{aligned} \quad (\text{S142})$$

at $\theta = \phi$. The observations are fixed throughout this section; expectations and variances are over the particle-filter randomness.

(A6) Measurement-sensitivity decay. The simulator is C^1 in its incoming state. For a continuation $X_n^{\theta}(x_t, \omega_{t+1:n})$ from x_t , with $X_t^{\theta}(x_t) = x_t$,

$$\left\| D_{x_t} \log f_{Y_n|X_n}(y_n^* | X_n^{\theta}(x_t, \omega_{t+1:n}); \theta) \right\|_2 \leq C\kappa^{n-t}, \quad C < \infty, \quad 0 < \kappa < 1. \quad (\text{S143})$$

The bound is uniform in $t \leq n$, states, draws, and θ near ϕ . The parameter derivatives of the initial state, simulator, and log measurement density are bounded, holding incoming states fixed.

(A6-J) Joint-transition alternative. The log measurement density is C^1 in (x, θ) . The joint process (X_n^θ, D_n^θ) , $D_n^\theta = \partial_\theta X_n^\theta$, has transition densities satisfying

$$0 < \underline{M}_D \leq f_{(X_n, D_n)|(X_{n-1}, D_{n-1})}(x', v' | x, v; \theta) \leq \bar{M}_D < \infty \quad (\text{S144})$$

relative to a common finite measure on the invariant joint state space, uniformly in time, admissible pairs, and θ near ϕ .

The particle bounds assume either (A6) or (A6-J); the consistency results and population Lemmas S10 and S14 retain (A1)–(A5). Neither additional condition requires density evaluation by the algorithm. Their constants enter the particle bounds, while ϱ retains its physical-model definition. Under (A6), enlarge $G'(\theta)$ to bound the parameter derivatives coordinatewise, using Euclidean norms for state derivatives.

Sufficient condition for (A6). Suppose the parameter derivatives in (A6) are bounded, the spatial log measurement derivatives are bounded by L_g , and the one-step simulator Jacobians are bounded by L . If, for fixed $b \geq 1$ and $q \in (0, 1)$, $\|\partial X_{t+b}^\theta / \partial X_t^\theta\|_{\text{op}} \leq q$ uniformly with the draws fixed, then writing $n - t = mb + v$, $0 \leq v < b$, gives

$$\left\| D_{x_t} \log f_{Y_n|X_n}(y_n^* | X_n^\theta; \theta) \right\|_2 \leq L_g L^v q^m \leq C \kappa^{n-t}, \quad \kappa = q^{1/b} \in (0, 1). \quad (\text{S145})$$

Here $C = L_g \max_{0 \leq v < b} (L/\kappa)^v$; a zero block Jacobian permits any $q \in (0, 1)$.

Example: physical mixing without (A6). Consider the annulus $\mathcal{X} = \{x \in \mathbb{R}^2 : 1 \leq \|x\|_2 \leq 2\}$. Let R_u denote planar rotation by u , and let independent simulator draws satisfy $U_n \sim \text{Unif}[0, 2\pi]$ and $V_n \sim \text{Unif}[1, 2]$. Set

$$X_0^\theta = R_\theta x_0, \quad X_n^\theta = V_n R_{U_n} \frac{X_{n-1}^\theta}{\|X_{n-1}^\theta\|_2}, \quad \Pr(Y_n = 1 | X_n = x) = \frac{1}{2} + \frac{x_1}{4\|x\|_2}. \quad (\text{S146})$$

The physical transition forgets the incoming state in one step: its Lebesgue density on $\mathcal{X}$ is

$$f_{X_n|X_{n-1}}(x' | x; \theta) = \frac{1}{2\pi\|x'\|_2}, \quad \frac{1}{4\pi} \leq f_{X_n|X_{n-1}} \leq \frac{1}{2\pi}. \quad (\text{S147})$$

Both measurement probabilities lie in $[1/4, 3/4]$. For fixed draws, $X_n^\theta = R_\theta X_n^0$, so all parameter derivatives exist and are bounded; in particular the first two log measurement derivatives have magnitudes at most 1 and 2. The likelihood is smooth, and the MOP derivatives are bounded for each fixed horizon. Thus (A1)–(A5) hold.

Nevertheless, fix the observations $y_n^* = 1$. For $n > t$, the composition is

$$X_n^\theta(x, \omega_{t+1:n}) = V_n R_{U_{t+1} + \dots + U_n} \frac{x}{\|x\|_2}. \quad (\text{S148})$$

Its final angle is uniform, and a fixed-random-number change in the incoming angle is preserved. Differentiating the displayed measurement probability therefore gives

$$\mathbb{E} \left\| D_x \log f_{Y_n|X_n}(1 | X_n^\theta(x, \omega_{t+1:n}); \theta) \right\|_2^2 = \frac{1}{2\pi\|x\|_2^2} \int_0^{2\pi} \frac{\sin^2 u}{(2 + \cos u)^2} du > 0. \quad (\text{S149})$$

The right side is independent of $n - t$, so (A6) fails despite (A1)–(A5). Here $D_n^\theta = R_{\pi/2} X_n^\theta$, and the joint transition has the physical density relative to $dx' \delta_{R_{\pi/2} x'}(dv')$. Thus (A6-J) holds and the particle bounds still apply. In particular, (A6-J) does not imply (A6).

Differentiation and notation. By (A5), differentiation in θ holds the simulator draws and reference-filter resampling indices fixed and includes the dependence of X_n^θ on θ . Assumption (A3) gives, uniformly in n, j and θ near ϕ ,

$$\left\| \nabla_\theta \log g_{n,j}^{A,F,\theta} \right\|_\infty \leq G'(\theta). \quad (\text{S150})$$

The product rule and Equation (S150) give

$$\begin{aligned} & \left\| \nabla_\theta \prod_{s=1}^t G_s(X_s) \right\|_\infty \\ &= \left\| \prod_{s=1}^t G_s(X_s) \sum_{s=1}^t \nabla_\theta \log G_s(X_s) \right\|_\infty \\ &\leq \prod_{s=1}^t G_s(X_s) \sum_{s=1}^t \left\| \nabla_\theta \log G_s(X_s) \right\|_\infty \\ &\leq t \bar{G}^t G'(\theta). \end{aligned} \quad (\text{S151})$$

For fixed t , the mean-value theorem bounds the coordinatewise difference quotients by the integrable constant $t \bar{G}^t G'(\theta)$. Dominated convergence therefore gives

$$\nabla_\theta \mathbb{E}_U \left[\prod_{s=1}^t G_s(X_s) \right] = \mathbb{E}_U \left[\nabla_\theta \prod_{s=1}^t G_s(X_s) \right]. \quad (\text{S152})$$

At $\theta = \phi$, let ρ_t be the posterior law of the simulator history through time t , including its base variables. Its $X_{0,t}$ -marginal is the posterior path law, and

$$\rho_t(h) = \frac{\mathbb{E}_U [h \prod_{s=1}^t G_s(X_s)]}{\mathbb{E}_U [\prod_{s=1}^t G_s(X_s)]}. \quad (\text{S153})$$

Here $\mathbb{E}_U$ integrates over the base simulator variables before measurement weighting. Expectations in particle-error calculations are instead over the particle-filter law, including resampling. Future simulator draws are independent of the past.

Iterating the weight recursion at $\theta = \phi$ gives

$$\nabla_\theta \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} = \sum_{r=0}^{n-1} \alpha^r \nabla_\theta \log g_{n-r,j}^{A,F,\theta} \Big|_{\theta=\phi}. \quad (\text{S154})$$

Write

$$h_k = \nabla_\theta \log g_k^{A,F,\theta} \Big|_{\theta=\phi} \quad (\text{S155})$$

for the measurement gradient on an ancestral path. Following Chan et al. (2018), let

$$m(\xi_t) = \frac{1}{J} \sum_{j=1}^J \delta_{\xi_{t,j}}, \quad m(\xi_t)(h_k) = \frac{1}{J} \sum_{j=1}^J h_k(\xi_{t,j}), \quad 1 \leq k \leq t, \quad (\text{S156})$$

where $\xi_{t,j}$ is the simulator history along particle j 's ancestry through time t , including the base variables needed to evaluate h_k and its pathwise derivative contributions. Thus $m(\xi_t)(h_k)$ averages the time- k gradients over these paths.

At $\theta = \phi$, all weights equal one. Differentiating the two sample averages in the definition of $L_n^{A,\theta,\alpha}$ gives

$$\begin{aligned}\nabla_\theta \hat{\ell}_n^\alpha(\theta) &= \frac{1}{J} \sum_{j=1}^J \nabla_\theta \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \frac{1}{J} \sum_{j=1}^J \nabla_\theta \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} \\ &= \frac{1}{J} \sum_{j=1}^J \nabla_\theta \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \frac{\alpha}{J} \sum_{j=1}^J \nabla_\theta \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi},\end{aligned}\tag{S157}$$

where the second equality uses $w_{n,j}^{P,\theta} = (w_{n-1,j}^{F,\theta})^\alpha$. In the MOP- α recursion, the discounted-weight accumulation and process propagation preserve the index j and occur before the time- n resampling step. The resampling index k_j enters only afterward, through

$$w_{n,j}^{F,\theta} = w_{n,k_j}^{P,\theta} \frac{g_{n,k_j}^\theta}{g_{n,k_j}^\phi}.\tag{S158}$$

Consequently, the second average in Equation (S157) is pathwise

$$\frac{1}{J} \sum_{j=1}^J \nabla_\theta \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} = \frac{\alpha}{J} \sum_{j=1}^J \nabla_\theta \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi},\tag{S159}$$

which is the empirical average over the time- $(n-1)$ filtered genealogy. Substituting Equation (S154) into Equation (S157) and changing the index in the second sum give

$$\begin{aligned}\nabla_\theta \hat{\ell}_n^\alpha(\theta) &= \sum_{r=0}^{n-1} \alpha^r m(\xi_n)(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r m(\xi_{n-1})(h_{n-r}) \\ &= m(\xi_n)(h_n) + \sum_{r=1}^{n-1} \alpha^r \{m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r})\}.\end{aligned}\tag{S160}$$

Consequently, define

$$\begin{aligned}\Delta_{n,0}^J &= m(\xi_n)(h_n), & \Delta_{n,0} &= \rho_n(h_n), \\ \Delta_{n,r}^J &= m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r}), & \Delta_{n,r} &= \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}), \quad 1 \leq r < n.\end{aligned}\tag{S161}$$

For the population coefficients, Equations (S152) and (S155) give, at $\theta = \phi$,

$$\begin{aligned}\nabla_\theta \log \mathbb{E}_U \left[\prod_{s=1}^n G_s(X_s) \right] &= \frac{\mathbb{E}_U [\prod_{s=1}^n G_s(X_s) \sum_{s=1}^n h_s]}{\mathbb{E}_U [\prod_{s=1}^n G_s(X_s)]} \\ &= \sum_{s=1}^n \rho_n(h_s).\end{aligned}\tag{S162}$$

Subtracting the same identity at time $n - 1$ gives

$$\begin{aligned}
\nabla_{\theta} \ell_n(\theta) &= \sum_{s=1}^n \rho_n(h_s) - \sum_{s=1}^{n-1} \rho_{n-1}(h_s) \\
&= \rho_n(h_n) + \sum_{r=1}^{n-1} \{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\} \\
&= \sum_{r=0}^{n-1} \Delta_{n,r}.
\end{aligned} \tag{S163}$$

Finally, differentiating the two population weighted averages in the deterministic MOP- α recursion and using Equation (S152) gives

$$\begin{aligned}
\nabla_{\theta} \ell_n^{\alpha}(\theta) &= \sum_{r=0}^{n-1} \alpha^r \rho_n(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r \rho_{n-1}(h_{n-r}) \\
&= \rho_n(h_n) + \sum_{r=1}^{n-1} \alpha^r \{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\} \\
&= \sum_{r=0}^{n-1} \alpha^r \Delta_{n,r}.
\end{aligned} \tag{S164}$$

Lemma S10 (Lagwise first-derivative forgetting). *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), there are constants $C < \infty$ and $\varrho \in (0, 1)$ such that, for $0 \leq r < n$,*

$$\|\Delta_{n,r}\|_2 \leq C\sqrt{p} G'(\theta) \varrho^r. \tag{S165}$$

Proof. We use the posterior update and backward-kernel contraction.

Step 1: one-step posterior update. First suppose that $1 \leq r < n$, and put $k = n - r$. Fix a coordinate $i \in \{1, \dots, p\}$, and let $h_{k,i}$ be the corresponding coordinate of h_k in Equation (S155). By Equation (S150),

$$|h_{k,i}| \leq G'(\theta). \tag{S166}$$

Because $h_{k,i}$ is measurable with respect to the canonical extended path through time $k \leq n - 1$, Equation (S153) and conditional expectation with respect to the new base simulator variable give

$$\begin{aligned}
\rho_n(h_{k,i}) &= \frac{\mathbb{E}_U \left[h_{k,i} \prod_{s=1}^{n-1} G_s(X_s) \mathbb{E}_U \{G_n(X_n) \mid X_{n-1}\} \right]}{\mathbb{E}_U \left[\prod_{s=1}^{n-1} G_s(X_s) \mathbb{E}_U \{G_n(X_n) \mid X_{n-1}\} \right]} \\
&= \frac{\rho_{n-1} \{h_{k,i} M_n(G_n)\}}{\rho_{n-1} M_n(G_n)},
\end{aligned} \tag{S167}$$

where the first equality uses the independence of the new base simulator variable from the path through time $n - 1$. The second equality first uses the identity below and then divides its numerator and denominator by $\mathbb{E}_U \{\prod_{s=1}^{n-1} G_s(X_s)\}$. Using the manuscript's kernel notation, the required identity is

$$M_n(G_n)(x_{n-1}) = \int M_n(x_{n-1}, dx_n) G_n(x_n). \tag{S168}$$

The bounds on G_n and the fact that M_n is a Markov kernel imply for every x_{n-1} that

$$M_n(G_n)(x_{n-1}) \geq \underline{G} \int M_n(x_{n-1}, dx_n) = \underline{G}, \quad (\text{S169})$$

$$M_n(G_n)(x_{n-1}) \leq \bar{G} \int M_n(x_{n-1}, dx_n) = \bar{G}. \quad (\text{S170})$$

Step 2: backward-kernel contraction. For $s \geq 1$, the posterior backward kernel from X_s to X_{s-1} is

$$B_s(x_s, dx_{s-1}) = \frac{\pi_{s-1}(dx_{s-1})f_{X_s|X_{s-1}}(x_s | x_{s-1}; \theta)}{\int \pi_{s-1}(du)f_{X_s|X_{s-1}}(x_s | u; \theta)}. \quad (\text{S171})$$

Assumption (A2) gives the following minorization for every measurable set A :

$$\begin{aligned} B_s(x_s, A) &= \frac{\int_A \pi_{s-1}(dx_{s-1})f_{X_s|X_{s-1}}(x_s | x_{s-1}; \theta)}{\int \pi_{s-1}(du)f_{X_s|X_{s-1}}(x_s | u; \theta)} \\ &\geq \frac{\underline{M} \pi_{s-1}(A)}{\bar{M} \int \pi_{s-1}(du)} \\ &= \frac{\underline{M}}{\bar{M}} \pi_{s-1}(A). \end{aligned} \quad (\text{S172})$$

Put $\epsilon = \underline{M}/\bar{M}$. If $\epsilon < 1$, Equation (S172) permits the decomposition

$$B_s(x_s, \cdot) = \epsilon \pi_{s-1}(\cdot) + (1 - \epsilon) R_s(x_s, \cdot), \quad (\text{S173})$$

where

$$R_s(x_s, A) = \frac{B_s(x_s, A) - \epsilon \pi_{s-1}(A)}{1 - \epsilon} \quad (\text{S174})$$

is a Markov kernel. Hence, for every bounded real function f and any x_s, x'_s ,

$$\begin{aligned} B_s(f)(x_s) - B_s(f)(x'_s) &= (1 - \epsilon) \{R_s(f)(x_s) - R_s(f)(x'_s)\}, \\ |B_s(f)(x_s) - B_s(f)(x'_s)| &\leq (1 - \epsilon) \text{osc}(f). \end{aligned} \quad (\text{S175})$$

If $\epsilon = 1$, Equation (S172) implies $B_s(x_s, \cdot) = \pi_{s-1}(\cdot)$, so the left-hand side is zero. Taking the supremum over x_s, x'_s gives

$$\text{osc}\{B_s(f)\} \leq (1 - \epsilon) \text{osc}(f). \quad (\text{S176})$$

Step 3: project and propagate the time- k contribution. For the historical gradient $h_{k,i}$, define

$$\bar{h}_{k,i}(X_k) = \mathbb{E}_{\rho_k}(h_{k,i} | X_k). \quad (\text{S177})$$

To identify this conditional expectation under later posterior laws, for $t \geq k$ let

$$L_{k,t}(X_k) = \mathbb{E}_U \left\{ \prod_{s=k+1}^t G_s(X_s) \middle| X_k \right\}, \quad L_{k,k} = 1. \quad (\text{S178})$$

For every bounded function q of X_k , the Markov property and Equation (S153) give

$$\begin{aligned}\rho_t\{h_{k,i}q(X_k)\} &= \frac{\mathbb{E}_U \left[h_{k,i}q(X_k) \prod_{s=1}^k G_s(X_s) L_{k,t}(X_k) \right]}{\mathbb{E}_U \left[\prod_{s=1}^k G_s(X_s) L_{k,t}(X_k) \right]} \\ &= \frac{\mathbb{E}_U \left[\bar{h}_{k,i}(X_k) q(X_k) \prod_{s=1}^k G_s(X_s) L_{k,t}(X_k) \right]}{\mathbb{E}_U \left[\prod_{s=1}^k G_s(X_s) L_{k,t}(X_k) \right]} \\ &= \rho_t\{\bar{h}_{k,i}(X_k) q(X_k)\}.\end{aligned}\tag{S179}$$

The second equality is the defining conditional-expectation identity in Equation (S177), multiplied by the X_k -measurable function $q(X_k)L_{k,t}(X_k)$. Hence

$$\mathbb{E}_{\rho_t}(h_{k,i} \mid X_k) = \bar{h}_{k,i}(X_k), \quad t \geq k.\tag{S180}$$

Repeated conditioning through the physical backward kernels therefore gives

$$\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = x_{n-1}\} = (B_{n-1}B_{n-2} \cdots B_{k+1}\bar{h}_{k,i})(x_{n-1}).\tag{S181}$$

There are $n - 1 - k = r - 1$ backward kernels in this display. Moreover, Equation (S166) gives

$$\begin{aligned}\|\bar{h}_{k,i}\|_\infty &\leq \|h_{k,i}\|_\infty \leq G'(\theta), \\ \text{osc}(\bar{h}_{k,i}) &\leq 2\|\bar{h}_{k,i}\|_\infty \\ &\leq 2G'(\theta).\end{aligned}\tag{S182}$$

If $r = 1$, Equation (S181) contains no backward kernel. If $r \geq 2$, applying Equation (S176) successively gives

$$\begin{aligned}\text{osc}[\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = \cdot\}] \\ \leq 2G'(\theta) \begin{cases} 1, & r = 1, \\ (1 - \epsilon)^{r-1}, & r \geq 2. \end{cases}\end{aligned}\tag{S183}$$

Step 4: covariance identity and geometric bound. To finish, abbreviate the conditional expectation on the left of Equation (S183) by H , only for the next calculation. Since the X_{n-1} -marginal of ρ_{n-1} is π_{n-1} , the tower property gives

$$\rho_{n-1}(h_{k,i}) = \pi_{n-1}(H),\tag{S184}$$

$$\rho_{n-1}\{h_{k,i}M_n(G_n)\} = \pi_{n-1}\{HM_n(G_n)\}.\tag{S185}$$

Substituting Equations (S184)–(S185) into Equation (S167) and then subtracting $\rho_{n-1}(h_{k,i})$ gives the exact covariance identity

$$\begin{aligned}\Delta_{n,r,i} &= \rho_n(h_{k,i}) - \rho_{n-1}(h_{k,i}) \\ &= \frac{\pi_{n-1}\{HM_n(G_n)\} - \pi_{n-1}(H)\pi_{n-1}M_n(G_n)}{\pi_{n-1}M_n(G_n)}\end{aligned}$$

$$= \frac{\pi_{n-1}[\{H - \pi_{n-1}(H)\}\{M_n(G_n) - \pi_{n-1}M_n(G_n)\}]}{\pi_{n-1}M_n(G_n)}. \quad (\text{S186})$$

For any probability measure μ and bounded real f ,

$$|f(x) - \mu(f)| \leq \text{osc}(f). \quad (\text{S187})$$

The first positive lag also inherits one contraction from the forward transition. To see this directly, fix x^* . Assumption (A2) gives, for every measurable set A ,

$$\begin{aligned} M_n(x, A) &= \int_A f_{X_n|X_{n-1}}(x_n | x; \theta) dx_n \\ &\geq \underline{M} \int_A dx_n \\ &\geq \frac{\underline{M}}{\overline{M}} \int_A f_{X_n|X_{n-1}}(x_n | x^*; \theta) dx_n \\ &= \epsilon M_n(x^*, A). \end{aligned} \quad (\text{S188})$$

If $\epsilon < 1$, Equation (S188) permits the decomposition

$$M_n(x, \cdot) = \epsilon M_n(x^*, \cdot) + (1 - \epsilon) R_n(x, \cdot), \quad (\text{S189})$$

where R_n is a Markov kernel. Therefore, for any x, x' ,

$$\begin{aligned} M_n(G_n)(x) - M_n(G_n)(x') &= (1 - \epsilon)\{R_n(G_n)(x) - R_n(G_n)(x')\}, \\ |M_n(G_n)(x) - M_n(G_n)(x')| &\leq (1 - \epsilon) \text{osc}(G_n) \\ &\leq (1 - \epsilon)(\bar{G} - \underline{G}). \end{aligned} \quad (\text{S190})$$

Taking the supremum over x, x' gives

$$\text{osc}\{M_n(G_n)\} \leq (1 - \epsilon)(\bar{G} - \underline{G}). \quad (\text{S191})$$

If $\epsilon = 1$, Equation (S188) implies that $M_n(x, \cdot)$ does not depend on x , so the left side of Equation (S191) is zero and the same bound holds. Taking absolute values in Equation (S186), using Equations (S187)–(S191), and then using Equation (S183), yields

$$\begin{aligned} |\Delta_{n,r,i}| &\leq \frac{\text{osc}(H) \text{osc}\{M_n(G_n)\}}{\underline{G}} \\ &\leq \frac{2(\bar{G} - \underline{G})}{\underline{G}} G'(\theta) \begin{cases} (1 - \epsilon), & r = 1, \\ (1 - \epsilon)^r, & r \geq 2. \end{cases} \end{aligned} \quad (\text{S192})$$

For both cases, take

$$\varrho = 1 - \frac{\epsilon}{2} = 1 - \frac{\underline{M}}{2\overline{M}} \in (0, 1). \quad (\text{S193})$$

Since $1 - \epsilon \leq \varrho$, the two cases in Equation (S192) are both bounded by ϱ^r . Summing the squared coordinate bounds and taking square roots give

$$\|\Delta_{n,r}\|_2 = \left\{ \sum_{i=1}^p |\Delta_{n,r,i}|^2 \right\}^{1/2}$$

$$\leq C\sqrt{p}G'(\theta)\varrho^r, \quad 1 \leq r < n. \quad (\text{S194})$$

Step 5: the zero-lag case. Finally, when $r = 0$, Equation (S161) and Equation (S150) give

$$\begin{aligned} \|\Delta_{n,0}\|_2 &= \|\rho_n(h_n)\|_2 \\ &\leq \rho_n(\|h_n\|_2) \\ &\leq \sqrt{p}G'(\theta) \\ &\leq C\sqrt{p}G'(\theta)\varrho^0. \end{aligned} \quad (\text{S195})$$

Equations (S194) and (S195) prove Equation (S165). If $\epsilon = 1$, Equation (S186) and the zero oscillation in Equation (S191) show that $\Delta_{n,r} = 0$ for every $r \geq 1$, so any fixed $\varrho \in (0, 1)$ proves the same result. The chosen ϱ depends only on the model minorization bounds, independently of N, J, α, k . The factor ϱ^r includes the first positive lag. This argument bounds linear posterior expectations; particle covariances are treated in Lemma S13. $\square$

Lemma S11 (Uniform weak bias of current-particle averages). *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), with multinomial resampling, there is $C < \infty$ such that, for every bounded real function f of the current particle location and every $t \geq 1$,*

$$|\mathbb{E}m(\xi_t)(f) - \rho_t(f)| \leq \frac{C}{J} \text{osc}(f). \quad (\text{S196})$$

The constant is independent of t and J . Under (A6-J), the same conclusion holds for a bounded function of the current pair $(X_t, \partial_\theta X_t)$, with constants depending on the joint-transition bounds. Under (A6), Lemma S13 handles h_t by decomposing its simulator history.

Proof. We telescope the bias over particle updates, using semigroup contraction.

Step 1: prediction-to-filtering reduction. First take f to be a function of the physical location, in the Feynman–Kac model defined in Section S3. The current-location marginal of ρ_t is the filtering distribution, and hence

$$\rho_t(f) = \pi_t(f) = \Psi_t(\eta_t)(f). \quad (\text{S197})$$

Conditionally on the prediction cloud η_t^J , multinomial resampling draws the filtering particles independently from $\Psi_t(\eta_t^J)$. Therefore

$$\begin{aligned} \mathbb{E}\{m(\xi_t)(f) \mid \eta_t^J\} &= \frac{1}{J} \sum_{j=1}^J \mathbb{E}\{f(x_{t,j}^F) \mid \eta_t^J\} \\ &= \frac{1}{J} \sum_{j=1}^J \Psi_t(\eta_t^J)(f) \\ &= \Psi_t(\eta_t^J)(f). \end{aligned} \quad (\text{S198})$$

Step 2: normalized-semigroup contraction. Take $\Phi_{t,t}$ to be the identity. For $1 \leq p \leq t$, define

$$Q_{t,t}(f)(x) = G_t(x)f(x),$$

$$Q_{p,t}(f)(x) = G_p(x)M_{p+1}\{Q_{p+1,t}(f)\}(x), \quad p < t. \quad (\text{S199})$$

The definitions of $\Phi_{p,t}$ and Ψ_t give, for every probability measure μ ,

$$\Psi_t\{\Phi_{p,t}(\mu)\}(f) = \frac{\mu Q_{p,t}(f)}{\mu Q_{p,t}(1)}. \quad (\text{S200})$$

For $p < t$,

$$Q_{p,t}(1)(x) = G_p(x) \int f_{X_{p+1}|X_p}(u \mid x; \theta) Q_{p+1,t}(1)(u) du. \quad (\text{S201})$$

Thus, for every x, x' ,

$$\begin{aligned} \frac{Q_{p,t}(1)(x)}{Q_{p,t}(1)(x')} &\leq \frac{\bar{G}\bar{M} \int Q_{p+1,t}(1)(u) du}{\underline{G}\underline{M} \int Q_{p+1,t}(1)(u) du} \\ &= \frac{\bar{G}\bar{M}}{\underline{G}\underline{M}}. \end{aligned} \quad (\text{S202})$$

For $p = t$, the same ratio is at most $\bar{G}/\underline{G}$. Put

$$R = \frac{\bar{G}\bar{M}}{\underline{G}\underline{M}} \geq 1. \quad (\text{S203})$$

Then

$$\sup_{x,x'} \frac{Q_{p,t}(1)(x)}{Q_{p,t}(1)(x')} \leq R, \quad 1 \leq p \leq t. \quad (\text{S204})$$

For $p \leq s < t$, the normalized forward kernel is

$$P_{s,s+1}^{(t)}(x, A) = \frac{\int_A f_{X_{s+1}|X_s}(u \mid x; \theta) Q_{s+1,t}(1)(u) du}{\int f_{X_{s+1}|X_s}(u \mid x; \theta) Q_{s+1,t}(1)(u) du}. \quad (\text{S205})$$

Define, only within this proof,

$$\nu_{s,t}(A) = \frac{\int_A Q_{s+1,t}(1)(u) du}{\int Q_{s+1,t}(1)(u) du}. \quad (\text{S206})$$

Assumption (A2) gives

$$\begin{aligned} P_{s,s+1}^{(t)}(x, A) &\geq \frac{\underline{M} \int_A Q_{s+1,t}(1)(u) du}{\bar{M} \int Q_{s+1,t}(1)(u) du} \\ &= \frac{\underline{M}}{\bar{M}} \nu_{s,t}(A). \end{aligned} \quad (\text{S207})$$

Consequently, by Equation (S193), the Dobrushin coefficient of every kernel in Equation (S205) is at most ϱ . The factor $G_s(x)$ cancels from the normalized conditional expectation, and hence

$$\frac{Q_{p,t}(f)}{Q_{p,t}(1)} = P_{p,p+1}^{(t)} P_{p+1,p+2}^{(t)} \cdots P_{t-1,t}^{(t)} f. \quad (\text{S208})$$

It follows that

$$\text{osc} \left\{ \frac{Q_{p,t}(f)}{Q_{p,t}(1)} \right\} \leq \varrho^{t-p} \text{osc}(f). \quad (\text{S209})$$

Step 3: one complete particle update. For one selection–mutation update, if $p = 1$, put $\mu = \eta_1 = \pi_0 M_1$. For $p \geq 2$, condition on η_{p-1}^J and put

$$\mu = \Phi_p(\eta_{p-1}^J). \quad (\text{S210})$$

In either case, the particles forming η_p^J are conditionally independent with common law μ . Define the normalizer q and centered fluctuation d by

$$q = \frac{Q_{p,t}(1)}{\mu Q_{p,t}(1)}, \quad a = \frac{\mu Q_{p,t}(f)}{\mu Q_{p,t}(1)}, \quad d = \frac{Q_{p,t}(f - a)}{\mu Q_{p,t}(1)}. \quad (\text{S211})$$

Then

$$\mu(q) = 1, \quad \mu(d) = 0, \quad (\text{S212})$$

and Equation (S200) gives

$$\begin{aligned} & \Psi_t\{\Phi_{p,t}(\eta_p^J)\}(f) - \Psi_t\{\Phi_{p,t}(\mu)\}(f) \\ &= \frac{\eta_p^J Q_{p,t}(f)}{\eta_p^J Q_{p,t}(1)} - a \\ &= \frac{\eta_p^J Q_{p,t}(f - a)}{\eta_p^J Q_{p,t}(1)} \\ &= \frac{\eta_p^J(d)}{\eta_p^J(q)}. \end{aligned} \quad (\text{S213})$$

Equation (S204) implies

$$R^{-1} \leq q(x) \leq R, \quad \eta_p^J(q) \geq R^{-1}, \quad \|q - 1\|_\infty \leq R. \quad (\text{S214})$$

Writing

$$F = \frac{Q_{p,t}(f)}{Q_{p,t}(1)}, \quad (\text{S215})$$

we have $a = \mu(qF)$ and $\mu(q) = 1$. Thus a is an average of F under the probability measure $q d\mu$, so

$$\inf F \leq a \leq \sup F. \quad (\text{S216})$$

Since $d = q(F - a)$, Equations (S214), (S216), and (S209) give

$$\|d\|_\infty \leq R \text{osc}(F)$$

$$\leq R\varrho^{t-p} \text{osc}(f). \quad (\text{S217})$$

Conditional independence and $\mu(d) = 0$ give

$$\begin{aligned} \mathbb{E}\{\eta_p^J(d) \mid \mu\} &= 0, \\ \mathbb{E}\{\eta_p^J(d)^2 \mid \mu\} &= \frac{1}{J}\mu(d^2), \\ \mathbb{E}\{(\eta_p^J(q) - 1)^2 \mid \mu\} &= \frac{1}{J}\mu\{(q-1)^2\}. \end{aligned} \quad (\text{S218})$$

Consequently,

$$\begin{aligned} &\mathbb{E} \left\{ \frac{\eta_p^J(d)}{\eta_p^J(q)} \middle| \mu \right\} \\ &= -\mathbb{E} \left\{ \frac{\eta_p^J(d)\{\eta_p^J(q) - 1\}}{\eta_p^J(q)} \middle| \mu \right\}. \end{aligned} \quad (\text{S219})$$

Conditional Cauchy–Schwarz and Equations (S214)–(S218) yield

$$\begin{aligned} &\left| \mathbb{E} \left\{ \frac{\eta_p^J(d)}{\eta_p^J(q)} \middle| \mu \right\} \right| \\ &\leq R [\mathbb{E}\{\eta_p^J(d)^2 \mid \mu\} \mathbb{E}\{(\eta_p^J(q) - 1)^2 \mid \mu\}]^{1/2} \\ &= \frac{R}{J} [\mu(d^2)\mu\{(q-1)^2\}]^{1/2} \\ &\leq \frac{C}{J} \varrho^{t-p} \text{osc}(f). \end{aligned} \quad (\text{S220})$$

Step 4: telescope over update times. Since $\eta_t = \Phi_{1,t}(\eta_1)$,

$$\begin{aligned} &\mathbb{E}\Psi_t(\eta_t^J)(f) - \Psi_t(\eta_t)(f) \\ &= \mathbb{E} [\Psi_t\{\Phi_{1,t}(\eta_1^J)\}(f) - \Psi_t\{\Phi_{1,t}(\eta_1)\}(f)] \\ &\quad + \sum_{p=2}^t \mathbb{E} [\Psi_t\{\Phi_{p,t}(\eta_p^J)\}(f) - \Psi_t\{\Phi_{p,t}(\Phi_p(\eta_{p-1}^J))\}(f)]. \end{aligned} \quad (\text{S221})$$

Indeed, for $2 \leq p \leq t$,

$$\Phi_{p,t} \circ \Phi_p = \Phi_{p-1,t}, \quad (\text{S222})$$

so consecutive terms cancel. The only terms left are $\Psi_t(\eta_t^J)(f)$ and $-\Psi_t(\eta_t)(f)$.

Apply Equation (S220) to every summand in Equation (S221). Then

$$\begin{aligned} |\mathbb{E}\Psi_t(\eta_t^J)(f) - \Psi_t(\eta_t)(f)| &\leq \frac{C \text{osc}(f)}{J} \sum_{p=1}^t \varrho^{t-p} \\ &\leq \frac{C}{J(1-\varrho)} \text{osc}(f). \end{aligned} \quad (\text{S223})$$

Absorb $(1 - \varrho)^{-1}$ into C and combine this display with Equations (S197) and (S198).

For (A6-J), apply the same calculation to the pair $(X_t, \partial_\theta X_t)$. In every display replace the physical transition density by that in Equation (S144), du by the common reference measure, $(\underline{M}, \bar{M})$ by $(\underline{M}_D, \bar{M}_D)$, and the contraction factor by $1 - \underline{M}_D/(2\bar{M}_D)$. The potential remains $G_t(X_t)$, so its bounds and all ratio and conditional-resampling identities are unchanged. For each coordinate, the current measurement gradient is

$$h_{t,i}(x, v) = \partial_{\theta_i} \log f_{Y_t|X_t}(y_t^* | x; \theta) + \sum_{a=1}^d \partial_{x_a} \log f_{Y_t|X_t}(y_t^* | x; \theta) v_{ai}, \quad \theta = \phi. \quad (\text{S224})$$

It is bounded by (A3) on the admissible joint state space. $\square$

Lemma S12 (Full-score summation-by-parts identities). *Suppose $\theta = \phi$ and $\alpha \in [0, 1]$. Then the particle and population scores satisfy Equations (S228) and (S229) below. Moreover, for every coordinate i and $t \leq N$,*

$$\left\| \sum_{k=1}^t \alpha^{t-k} h_{k,i} \right\|_\infty \leq G'(\theta) \sum_{k=1}^t \alpha^{t-k}. \quad (\text{S225})$$

Proof. Step 1: particle summation by parts. Sum Equation (S160) over $n = 1, \dots, N$ and put $k = n - r$. Since the sums are finite,

$$\nabla_\theta \hat{\ell}^\alpha(\theta) = \sum_{k=1}^N \left[m(\xi_k)(h_k) + \sum_{t=k+1}^N \alpha^{t-k} \{m(\xi_t)(h_k) - m(\xi_{t-1})(h_k)\} \right]. \quad (\text{S226})$$

For each fixed k , collecting the coefficient of every $m(\xi_t)(h_k)$ gives

$$\begin{aligned} & m(\xi_k)(h_k) + \sum_{t=k+1}^N \alpha^{t-k} \{m(\xi_t)(h_k) - m(\xi_{t-1})(h_k)\} \\ &= (1 - \alpha) \sum_{t=k}^{N-1} \alpha^{t-k} m(\xi_t)(h_k) + \alpha^{N-k} m(\xi_N)(h_k). \end{aligned} \quad (\text{S227})$$

Indeed, the coefficients are $1 - \alpha$ at $t = k$, $\alpha^{t-k} - \alpha^{t+1-k} = (1 - \alpha)\alpha^{t-k}$ for $k < t < N$, and α^{N-k} at $t = N$. Interchanging the finite sums once more yields

$$\nabla_\theta \hat{\ell}^\alpha(\theta) = (1 - \alpha) \sum_{t=1}^{N-1} m(\xi_t) \left(\sum_{k=1}^t \alpha^{t-k} h_k \right) + m(\xi_N) \left(\sum_{k=1}^N \alpha^{N-k} h_k \right). \quad (\text{S228})$$

Step 2: population identity and uniform bound. The same algebra applied to Equation (S164) gives

$$\nabla_\theta \ell^\alpha(\theta) = (1 - \alpha) \sum_{t=1}^{N-1} \rho_t \left(\sum_{k=1}^t \alpha^{t-k} h_k \right) + \rho_N \left(\sum_{k=1}^N \alpha^{N-k} h_k \right). \quad (\text{S229})$$

For every coordinate i , Equation (S150) gives

$$\left\| \sum_{k=1}^t \alpha^{t-k} h_{k,i} \right\|_{\infty} \leq G'(\theta) \sum_{k=1}^t \alpha^{t-k}. \quad (\text{S230})$$

Equation (S230) is identical to Equation (S225), completing the proof. $\square$

Claim S1 (Conditional empirical moments). *Let $\mathcal{F}$ be a sigma-field and let μ be an $\mathcal{F}$ -measurable probability measure. Suppose that, conditionally on $\mathcal{F}$, $X_1, \dots, X_J$ are independent with common law μ . The following bounds hold for bounded $\mathcal{F}$ -measurable real functions f, f_1, f_2 :*

$$\mathbb{E} \left\{ \left. \frac{1}{J} \sum_{j=1}^J \{f(X_j) - \mu(f)\} \right| \mathcal{F} \right\} = 0, \quad (\text{S231})$$

$$\mathbb{E} \left\{ \left. \left[\frac{1}{J} \sum_{j=1}^J \{f(X_j) - \mu(f)\} \right]^2 \right| \mathcal{F} \right\} = \frac{\mu\{(f - \mu f)^2\}}{J} \leq \frac{\text{osc}(f)^2}{4J}, \quad (\text{S232})$$

$$\begin{aligned} \mathbb{E} \left\{ \left. \left[\frac{1}{J} \sum_{j=1}^J \{f(X_j) - \mu(f)\} \right]^4 \right| \mathcal{F} \right\} &= \frac{\mu\{(f - \mu f)^4\}}{J^3} + \frac{3(J-1)}{J^3} \mu\{(f - \mu f)^2\}^2 \\ &\leq \frac{\mu\{(f - \mu f)^4\}}{J^3} + \frac{3}{J^2} \mu\{(f - \mu f)^2\}^2 \\ &\leq \frac{3 \text{osc}(f)^4}{J^2}. \end{aligned} \quad (\text{S233})$$

Consequently, for bounded f_1 and f_2 ,

$$\begin{aligned} &\left\| \left[\frac{1}{J} \sum_{j=1}^J \{f_1(X_j) - \mu(f_1)\} \right] \left[\frac{1}{J} \sum_{j=1}^J \{f_2(X_j) - \mu(f_2)\} \right] \right\|_{L^2|\mathcal{F}} \\ &\leq \frac{\sqrt{3}}{J} \text{osc}(f_1) \text{osc}(f_2). \end{aligned} \quad (\text{S234})$$

Proof. Put $Y_j = f(X_j) - \mu(f)$. Conditional independence and $\mathbb{E}(Y_j | \mathcal{F}) = 0$ give Equations (S231) and (S232). They also give

$$\begin{aligned} &\mathbb{E} \left\{ \left. \left(\frac{1}{J} \sum_{j=1}^J Y_j \right)^4 \right| \mathcal{F} \right\} \\ &= \frac{1}{J^4} \{ J \mu\{(f - \mu f)^4\} + 3J(J-1) \mu\{(f - \mu f)^2\}^2 \} \\ &= \frac{\mu\{(f - \mu f)^4\}}{J^3} + \frac{3(J-1)}{J^3} \mu\{(f - \mu f)^2\}^2 \\ &\leq \frac{3}{J^2} \text{osc}(f)^4, \end{aligned} \quad (\text{S235})$$

which is Equation (S233). Conditional Cauchy–Schwarz applied to the two fourth moments proves Equation (S234). $\square$

Claim S2 (Normalized-ratio perturbations). *Let μ and ν be probability measures. For $l \in \{0, 1\}$, let $a_l > 0$ and b_l be bounded functions such that $\mu(a_l) = 1$, and put*

$$\mathcal{R}_l(\lambda) = \frac{\lambda(b_l)}{\lambda(a_l)}, \quad D_\mu \mathcal{R}_l = b_l - \mathcal{R}_l(\mu)a_l. \quad (\text{S236})$$

Then

$$\mathcal{R}_l(\nu) - \mathcal{R}_l(\mu) = (\nu - \mu)D_\mu \mathcal{R}_l - \frac{(\nu - \mu)(a_l)}{\nu(a_l)}(\nu - \mu)D_\mu \mathcal{R}_l, \quad (\text{S237})$$

and hence

$$\begin{aligned} & \{\mathcal{R}_1(\nu) - \mathcal{R}_0(\nu)\} - \{\mathcal{R}_1(\mu) - \mathcal{R}_0(\mu)\} \\ & - (\nu - \mu)\{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\} \\ & = -\frac{(\nu - \mu)(a_1 - a_0)}{\nu(a_0)\nu(a_1)}(\nu - \mu)D_\mu \mathcal{R}_1 \\ & - \frac{(\nu - \mu)(a_0)}{\nu(a_0)}(\nu - \mu)\{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\}. \end{aligned} \quad (\text{S238})$$

If $\inf_{l \in \{0, 1\}} \nu(a_l) \geq C^{-1}$, then the absolute value of the left side of Equation (S238) is at most

$$\begin{aligned} & C^2 |(\nu - \mu)(a_1 - a_0)| |(\nu - \mu)D_\mu \mathcal{R}_1| \\ & + C |(\nu - \mu)(a_0)| |(\nu - \mu)\{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\}|. \end{aligned} \quad (\text{S239})$$

Proof. Since $\mu(a_l) = 1$,

$$\begin{aligned} \mathcal{R}_l(\nu) - \mathcal{R}_l(\mu) &= \frac{\nu(b_l) - \mathcal{R}_l(\mu)\nu(a_l)}{\nu(a_l)} \\ &= \frac{(\nu - \mu)D_\mu \mathcal{R}_l}{\nu(a_l)} \\ &= (\nu - \mu)D_\mu \mathcal{R}_l - \frac{(\nu - \mu)(a_l)}{\nu(a_l)}(\nu - \mu)D_\mu \mathcal{R}_l, \end{aligned} \quad (\text{S240})$$

which proves Equation (S237). Moreover,

$$\begin{aligned} & \frac{(\nu - \mu)(a_1)}{\nu(a_1)} - \frac{(\nu - \mu)(a_0)}{\nu(a_0)} \\ &= \left(1 - \frac{1}{\nu(a_1)}\right) - \left(1 - \frac{1}{\nu(a_0)}\right) \\ &= \frac{\nu(a_1) - \nu(a_0)}{\nu(a_0)\nu(a_1)} \\ &= \frac{(\nu - \mu)(a_1 - a_0)}{\nu(a_0)\nu(a_1)}. \end{aligned} \quad (\text{S241})$$

Subtracting the cases $l = 0$ and $l = 1$ in Equation (S237) now proves Equation (S238). Taking absolute values there and using $\nu(a_0)^{-1} \leq C$ and $\nu(a_1)^{-1} \leq C$ proves Equation (S239). This exact identity is the paired specialization of the normalized Feynman–Kac perturbation calculation in Equations (3.1)–(3.4) and Appendix A.3 of Del Moral and Rio (2011). $\square$

Lemma S13 (Particle error and variance of the DMOP- α score). *Suppose $J \geq 2$, $\theta = \phi$, $\alpha \in [0, 1]$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Suppose also that either (A6) or (A6-J) holds. Then*

$$\text{Var} \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\} \lesssim \frac{NpG'(\theta)^2(1 + \log J)^2}{J} \left(\sum_{r=0}^{N-1} \alpha^r \right)^2, \quad (\text{S242})$$

$$\left\| \mathbb{E} \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2 \lesssim \frac{N\sqrt{p}G'(\theta)}{J} \sum_{r=0}^{N-1} \alpha^r, \quad (\text{S243})$$

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2^2 \lesssim pG'(\theta)^2 \left(\frac{N(1 + \log J)^2}{J} + \frac{N^2}{J^2} \right) \left(\sum_{r=0}^{N-1} \alpha^r \right)^2. \quad (\text{S244})$$

The finite geometric sum is

$$\sum_{r=0}^{N-1} \alpha^r = \begin{cases} \frac{1 - \alpha^N}{1 - \alpha}, & 0 \leq \alpha < 1, \\ N, & \alpha = 1. \end{cases} \quad (\text{S245})$$

The constants suppressed by $\lesssim$ depend only on the fixed model and stability bounds and are independent of N, J , and α . As elsewhere, $\text{Var}(Z) = \mathbb{E}\|Z - \mathbb{E}Z\|_2^2$ for a random vector Z .

Proof. We bound lag-truncated scores uniformly in their starting time, then use their geometric mixture to recover DMOP- α . Under (A6), the chain rule first decomposes h_s by the age of its simulator contributions.

Step 1: auxiliary lag truncations. For $0 \leq k < N$, define

$$\nabla_{\theta} \hat{s}_k^1(\theta) = \sum_{n=1}^N \sum_{r=0}^{\min(k, n-1)} \Delta_{n,r}^J, \quad \nabla_{\theta} s_k^1(\theta) = \sum_{n=1}^N \sum_{r=0}^{\min(k, n-1)} \Delta_{n,r}. \quad (\text{S246})$$

For a fixed creation time s , the terms in the first sum telescope:

$$\begin{aligned} m(\xi_s)(h_s) + \sum_{n=s+1}^{\min(N, s+k)} \{m(\xi_n)(h_s) - m(\xi_{n-1})(h_s)\} \\ = m(\xi_{\min(N, s+k)})(h_s). \end{aligned} \quad (\text{S247})$$

Summing Equation (S247) over $s = 1, \dots, N$ gives

$$\nabla_{\theta} \hat{s}_k^1(\theta) = \sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N) \left(\sum_{s=N-k}^N h_s \right), \quad (\text{S248})$$

$$\nabla_{\theta} s_k^1(\theta) = \sum_{s=1}^{N-k-1} \rho_{s+k}(h_s) + \rho_N \left(\sum_{s=N-k}^N h_s \right). \quad (\text{S249})$$

An empty sum in these displays is zero.

Step 1a: chain-rule decomposition. All derivatives in this step are evaluated at $\theta = \phi$, with reference resampling indices fixed. Write $X_u^\theta = \text{process}_u(X_{u-1}^\theta; \theta, \omega_u)$, and let $\partial_{\theta_i} \text{process}_u$ denote its partial derivative holding X_{u-1}^θ fixed. The chain rule, iterated from time zero, gives

$$\partial_{\theta_i} X_s^\theta = \frac{\partial X_s^\theta}{\partial X_0^\theta} \partial_{\theta_i} X_0^\theta + \sum_{u=1}^s \frac{\partial X_s^\theta}{\partial X_u^\theta} \partial_{\theta_i} \text{process}_u(X_{u-1}^\theta; \theta, \omega_u). \quad (\text{S250})$$

Under (A6), decompose the coordinate- i measurement gradient at time s by the age of its chain-rule contributions. The superscript $[r]$ is an age index, not a power or derivative order: for $1 \leq r < s$, $h_{s,i}^{[r]}$ is the contribution from the simulator's partial parameter derivative at time $s-r$, propagated to the measurement at time s . This age differs from the discount lag $n-s$, which gives the coefficient α^{n-s} when the time- s measurement gradient contributes at time n . Define

$$\begin{aligned} h_{s,i}^{[0]} &= \partial_{\theta_i} \log f_{Y_s|X_s}(y_s^* | x; \theta) \Big|_{x=X_s^\theta} + D_x \log f_{Y_s|X_s}(y_s^* | x; \theta) \Big|_{x=X_s^\theta} \partial_{\theta_i} \text{process}_s(X_{s-1}^\theta; \theta, \omega_s), \\ h_{s,i}^{[r]} &= D_x \log f_{Y_s|X_s}(y_s^* | X_s^\theta(x, \omega_{s-r+1:s}); \theta) \Big|_{x=X_{s-r}^\theta} \partial_{\theta_i} \text{process}_{s-r}(X_{s-r-1}^\theta; \theta, \omega_{s-r}), \quad 1 \leq r < s, \\ h_{s,i}^{[s]} &= D_x \log f_{Y_s|X_s}(y_s^* | X_s^\theta(x, \omega_{1:s}); \theta) \Big|_{x=X_0^\theta} \partial_{\theta_i} X_0^\theta. \end{aligned} \quad (\text{S251})$$

Here $h_{s,i}^{[0]}$ combines the direct parameter derivative of the log measurement density, holding x fixed, with the contribution from the time- s simulator's partial parameter derivative. The endpoint $h_{s,i}^{[s]}$ is the contribution from the initial-state derivative. Multiplying Equation (S250) by the spatial log measurement derivative gives

$$h_{s,i} = \sum_{r=0}^s h_{s,i}^{[r]}, \quad \|h_{s,i}^{[r]}\|_\infty \leq CG'(\theta) \kappa^r. \quad (\text{S252})$$

For $r = 0$ use (A6) with $t = s$ and the bounded partial derivatives; for $1 \leq r < s$ use it with $t = s - r$; for $r = s$ use $t = 0$ and the initial-gradient bound. For $r < s$, the displayed function depends only on X_{s-r-1} and the simulator variables from $s - r$ through s . For $r = s$ it also uses the initial simulator draw, whose particle replicates are independent. Set $h_{s,i}^{[r]} = 0$ for $r > s$.

Under (A6-J), take instead $h_{s,i}^{[0]} = h_{s,i}$, the bounded current function in Equation (S224), and $h_{s,i}^{[r]} = 0$ for $r \geq 1$. Use the joint pair as the state throughout the particle calculation. The sums over r below then contain only $r = 0$; κ is needed only under (A6).

Step 1b: finite-window particle error. Let f be a bounded real function of the state at time a and the subsequent simulator variables through time $b \geq a$, with no earlier history. Write $Q_{a,b}$ for the path lift of the semigroup in Equation (S199). First suppose $a \geq 1$. Under (A6), the starting cloud is the physical cloud; under (A6-J), it is the joint cloud. Adding and subtracting the conditional population continuation gives

$$m(\xi_b)(f) - \rho_b(f) = \left\{ m(\xi_b)(f) - \frac{\eta_a^J Q_{a,b}(f)}{\eta_a^J Q_{a,b}(1)} \right\} + \left\{ \frac{\eta_a^J Q_{a,b}(f)}{\eta_a^J Q_{a,b}(1)} - \frac{\eta_a Q_{a,b}(f)}{\eta_a Q_{a,b}(1)} \right\}. \quad (\text{S253})$$

For the first brace, condition on η_a^J . Let $\nu = m(\xi_b^P)$ and $\mu = \Phi_{a,b}(\eta_a^J)$ on the prediction window. The first prediction following selection from η_a^J has conditionally independent particles

with common law $\Phi_{a,a+1}(\eta_a^J)$. To apply Theorems 3.3 and 3.5 of Chan et al. (2018), first note that future normalizers depend on the current physical state and satisfy Equation (S204). For a bounded function $\psi(X_b, \omega_b)$ of the last state and its innovation, its normalized continuation, for $b > a$, is

$$\begin{aligned} \frac{Q_{a,b}(\psi)}{Q_{a,b}(1)} &= P_{a,a+1}^{(b)} \cdots P_{b-2,b-1}^{(b)} \left[\frac{\mathbb{E}\{G_b(X_b)\psi(X_b, \omega_b) \mid X_{b-1} = \cdot\}}{M_b(G_b)} \right], \\ \text{osc} \left\{ \frac{Q_{a,b}(\psi)}{Q_{a,b}(1)} \right\} &\leq \varrho^{b-a-1} \text{osc}(\psi). \end{aligned} \quad (\text{S254})$$

The product is empty when $b = a + 1$. The bracket is a conditional average of ψ , and the other kernels contract as in Equation (S209). Thus the innovation-augmented marginal has a summable forgetting bound, with a one-step delay. For prediction histories, omit the terminal G_b : the final bracket becomes $\mathbb{E}[\psi(X_b, \omega_b) \mid X_{b-1} = \cdot]$, and the same contraction argument gives the corresponding summable bound. Apply the path construction of Chan et al. (2018) to this marginal. Under (A6-J), use the joint marginal and its transition bounds instead, as in Lemma S11. The cited theorems now apply over at most $b - a$ updates, uniformly in a . For the second moment, put $\bar{f} = f - \mu(f)$ and choose c so that $0 \leq \bar{f} + c \leq \text{osc}(f)$. Then $\mu(\bar{f} + c) = c$, and

$$\begin{aligned} \mathbb{E}\{|\nu(f) - \mu(f)|^2 \mid \eta_a^J\} &= \frac{1}{J^2} \sum_j \mathbb{E}\{\bar{f}(\xi_{b,j}^P)^2 \mid \eta_a^J\} + \frac{1}{J^2} \sum_{j \neq l} \mathbb{E}\{\bar{f}(\xi_{b,j}^P) \bar{f}(\xi_{b,l}^P) \mid \eta_a^J\}, \\ \mathbb{E}\{\bar{f}(\xi_{b,j}^P) \bar{f}(\xi_{b,l}^P) \mid \eta_a^J\} &= \mathbb{E}\{(\bar{f} + c)(\xi_{b,j}^P)(\bar{f} + c)(\xi_{b,l}^P) \mid \eta_a^J\} - c^2 \\ &\quad - c\{\mathbb{E}[(\bar{f} + c)(\xi_{b,j}^P) \mid \eta_a^J] - c\} \\ &\quad - c\{\mathbb{E}[(\bar{f} + c)(\xi_{b,l}^P) \mid \eta_a^J] - c\}. \end{aligned} \quad (\text{S255})$$

The diagonal terms are at most $\text{osc}(f)^2$. Theorem 3.5 bounds the first difference in the second identity by $C(b-a) \text{osc}(f)^2/J$; Theorem 3.3 bounds each remaining difference by $C(b-a) \text{osc}(f)/J$, and $|c| \leq \text{osc}(f)$. Thus, initially for $J \geq c_0(b-a+1)$,

$$\mathbb{E}\{|\nu(f) - \mu(f)|^2 \mid \eta_a^J\} \leq \frac{C(b-a+1)}{J} \text{osc}(f)^2, \quad (\text{S256})$$

$$|\mathbb{E}\{\nu(f) - \mu(f) \mid \eta_a^J\}| \leq \frac{C(b-a+1)}{J} \text{osc}(f). \quad (\text{S257})$$

When $a = b$, $\nu = \mu = \eta_a^J$, so both prediction errors are zero.

To pass to filtering, put $d = G_b\{f - \Psi_b(\mu)(f)\}$. Then $\mu(d) = 0$ and the exact ratio expansion is

$$\begin{aligned} \Psi_b(\nu)(f) - \Psi_b(\mu)(f) &= \frac{(\nu - \mu)(d)}{\nu(G_b)} \\ &= \frac{(\nu - \mu)(d)}{\mu(G_b)} - \frac{(\nu - \mu)(d)(\nu - \mu)(G_b)}{\nu(G_b)\mu(G_b)}. \end{aligned} \quad (\text{S258})$$

Both denominators are at least $\underline{G}$, and $\text{osc}(d) \leq 2\bar{G} \text{osc}(f)$. Equations (S256) and (S257) consequently give

$$\mathbb{E}\{|\Psi_b(\nu)(f) - \Psi_b(\mu)(f)|^2 \mid \eta_a^J\} \leq \frac{C(b-a+1)}{J} \text{osc}(f)^2,$$

$$\begin{aligned}
|\mathbb{E}\{\Psi_b(\nu)(f) - \Psi_b(\mu)(f) \mid \eta_a^J\}| &\leq \frac{|\mathbb{E}\{(\nu - \mu)(d) \mid \eta_a^J\}|}{\underline{G}} + \frac{\|(\nu - \mu)(d)\|_{L^2(\cdot|\eta_a^J)} \|(\nu - \mu)(G_b)\|_{L^2(\cdot|\eta_a^J)}}{\underline{G}^2} \\
&\leq \frac{C(b-a+1)}{J} \text{osc}(f).
\end{aligned} \tag{S259}$$

Finally, conditional multinomial resampling gives

$$\begin{aligned}
\mathbb{E}\{m(\xi_b)(f) - \Psi_b(\nu)(f) \mid \xi_b^P\} &= 0, \\
\mathbb{E}\{|m(\xi_b)(f) - \Psi_b(\nu)(f)|^2 \mid \xi_b^P\} &\leq \text{osc}(f)^2/J.
\end{aligned} \tag{S260}$$

Adding this centered term proves the same orders for the first brace of Equation (S253).

For the second brace, put

$$q = \frac{Q_{a,b}(1)}{\eta_a Q_{a,b}(1)}, \quad d = \frac{Q_{a,b}\{f - \rho_b(f)\}}{\eta_a Q_{a,b}(1)}. \tag{S261}$$

These functions of the starting state satisfy

$$\eta_a(q) = 1, \quad \eta_a(d) = 0, \quad R^{-1} \leq q \leq R, \quad \|d\|_\infty \leq R \text{osc}(f). \tag{S262}$$

Karjalainen et al.'s Lemma 2.3 applies to these bounded current prediction functions. The prediction weak-bias bound follows from Lemma S11, since for $a \geq 2$

$$\begin{aligned}
\mathbb{E}\eta_a^J(u) - \eta_a(u) &= \mathbb{E}m(\xi_{a-1})(M_a u) - \rho_{a-1}(M_a u), \\
|\mathbb{E}\eta_a^J(u) - \eta_a(u)| &\leq C \text{osc}(u)/J.
\end{aligned} \tag{S263}$$

At initialization the corresponding empirical averages are independent and unbiased. Applying these bounds to d and q yields

$$\mathbb{E} \left| \frac{\eta_a^J(d)}{\eta_a^J(q)} \right|^2 \leq R^2 \mathbb{E} |\eta_a^J(d)|^2 \leq C \text{osc}(f)^2/J, \tag{S264}$$

$$\begin{aligned}
\left| \frac{\mathbb{E} \eta_a^J(d)}{\eta_a^J(q)} \right| &= \left| \mathbb{E} \eta_a^J(d) - \mathbb{E} \frac{\eta_a^J(d) \{\eta_a^J(q) - 1\}}{\eta_a^J(q)} \right| \\
&\leq |\mathbb{E} \eta_a^J(d)| + R \{\mathbb{E} |\eta_a^J(d)|^2 \mathbb{E} |\eta_a^J(q) - 1|^2\}^{1/2} \leq C \text{osc}(f)/J.
\end{aligned} \tag{S265}$$

Combine the two braces using $|u+v|^2 \leq 2|u|^2 + 2|v|^2$. For $a = 0$ and $b \geq 1$, the time-one prediction histories, including the independent initial draw and its gradient when present, are unconditionally independent with their common population prediction law. Apply the same diagonal-off-diagonal calculation to these histories over the remaining $b-1$ updates, followed by the displayed Bayes-ratio and final resampling steps. The normalizer and marginal-forgetting bounds above still apply, because future transitions depend on these histories only through X_1 under (A6), or the joint pair under (A6-J). This gives bounds $Cb \text{osc}(f)^2/J$ and $Cb \text{osc}(f)/J$. For $a = b = 0$, the empirical average is unbiased and its variance is at most $\text{osc}(f)^2/J$. For $J < c_0(b-a+1)$ the pathwise bound $2\|f\|_\infty$ gives the same orders after a constant shift of f and enlargement of C . For every $J \geq 2$ we have therefore proved

$$\|m(\xi_b)(f) - \rho_b(f)\|_{L^2} \leq C \text{osc}(f) \sqrt{(b-a+1)/J},$$

$$|\mathbb{E}m(\xi_b)(f) - \rho_b(f)| \leq C \text{osc}(f)(b - a + 1)/J. \quad (\text{S266})$$

Step 1c: sum over gradient ages. Apply Equation (S266) to $f = h_{s,i}^{[r]}$ at $b = s + q$. Under (A6) start at $a = \max(0, s - r - 1)$; when $r = s$ include the independent initial draw. The window contains at most $q + r + 2$ times. Under (A6-J), use only $r = 0$ and start at s . Thus

$$\begin{aligned} \|m(\xi_{s+q})(h_{s,i}^{[r]}) - \rho_{s+q}(h_{s,i}^{[r]})\|_{L^2} &\leq \frac{CG'(\theta)\kappa^r}{\sqrt{J}}\sqrt{q+r+2}, \\ |\mathbb{E}m(\xi_{s+q})(h_{s,i}^{[r]}) - \rho_{s+q}(h_{s,i}^{[r]})| &\leq \frac{CG'(\theta)\kappa^r}{J}(q+r+2). \end{aligned} \quad (\text{S267})$$

Under (A6-J), only the $r = 0$ inequalities are used. Since $q + r + 2 \leq (q + 1)(r + 2)$,

$$\begin{aligned} \sum_{r=0}^s \kappa^r \sqrt{q+r+2} &\leq \sqrt{q+1} \sum_{r=0}^{\infty} \kappa^r \sqrt{r+2} \leq C_\kappa \sqrt{q+1}, \\ \sum_{r=0}^s \kappa^r (q+r+2) &\leq \frac{q+2}{1-\kappa} + \frac{\kappa}{(1-\kappa)^2} \leq C_\kappa (q+1). \end{aligned} \quad (\text{S268})$$

Minkowski's inequality, the triangle inequality for expectations, and Equation (S252) give

$$\mathbb{E}|m(\xi_{s+q})(h_{s,i}) - \rho_{s+q}(h_{s,i})|^2 \leq C(q+1)G'(\theta)^2/J, \quad (\text{S269})$$

$$|\mathbb{E}m(\xi_{s+q})(h_{s,i}) - \rho_{s+q}(h_{s,i})| \leq C(q+1)G'(\theta)/J. \quad (\text{S270})$$

Taking $q = 0$ proves the uniform J^{-1} weak-bias bound for h_s .

For the terminal block, put $s = N - q$ and apply Minkowski:

$$\begin{aligned} &\left\| m(\xi_N) \left(\sum_{s=N-k}^N h_{s,i} \right) - \rho_N \left(\sum_{s=N-k}^N h_{s,i} \right) \right\|_{L^2} \\ &\leq \sum_{q=0}^k \|m(\xi_N)(h_{N-q,i}) - \rho_N(h_{N-q,i})\|_{L^2} \leq \frac{CG'(\theta)}{\sqrt{J}} \sum_{q=0}^k \sqrt{q+1} \leq \frac{CG'(\theta)(k+1)^{3/2}}{\sqrt{J}}. \end{aligned} \quad (\text{S271})$$

Consequently,

$$\mathbb{E} \left| m(\xi_N) \left(\sum_{s=N-k}^N h_{s,i} \right) - \rho_N \left(\sum_{s=N-k}^N h_{s,i} \right) \right|^2 \leq C(k+1)^3 G'(\theta)^2/J, \quad (\text{S272})$$

$$\left| \mathbb{E}m(\xi_N) \left(\sum_{s=N-k}^N h_{s,i} \right) - \rho_N \left(\sum_{s=N-k}^N h_{s,i} \right) \right| \leq \frac{CG'(\theta)}{J} \sum_{q=0}^k (q+1) \leq C(k+1)^2 G'(\theta)/J. \quad (\text{S273})$$

Step 1d: covariance summation. Fix r and consider the interior terms $m(\xi_{s+k})(h_{s,i}^{[r]})$, $1 \leq s < N - k$. Under (A6) these depend on the physical cloud at $\max(0, s - r - 1)$ and subsequent variables through $s + k$. Under (A6-J) the $r = 0$ terms depend on the joint cloud at s and the

next k updates. Write $a(u)$ for the mixing coefficient bound in Equation (S131), using the physical cloud in the first case and the joint cloud in the second. Equation (S267) and Cauchy–Schwarz give

$$|\text{Cov}\{m(\xi_{s+k})(h_{s,i}^{[r]}), m(\xi_{t+k})(h_{t,i}^{[r]})\}| \leq \frac{CG'(\theta)^2 \kappa^{2r} (k+r+2)}{J}. \quad (\text{S274})$$

For $t-s > k+r+2$, the prediction clouds separating the windows are at least $t-s-k-r-2$ transitions apart, allowing one transition to absorb the past endpoint resampling. The centered variables have L^4 norms at most $2CG'(\theta)\kappa^r$ by their pathwise bounds, so Davydov's inequality gives

$$|\text{Cov}\{m(\xi_{s+k})(h_{s,i}^{[r]}), m(\xi_{t+k})(h_{t,i}^{[r]})\}| \leq CG'(\theta)^2 \kappa^{2r} a(t-s-k-r-2)^{1/2}. \quad (\text{S275})$$

Under (A6-J), the larger gap $t-s-k-1$ is available; the weaker displayed bound with $r=0$ suffices. Combining these inequalities gives

$$\begin{aligned} & \sum_{t=s+1}^{N-k-1} |\text{Cov}\{m(\xi_{s+k})(h_{s,i}^{[r]}), m(\xi_{t+k})(h_{t,i}^{[r]})\}| \\ & \leq CG'(\theta)^2 \kappa^{2r} \left\{ \frac{(k+r+2)^2}{J} + \sum_{u=1}^{\infty} \min\left(\frac{k+r+2}{J}, a(u)^{1/2}\right) \right\}. \end{aligned} \quad (\text{S276})$$

Choose $u_0 \leq C(1+\log J)^2$ so that the geometric bound for $a(u_0)^{1/2}$ is at most J^{-1} . The geometric tail has sum at most $C(1+\log J)/J$, and hence

$$\sum_{u=1}^{\infty} \min\left(\frac{k+r+2}{J}, a(u)^{1/2}\right) \leq \frac{u_0(k+r+2)}{J} + \frac{C(1+\log J)}{J} \leq \frac{C(k+r+2)(1+\log J)^2}{J}. \quad (\text{S277})$$

Retaining this logarithmic factor, expanding the variance as the diagonal sum plus twice the covariance sum gives

$$\text{Var} \left\{ \sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_{s,i}^{[r]}) \right\} \lesssim \frac{NG'(\theta)^2 \kappa^{2r} (k+r+2)^2 (1+\log J)^2}{J}. \quad (\text{S278})$$

Center each age sum and apply Minkowski:

$$\begin{aligned} & \left\| \sum_{s=1}^{N-k-1} \{m(\xi_{s+k})(h_{s,i}) - \mathbb{E}m(\xi_{s+k})(h_{s,i})\} \right\|_{L^2} \\ & \leq \sum_{r=0}^N \left\| \sum_{s=1}^{N-k-1} \{m(\xi_{s+k})(h_{s,i}^{[r]}) - \mathbb{E}m(\xi_{s+k})(h_{s,i}^{[r]})\} \right\|_{L^2} \\ & \lesssim G'(\theta)(1+\log J) \sqrt{N/J} \sum_{r=0}^N \kappa^r (k+r+2) \leq C_\kappa G'(\theta)(1+\log J) \sqrt{N/J} (k+1). \end{aligned} \quad (\text{S279})$$

Again the joint alternative uses only $r=0$. The terminal variance is bounded by its second moment in Equation (S272); since $k+1 \leq N$,

$$\text{Var} \left\{ m(\xi_N) \left(\sum_{s=N-k}^N h_{s,i} \right) \right\} \leq C(k+1)^3 G'(\theta)^2 / J \leq CN(k+1)^2 G'(\theta)^2 / J. \quad (\text{S280})$$

Use $\text{Var}(A + B) \leq 2\text{Var}(A) + 2\text{Var}(B)$ in Equation (S248) and sum coordinates:

$$\text{Var}\{\nabla_{\theta} \hat{s}_k^1(\theta)\} \lesssim NpG'(\theta)^2(k+1)^2(1 + \log J)^2/J. \quad (\text{S281})$$

Finally, the one-input and terminal bias bounds yield

$$\begin{aligned} \|\mathbb{E}\nabla_{\theta} \hat{s}_k^1(\theta) - \nabla_{\theta} s_k^1(\theta)\|_2 &\leq \frac{C\sqrt{p}G'(\theta)}{J} \{(N-k-1)(k+1) + (k+1)^2\} \\ &= \frac{CN\sqrt{p}G'(\theta)(k+1)}{J}. \end{aligned} \quad (\text{S282})$$

Step 2: geometric mixture. For $0 \leq k < N-1$, put

$$\beta_k = (1 - \alpha)\alpha^k, \quad \beta_{N-1} = \alpha^{N-1}. \quad (\text{S283})$$

These nonnegative numbers sum to one. Moreover, for every $0 \leq r < N$,

$$\begin{aligned} \sum_{k=r}^{N-1} \beta_k &= \sum_{k=r}^{N-2} (1 - \alpha)\alpha^k + \alpha^{N-1} \\ &= \alpha^r(1 - \alpha^{N-1-r}) + \alpha^{N-1} \\ &= \alpha^r. \end{aligned} \quad (\text{S284})$$

Equations (S160), (S164), and (S246) therefore give the pathwise identities

$$\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) = \sum_{k=0}^{N-1} \beta_k \nabla_{\theta} \hat{s}_k^1(\theta), \quad \nabla_{\theta} \ell^{\alpha}(\theta) = \sum_{k=0}^{N-1} \beta_k \nabla_{\theta} s_k^1(\theta). \quad (\text{S285})$$

The coefficient of each $\Delta_{n,r}^J$ or $\Delta_{n,r}$ is the tail sum in Equation (S284). For $N \geq 2$, the mixture weight on $\nabla_{\theta} \hat{s}_0^1(\theta)$ is $\beta_0 = 1 - \alpha$. Every truncated score contains $\Delta_{n,0}^J$, so its coefficient is

$$\sum_{k=0}^{N-1} \beta_k = 1. \quad (\text{S286})$$

In particular, at $\alpha = 0$,

$$\nabla_{\theta} \hat{\ell}_n^0(\theta) = \Delta_{n,0}^J, \quad \nabla_{\theta} \ell_n^0(\theta) = \Delta_{n,0}. \quad (\text{S287})$$

For $N = 1$, the only mixture weight is $\beta_0 = 1$ for every α .

The mean lag in this censored geometric mixture satisfies

$$\begin{aligned} \sum_{k=0}^{N-1} \beta_k(k+1) &= \sum_{r=0}^{N-1} \sum_{k=r}^{N-1} \beta_k \\ &= \sum_{r=0}^{N-1} \alpha^r. \end{aligned} \quad (\text{S288})$$

Center the first identity in Equation (S285). Minkowski's inequality, Equation (S281), and Equation (S288) give

$$\begin{aligned}
\left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \mathbb{E} \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\|_{L^2} &\leq \sum_{k=0}^{N-1} \beta_k \left\| \nabla_{\theta} \hat{s}_k^1(\theta) - \mathbb{E} \nabla_{\theta} \hat{s}_k^1(\theta) \right\|_{L^2} \\
&\lesssim C(1 + \log J) \sqrt{\frac{NpG'(\theta)^2}{J}} \sum_{k=0}^{N-1} \beta_k (k+1) \\
&\lesssim C(1 + \log J) \sqrt{\frac{NpG'(\theta)^2}{J}} \sum_{r=0}^{N-1} \alpha^r.
\end{aligned} \tag{S289}$$

Squaring proves Equation (S242). Taking expectations in Equation (S285), applying the triangle inequality and Equation (S282), and then using Equation (S288) give

$$\begin{aligned}
\left\| \mathbb{E} \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2 &\leq \sum_{k=0}^{N-1} \beta_k \left\| \mathbb{E} \nabla_{\theta} \hat{s}_k^1(\theta) - \nabla_{\theta} s_k^1(\theta) \right\|_2 \\
&\leq \frac{CN\sqrt{p}G'(\theta)}{J} \sum_{k=0}^{N-1} \beta_k (k+1) \\
&= \frac{CN\sqrt{p}G'(\theta)}{J} \sum_{r=0}^{N-1} \alpha^r,
\end{aligned} \tag{S290}$$

which proves Equation (S243). Finally, for a random vector Z and deterministic z ,

$$\mathbb{E} \|Z - z\|_2^2 = \text{Var}(Z) + \|\mathbb{E}Z - z\|_2^2. \tag{S291}$$

Apply this identity with $Z = \nabla_{\theta} \hat{\ell}^{\alpha}(\theta)$ and $z = \nabla_{\theta} \ell^{\alpha}(\theta)$, and substitute Equations (S242) and (S243). This proves Equation (S244). $\square$

Corollary S1 (Particle bias of the DMOP- α score). *Under the conditions of Lemma S13, Equation (S243) holds.*

Proof. This is the second conclusion of Lemma S13. $\square$

Lemma S14 (Population discounting bias). *Under the conditions of Lemma S10, suppose $\theta = \phi$ and $\alpha \in [0, 1]$. Then*

$$\begin{aligned}
\left\| \nabla_{\theta} \ell_n^{\alpha}(\theta) - \nabla_{\theta} \ell_n(\theta) \right\|_2 &\leq C\sqrt{p}G'(\theta) \sum_{r=1}^{n-1} (1 - \alpha^r) \varrho^r \\
&\leq C\sqrt{p}G'(\theta) \frac{\varrho(1 - \alpha)}{(1 - \varrho)(1 - \alpha\varrho)}.
\end{aligned} \tag{S292}$$

For the full score,

$$\left\| \nabla_{\theta} \ell(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2$$

$$\begin{aligned}
&\leq C\sqrt{p}G'(\theta)\varrho(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)(\alpha\varrho)^q \\
&= C\sqrt{p}G'(\theta)\varrho(1-\alpha)\frac{(N-1)-N\alpha\varrho+(\alpha\varrho)^N}{(1-\alpha\varrho)^2} \\
&\leq C(N-1)\sqrt{p}G'(\theta)\frac{\varrho(1-\alpha)}{1-\alpha\varrho}.
\end{aligned} \tag{S293}$$

For $N = 1$, the finite sum is empty, the displayed fraction is zero, and the full-score difference is zero. The dependence on the transition bounds can be sharpened further. With the existing $\varrho = 1 - \underline{M}/(2\bar{M})$ from Equation (S193), for $\frac{1}{2} < \varrho < 1$,

$$\begin{aligned}
&\|\nabla_{\theta}\ell(\theta) - \nabla_{\theta}\ell^{\alpha}(\theta)\|_2 \\
&\leq 2\sqrt{p}G'(\theta)\frac{1 - \sqrt{2(1-\varrho)}}{1 + \sqrt{2(1-\varrho)}}(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)\left\{\alpha\frac{2\varrho-1}{3-2\varrho}\right\}^q \\
&\leq 2(N-1)\sqrt{p}G'(\theta)\frac{1 - \sqrt{2(1-\varrho)}}{1 + \sqrt{2(1-\varrho)}}\frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)}.
\end{aligned} \tag{S294}$$

If every transition is independent of its starting location, in particular if $\varrho = \frac{1}{2}$, the population discounting bias is zero for every N and α .

Proof. Step 1: exact lagwise difference. Equation (S164) at α and at $\alpha = 1$ gives

$$\begin{aligned}
\nabla_{\theta}\ell_n(\theta) - \nabla_{\theta}\ell_n^{\alpha}(\theta) &= \sum_{r=0}^{n-1}(1-\alpha^r)\Delta_{n,r} \\
&= \sum_{r=1}^{n-1}(1-\alpha^r)\Delta_{n,r},
\end{aligned} \tag{S295}$$

The coefficient of $\Delta_{n,0}$ is one in both scores, so its difference is zero. In particular, at the left endpoint Equation (S287) gives

$$\nabla_{\theta}\ell_n^0(\theta) = \Delta_{n,0}, \quad \nabla_{\theta}\ell_n(\theta) - \nabla_{\theta}\ell_n^0(\theta) = \sum_{r=1}^{n-1}\Delta_{n,r}. \tag{S296}$$

Thus DMOP-0 retains the zero-lag contribution with coefficient one and retains no positive-lag contribution.

Step 2: apply forgetting and sum. Taking norms, applying the triangle inequality and Lemma S10, and then extending the nonnegative scalar sum to infinity give

$$\begin{aligned}
\|\nabla_{\theta}\ell_n(\theta) - \nabla_{\theta}\ell_n^{\alpha}(\theta)\|_2 &\leq \sum_{r=1}^{n-1}(1-\alpha^r)\|\Delta_{n,r}\|_2 \\
&\leq C\sqrt{p}G'(\theta)\sum_{r=1}^{n-1}(1-\alpha^r)\varrho^r
\end{aligned}$$

$$\begin{aligned}
&\leq C\sqrt{p}G'(\theta)\sum_{r=1}^{\infty}(1-\alpha^r)\varrho^r \\
&= C\sqrt{p}G'(\theta)\left\{\sum_{r=1}^{\infty}\varrho^r - \sum_{r=1}^{\infty}(\alpha\varrho)^r\right\} \\
&= C\sqrt{p}G'(\theta)\left\{\frac{\varrho}{1-\varrho} - \frac{\alpha\varrho}{1-\alpha\varrho}\right\} \\
&= C\sqrt{p}G'(\theta)\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \tag{S297}
\end{aligned}$$

Step 3: exact summation by parts for the full score. Reindexing Equation (S164) by $s = n - r$ shows that the contribution of h_s to the full DMOP- α population score is

$$\begin{aligned}
&\rho_s(h_s) + \sum_{q=1}^{N-s} \alpha^q \{\rho_{s+q}(h_s) - \rho_{s+q-1}(h_s)\} \\
&= \rho_s(h_s) + \sum_{q=1}^{N-s} \alpha^q \rho_{s+q}(h_s) - \sum_{q=0}^{N-s-1} \alpha^{q+1} \rho_{s+q}(h_s) \\
&= (1-\alpha) \sum_{q=0}^{N-s-1} \alpha^q \rho_{s+q}(h_s) + \alpha^{N-s} \rho_N(h_s). \tag{S298}
\end{aligned}$$

The contribution of h_s to the exact score in Equation (S162) is $\rho_N(h_s)$. Subtracting Equation (S298) from this contribution gives

$$\begin{aligned}
&\rho_N(h_s) - \rho_s(h_s) - \sum_{q=1}^{N-s} \alpha^q \{\rho_{s+q}(h_s) - \rho_{s+q-1}(h_s)\} \\
&= (1-\alpha) \sum_{q=0}^{N-s-1} \alpha^q \{\rho_N(h_s) - \rho_{s+q}(h_s)\}. \tag{S299}
\end{aligned}$$

Summing Equation (S299) over $s = 1, \dots, N-1$ yields the exact identity

$$\nabla_{\theta} \ell(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) = (1-\alpha) \sum_{s=1}^{N-1} \sum_{q=0}^{N-s-1} \alpha^q \{\rho_N(h_s) - \rho_{s+q}(h_s)\}. \tag{S300}$$

Step 4: contract each partial-smoother comparison. Fix $s \leq N-1$, $0 \leq q \leq N-s-1$, and a coordinate $i \in \{1, \dots, p\}$. The Markov property implies that observations after time $s+q$ are conditionally independent of the path through time s given X_{s+q} . Hence the following two conditional expectations are the same function of x :

$$\begin{aligned}
&\mathbb{E}_{\rho_N}\{h_{s,i} \mid X_{s+q} = x\} = \mathbb{E}_{\rho_{s+q}}\{h_{s,i} \mid X_{s+q} = x\} \\
&= \begin{cases} \bar{h}_{s,i}(x), & q = 0, \\ (B_{s+q}B_{s+q-1} \cdots B_{s+1}\bar{h}_{s,i})(x), & q \geq 1. \end{cases} \tag{S301}
\end{aligned}$$

Here $\bar{h}_{s,i}$ and B_t are already defined in Equations (S177) and (S171). We first sharpen the bound on the change of a probability expectation under a positive likelihood.

For a probability measure μ , a positive function L with $0 < a \leq L \leq b$, and a bounded real function f , suppose first that $a < b$. Convexity of the absolute value gives, pointwise,

$$|L - \mu(L)| \leq \frac{b-L}{b-a} \{\mu(L) - a\} + \frac{L-a}{b-a} \{b - \mu(L)\}. \quad (\text{S302})$$

Subtract the midpoint of the range of f inside $\mu[f\{L - \mu(L)\}]$, and then integrate Equation (S302). Since $\mu\{L - \mu(L)\} = 0$, this gives

$$\begin{aligned} \left| \frac{\mu(fL)}{\mu(L)} - \mu(f) \right| &\leq \frac{\text{osc}(f)}{2\mu(L)} \mu|L - \mu(L)| \\ &\leq \text{osc}(f) \frac{\{b - \mu(L)\}\{\mu(L) - a\}}{\mu(L)(b-a)} \\ &= \text{osc}(f) \frac{a + b - \mu(L) - ab/\mu(L)}{b-a} \\ &\leq \text{osc}(f) \frac{\sqrt{b} - \sqrt{a}}{\sqrt{b} + \sqrt{a}}. \end{aligned} \quad (\text{S303})$$

The last inequality is $\mu(L) + ab/\mu(L) \geq 2\sqrt{ab}$. When $a = b$, L is constant and the difference is zero.

Equation (S193) gives $\underline{M}/\bar{M} = 2(1 - \varrho)$ and $\frac{1}{2} \leq \varrho < 1$. For $\frac{1}{2} < \varrho < 1$, apply Equation (S303) to the two backward kernels at x and x' . In this application set

$$L(u) = \frac{f_{X_t|X_{t-1}}(x | u; \theta)}{f_{X_t|X_{t-1}}(x' | u; \theta)}. \quad (\text{S304})$$

Equation (S171) and (A2) give

$$2(1 - \varrho) \leq L(u) \leq \{2(1 - \varrho)\}^{-1}, \quad B_t(f)(x) = \frac{B_t(fL)(x')}{B_t(L)(x')}. \quad (\text{S305})$$

Taking $\mu = B_t(x', \cdot)$, $a = 2(1 - \varrho)$, and $b = \{2(1 - \varrho)\}^{-1}$ in Equation (S303) therefore gives

$$\text{osc}\{B_t(f)\} \leq \frac{2\varrho - 1}{3 - 2\varrho} \text{osc}(f). \quad (\text{S306})$$

Applying this inequality to the q kernels in Equation (S301), starting from $\text{osc}(\bar{h}_{s,i}) \leq 2G'(\theta)$, yields

$$\text{osc}[x \mapsto \mathbb{E}_{\rho_N}\{h_{s,i} | X_{s+q} = x\}] \leq 2G'(\theta) \left(\frac{2\varrho - 1}{3 - 2\varrho} \right)^q \leq 2G'(\theta)\varrho^q. \quad (\text{S307})$$

This also covers $q = 0$, when the product contains no backward kernel.

Next, the future likelihood already defined in Equation (S178) satisfies, since $s + q < N$,

$$L_{s+q,N}(x) = \int M_{s+q+1}(x, du) G_{s+q+1}(u) L_{s+q+1,N}(u),$$

$$L_{s+q,N}(x) \geq 2(1 - \varrho)L_{s+q,N}(x') \quad \text{for every } x, x'. \quad (\text{S308})$$

The second line follows by applying $M_{s+q+1}(x, \cdot) \geq 2(1 - \varrho)M_{s+q+1}(x', \cdot)$ to the nonnegative integrand. Thus its supremum divided by its infimum is at most $\{2(1 - \varrho)\}^{-1}$. Apply Equation (S303) with $\mu = \pi_{s+q}$ and $L = L_{s+q,N}$ to obtain

$$\left| \frac{\pi_{s+q}(fL_{s+q,N})}{\pi_{s+q}(L_{s+q,N})} - \pi_{s+q}(f) \right| \leq \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} \text{osc}(f). \quad (\text{S309})$$

The tower property and the common conditional expectation in Equation (S301) identify the two quantities being compared:

$$\begin{aligned} \rho_N(h_{s,i}) &= \frac{\rho_{s+q}[\mathbb{E}_{\rho_{s+q}}\{h_{s,i} \mid X_{s+q}\}L_{s+q,N}(X_{s+q})]}{\rho_{s+q}\{L_{s+q,N}(X_{s+q})\}}, \\ \rho_{s+q}(h_{s,i}) &= \rho_{s+q}[\mathbb{E}_{\rho_{s+q}}\{h_{s,i} \mid X_{s+q}\}]. \end{aligned} \quad (\text{S310})$$

Consequently, Equations (S307)–(S310) give

$$\begin{aligned} |\rho_N(h_{s,i}) - \rho_{s+q}(h_{s,i})| &\leq 2G'(\theta) \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} \left(\frac{2\varrho - 1}{3 - 2\varrho} \right)^q \\ &\leq 2G'(\theta)\varrho^{q+1}. \end{aligned} \quad (\text{S311})$$

For the second inequality, each of $(1 - \sqrt{2(1 - \varrho)})/(1 + \sqrt{2(1 - \varrho)})$ and $(2\varrho - 1)/(3 - 2\varrho)$ is at most $2\varrho - 1 \leq \varrho$. Summing the squared coordinate bounds and taking square roots gives

$$\begin{aligned} \|\rho_N(h_s) - \rho_{s+q}(h_s)\|_2 &\leq 2\sqrt{p}G'(\theta) \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} \left(\frac{2\varrho - 1}{3 - 2\varrho} \right)^q \\ &\leq 2\sqrt{p}G'(\theta)\varrho^{q+1}. \end{aligned} \quad (\text{S312})$$

If every transition is independent of its starting location, as (A2) implies when $\varrho = \frac{1}{2}$, then $L_{s+q,N}$ is constant, Equation (S310) has equal right-hand sides, and the full bias is zero.

Step 5: sum the boundary comparisons. Insert Equation (S312) into Equation (S300), apply the triangle inequality, and reverse the two finite sums. This gives

$$\begin{aligned} &\|\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2 \\ &\leq 2\sqrt{p}G'(\theta)\varrho(1 - \alpha) \sum_{s=1}^{N-1} \sum_{q=0}^{N-s-1} (\alpha\varrho)^q \\ &= 2\sqrt{p}G'(\theta)\varrho(1 - \alpha) \sum_{q=0}^{N-2} (N - 1 - q)(\alpha\varrho)^q \\ &\leq 2(N - 1)\sqrt{p}G'(\theta)\varrho(1 - \alpha) \sum_{q=0}^{\infty} (\alpha\varrho)^q \\ &= 2(N - 1)\sqrt{p}G'(\theta) \frac{\varrho(1 - \alpha)}{1 - \alpha\varrho}. \end{aligned} \quad (\text{S313})$$

Finally, multiplying the finite sum by $1 - \alpha\varrho$ gives

$$\begin{aligned}
& (1 - \alpha\varrho) \sum_{q=0}^{N-2} (N - 1 - q)(\alpha\varrho)^q \\
&= (N - 1) - \sum_{q=1}^{N-1} (\alpha\varrho)^q \\
&= (N - 1) - \frac{\alpha\varrho\{1 - (\alpha\varrho)^{N-1}\}}{1 - \alpha\varrho}.
\end{aligned} \tag{S314}$$

Multiplying Equation (S314) once more by $1 - \alpha\varrho$ and collecting terms therefore gives

$$\sum_{q=0}^{N-2} (N - 1 - q)(\alpha\varrho)^q = \frac{(N - 1) - N\alpha\varrho + (\alpha\varrho)^N}{(1 - \alpha\varrho)^2}. \tag{S315}$$

Equations (S313) and (S315) prove Equation (S293). To obtain the sharper bound, retain the first line of Equation (S312) in the same finite sum. For $\frac{1}{2} < \varrho < 1$,

$$\begin{aligned}
& \|\nabla_{\theta}\ell(\theta) - \nabla_{\theta}\ell^{\alpha}(\theta)\|_2 \\
& \leq 2\sqrt{p}G'(\theta) \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} (1 - \alpha) \sum_{q=0}^{N-2} (N - 1 - q) \left\{ \alpha \frac{2\varrho - 1}{3 - 2\varrho} \right\}^q \\
& \leq 2(N - 1)\sqrt{p}G'(\theta) \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} (1 - \alpha) \sum_{q=0}^{\infty} \left\{ \alpha \frac{2\varrho - 1}{3 - 2\varrho} \right\}^q \\
& = 2(N - 1)\sqrt{p}G'(\theta) \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} \frac{(1 - \alpha)(3 - 2\varrho)}{3 - 2\varrho - \alpha(2\varrho - 1)}.
\end{aligned} \tag{S316}$$

The ratio in the geometric series is strictly less than one even at $\alpha = 1$. This proves Equation (S294). $\square$

Remark 1 (Signed cancellation in the population bias). *Retaining signs gives a first-order expansion of the population bias. The score is a finite polynomial in α , so*

$$\begin{aligned}
\left. \frac{\partial}{\partial \alpha} \nabla_{\theta}\ell^{\alpha}(\theta) \right|_{\alpha=1} &= \sum_{n=1}^N \sum_{r=1}^{n-1} r \Delta_{n,r} \\
&= \sum_{s=1}^{N-1} \sum_{q=0}^{N-s-1} \{\rho_N(h_s) - \rho_{s+q}(h_s)\}.
\end{aligned} \tag{S317}$$

The first equality follows by differentiating Equation (S164) and summing over n . For the second, divide Equation (S300) by $1 - \alpha$ and let $\alpha \uparrow 1$; all sums are finite. Subtracting $(1 - \alpha)$ times Equation (S317) from Equation (S300) gives exactly

$$\nabla_{\theta}\ell(\theta) - \nabla_{\theta}\ell^{\alpha}(\theta) - (1 - \alpha) \left. \frac{\partial}{\partial \alpha} \nabla_{\theta}\ell^{\alpha}(\theta) \right|_{\alpha=1}$$

$$= -(1 - \alpha) \sum_{s=1}^{N-1} \sum_{q=0}^{N-s-1} (1 - \alpha^q) \{\rho_N(h_s) - \rho_{s+q}(h_s)\}. \quad (\text{S318})$$

Thus Equation (S312), the triangle inequality, and reversal of the finite sums give, for $\frac{1}{2} < \varrho < 1$,

$$\begin{aligned} & \left\| \nabla_{\theta} \ell(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) - (1 - \alpha) \frac{\partial}{\partial \alpha} \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_{\alpha=1} \Big|_2 \\ & \leq 2\sqrt{p} G'(\theta) \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} (1 - \alpha) \sum_{q=0}^{N-2} (N - 1 - q)(1 - \alpha^q) \left(\frac{2\varrho - 1}{3 - 2\varrho} \right)^q \\ & \leq 2(N - 1)\sqrt{p} G'(\theta) \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} (1 - \alpha) \left\{ \frac{3 - 2\varrho}{4(1 - \varrho)} - \frac{3 - 2\varrho}{3 - 2\varrho - \alpha(2\varrho - 1)} \right\} \\ & = (N - 1)\sqrt{p} G'(\theta) \frac{(1 - \sqrt{2(1 - \varrho)})^2 (3 - 2\varrho)(1 - \alpha)^2}{2(1 - \varrho)\{3 - 2\varrho - \alpha(2\varrho - 1)\}}. \end{aligned} \quad (\text{S319})$$

The last equality uses $2\varrho - 1 = (1 - \sqrt{2(1 - \varrho)})(1 + \sqrt{2(1 - \varrho)})$. For $\varrho = \frac{1}{2}$, the score difference and its signed derivative are both zero, so the same bound holds. In particular, the population bias is bounded by

$$\begin{aligned} & \|\nabla_{\theta} \ell(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta)\|_2 \\ & \leq (1 - \alpha) \left\| \sum_{n=1}^N \sum_{r=1}^{n-1} r \Delta_{n,r} \right\|_2 + (N - 1)\sqrt{p} G'(\theta) \frac{(1 - \sqrt{2(1 - \varrho)})^2 (3 - 2\varrho)(1 - \alpha)^2}{2(1 - \varrho)\{3 - 2\varrho - \alpha(2\varrho - 1)\}}. \end{aligned} \quad (\text{S320})$$

For every fixed observation sequence, if the signed sum vanishes, this gives $O\{N(1 - \alpha)^2\}$ bias and $O\{N^2(1 - \alpha)^4\}$ squared bias, with fixed model constants. The next example shows that the assumptions do not ensure this cancellation.

Remark 2 (Sharpness of the fixed-observation population order). *The powers of N and $1 - \alpha$ in Equation (S293) cannot be improved uniformly under the conditions of Lemma S14. To see this, take state space $\{0, 1\}$ with counting measure,*

$$X_0 \sim \text{Bernoulli}(3/4), \quad \Pr(X_n = X_{n-1} \mid X_{n-1}) = \frac{3}{4}, \quad (\text{S321})$$

fix $y_n^* = 1$ for every n , and, for $|\theta| \leq 1/4$, set

$$g_{\theta}(1 \mid 0) = \frac{1}{4}, \quad g_{\theta}(1 \mid 1) = \frac{9}{20}(1 + \theta), \quad (\text{S322})$$

with the probabilities at observation zero defined by complementation. At $\theta = \phi = 0$,

$$h_n(x) = x. \quad (\text{S323})$$

If the filtering probability of state one at time $n - 1$ is $3/4$, its one-step prediction and update are

$$\Pr(X_n = 1 \mid y_{1:n-1}^*) = \frac{3}{4} \frac{3}{4} + \frac{1}{4} \frac{1}{4} = \frac{5}{8},$$

$$\Pr(X_n = 1 \mid y_{1:n}^*) = \frac{(5/8)(9/20)}{(5/8)(9/20) + (3/8)(1/4)} = \frac{3}{4}. \quad (\text{S324})$$

The law of X_0 initializes this induction, so the filtered probability is $3/4$ at every time. The resulting time-homogeneous posterior backward kernel satisfies

$$B(1 \mid 1) = \frac{(3/4)(3/4)}{5/8} = \frac{9}{10}, \quad B(1 \mid 0) = \frac{(3/4)(1/4)}{3/8} = \frac{1}{2}. \quad (\text{S325})$$

Moreover, the next-observation likelihoods conditional on X_{n-1} are

$$M_n(G_n)(1) = \frac{3}{4} \frac{9}{20} + \frac{1}{4} \frac{1}{4} = \frac{2}{5}, \quad M_n(G_n)(0) = \frac{1}{4} \frac{9}{20} + \frac{3}{4} \frac{1}{4} = \frac{3}{10}. \quad (\text{S326})$$

Consequently,

$$\begin{aligned} \Pr_{\rho_n}(X_{n-1} = 1) &= \frac{(3/4)(2/5)}{(3/4)(2/5) + (1/4)(3/10)} = \frac{4}{5}, \\ \Pr_{\rho_{n-1}}(X_{n-1} = 1) &= \frac{3}{4}. \end{aligned} \quad (\text{S327})$$

Equation (S325) maps every affine function of the state to an affine function whose slope is multiplied by

$$B(1 \mid 1) - B(1 \mid 0) = \frac{2}{5}. \quad (\text{S328})$$

Applying the $r - 1$ backward kernels between X_{n-r} and X_{n-1} and then using Equation (S327) therefore gives, for $1 \leq r < n$,

$$\begin{aligned} \Delta_{n,r} &= \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}) \\ &= \left(\frac{4}{5} - \frac{3}{4}\right) \left(\frac{2}{5}\right)^{r-1} = \frac{1}{20} \left(\frac{2}{5}\right)^{r-1}. \end{aligned} \quad (\text{S329})$$

All these quantities are nonnegative. Equation (S295), summed over n , now implies

$$\begin{aligned} \nabla_{\theta} \ell(0) - \nabla_{\theta} \ell^{\alpha}(0) &= \frac{1}{20} \sum_{r=1}^{N-1} (N-r)(1-\alpha^r) \left(\frac{2}{5}\right)^{r-1} \\ &\geq \frac{(N-1)(1-\alpha)}{20}. \end{aligned} \quad (\text{S330})$$

Thus the squared population bias is at least $(N-1)^2(1-\alpha)^2/400$. This model satisfies the uniform transition and measurement bounds on $|\theta| \leq 1/4$, its simulator is independent of θ , and h_n is a bounded current function. A uniform improvement therefore requires additional assumptions, such as signed cancellation.

Theorem S5 (MSE bounds for DMOP- α). *Under the conditions of Lemmas S13 and S14, take the same $\varrho = 1 - \underline{M}/(2\bar{M}) \in [\frac{1}{2}, 1)$. For $\alpha \in [0, 1]$,*

$$\begin{aligned} & \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ & \lesssim pG'(\theta)^2 \left[\left(\frac{N(1 + \log J)^2}{J} + \frac{N^2}{J^2} \right) \left(\sum_{r=0}^{N-1} \alpha^r \right)^2 \right. \\ & \quad \left. + \frac{(1 - \sqrt{2(1 - \varrho)})^2 (1 - \alpha)^2}{(1 + \sqrt{2(1 - \varrho)})^2} \left\{ N - 1 + \sum_{q=1}^{N-2} (N - 1 - q) \left(\alpha \frac{2\varrho - 1}{3 - 2\varrho} \right)^q \right\}^2 \right]. \end{aligned} \quad (\text{S331})$$

For $0 \leq \alpha < 1$, this implies

$$\begin{aligned} & \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ & \lesssim pG'(\theta)^2 \left[\frac{N(1 + \log J)^2}{J(1 - \alpha)^2} + \frac{N^2}{J^2(1 - \alpha)^2} \right. \\ & \quad \left. + N^2 \left\{ \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} \frac{(1 - \alpha)(3 - 2\varrho)}{3 - 2\varrho - \alpha(2\varrho - 1)} \right\}^2 \right]. \end{aligned} \quad (\text{S332})$$

The finite sum beginning at $q = 1$ is empty for $N \leq 2$. At $\varrho = \frac{1}{2}$, the displayed population terms are zero. In particular, the two endpoints satisfy

$$\begin{aligned} & \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^0(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ & \lesssim pG'(\theta)^2 \left[\frac{N(1 + \log J)^2}{J} + \frac{N^2}{J^2} + (N - 1)^2 \left\{ \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} \right\}^2 \right], \end{aligned} \quad (\text{S333})$$

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^1(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \lesssim pG'(\theta)^2 \min \left\{ N^2, \frac{N^3}{J} \right\}. \quad (\text{S334})$$

At $\alpha = 0$, the variance is at most order $NpG'(\theta)^2(1 + \log J)^2/J$; at $\alpha = 1$, the population discounting bias is exactly zero. Remark 1 gives a complementary bound retaining signed cancellation.

Proof. Add and subtract $\nabla_{\theta} \ell^{\alpha}(\theta)$ and use $\|a + b\|_2^2 \leq 2\|a\|_2^2 + 2\|b\|_2^2$. This gives

$$\begin{aligned} & \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ & \leq 2\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2^2 + 2\left\| \nabla_{\theta} \ell^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2. \end{aligned} \quad (\text{S335})$$

Lemma S13 bounds the first term by

$$CpG'(\theta)^2 \left(\frac{N(1 + \log J)^2}{J} + \frac{N^2}{J^2} \right) \left(\sum_{r=0}^{N-1} \alpha^r \right)^2. \quad (\text{S336})$$

The sharper bound in Equation (S294) of Lemma S14 gives

$$\begin{aligned} & \|\nabla_{\theta}\ell^{\alpha}(\theta) - \nabla_{\theta}\ell(\theta)\|_2^2 \\ & \leq 4pG'(\theta)^2 \frac{(1 - \sqrt{2(1-\varrho)})^2(1-\alpha)^2}{(1 + \sqrt{2(1-\varrho)})^2} \left\{ N-1 + \sum_{q=1}^{N-2} (N-1-q) \left(\alpha \frac{2\varrho-1}{3-2\varrho} \right)^q \right\}^2. \end{aligned} \quad (\text{S337})$$

Here the $q = 0$ summand has been written separately as $N-1$. For $\varrho = \frac{1}{2}$, both sides of this last inequality are zero by Lemma S14. Substitution into Equation (S335), absorbing fixed numerical factors into $\lesssim$, proves the first bound. For $0 \leq \alpha < 1$,

$$\sum_{r=0}^{N-1} \alpha^r \leq \frac{1}{1-\alpha}. \quad (\text{S338})$$

Also,

$$\begin{aligned} & N-1 + \sum_{q=1}^{N-2} (N-1-q) \left(\alpha \frac{2\varrho-1}{3-2\varrho} \right)^q \\ & \leq (N-1) \left\{ 1 + \sum_{q=1}^{\infty} \left(\alpha \frac{2\varrho-1}{3-2\varrho} \right)^q \right\} \\ & = \frac{(N-1)(3-2\varrho)}{3-2\varrho - \alpha(2\varrho-1)}. \end{aligned} \quad (\text{S339})$$

The geometric ratio is in $[0, 1)$, including when $\varrho = \frac{1}{2}$. These inequalities and $N-1 \leq N$ prove Equation (S332). Alternatively, Equation (S320) may be substituted in Equation (S335), taking the smaller population bound without changing the particle contribution.

For $\alpha = 0$, the geometric sum is one and every summand with $q \geq 1$ is zero. Substitution in Equation (S331) gives Equation (S333); Equation (S242) gives the stated variance. For $\alpha = 1$, use the terminal-block estimate directly, without separately bounding and squaring the particle bias. Equation (S285) and the two telescopes in Equations (S248)–(S249), with $k = N-1$, give

$$\nabla_{\theta}\hat{\ell}^1(\theta) = m(\xi_N) \left(\sum_{s=1}^N h_s \right), \quad \nabla_{\theta}\ell^1(\theta) = \rho_N \left(\sum_{s=1}^N h_s \right) = \nabla_{\theta}\ell(\theta). \quad (\text{S340})$$

Consequently, Equation (S272) implies

$$\begin{aligned} & \mathbb{E} \left\| \nabla_{\theta}\hat{\ell}^1(\theta) - \nabla_{\theta}\ell(\theta) \right\|_2^2 \\ & = \sum_{i=1}^p \mathbb{E} \left| m(\xi_N) \left(\sum_{s=1}^N h_{s,i} \right) - \rho_N \left(\sum_{s=1}^N h_{s,i} \right) \right|^2 \leq \frac{CN^3 p G'(\theta)^2}{J}. \end{aligned} \quad (\text{S341})$$

Since $\|h_s\|_2 \leq \sqrt{p} G'(\theta)$, the same difference satisfies, pathwise,

$$\left\| \nabla_{\theta}\hat{\ell}^1(\theta) - \nabla_{\theta}\ell(\theta) \right\|_2 \leq m(\xi_N) \left(\left\| \sum_{s=1}^N h_s \right\|_2 \right) + \rho_N \left(\left\| \sum_{s=1}^N h_s \right\|_2 \right)$$

$$\leq 2N\sqrt{p}G'(\theta). \quad (\text{S342})$$

Squaring and combining with Equation (S341) proves Equation (S334). $\square$

Remark 3 (Neutral-model check). *To check the rates in Lemma S13, take iid Uniform[0, 1] states, constant weights at $\theta = \phi = 0$, and*

$$\Pr_{\theta}(Y_n = 1 \mid X_n = x) = \frac{1}{2} + \frac{\theta}{4}(2x - 1), \quad y_n^* = 1. \quad (\text{S343})$$

Then $h_n = (2X_n - 1)/2$ has variance $\sigma^2 = 1/12$, and direct conditional multinomial calculation gives

$$\text{Var} \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\} = \frac{\sigma^2}{J} \left[N(2 - J^{-1}) + \sum_{r=1}^{N-1} (N-r) \alpha^{2r} (1 - J^{-1})^{r+1} \right]. \quad (\text{S344})$$

For fixed $\alpha < 1$, the variance is of order N/J . Its genealogical contribution is of order $N/\{J(1 - \alpha^2)\}$ before horizon saturation, so the general factor $(1 - \alpha)^{-2}$ is loose in this model. At $\alpha = 1$ and $J/N \rightarrow \infty$, the same exact formula gives

$$\text{Var} \left\{ \nabla_{\theta} \hat{\ell}^1(\theta) \right\} \sim \frac{\sigma^2}{J} \left\{ 2N + \sum_{r=1}^{N-1} (N-r) \right\} = \frac{\sigma^2 N(N+3)}{2J}. \quad (\text{S345})$$

Thus the general N^3/J endpoint upper bound is also conservative in this model.

Remark 4 (Rates and normalization). *The finite particle term in Lemma S13 is*

$$pG'(\theta)^2 \left(\frac{N(1 + \log J)^2}{J} + \frac{N^2}{J^2} \right) \left(\sum_{r=0}^{N-1} \alpha^r \right)^2. \quad (\text{S346})$$

At $\alpha = 0$, the particle-MSE upper bound is $pG'(\theta)^2 \{N(1 + \log J)^2/J + N^2/J^2\}$. Its variance bound is $NpG'(\theta)^2(1 + \log J)^2/J$, whereas its MSE relative to the full score also contains the population term displayed in Equation (S333). Every fixed $\alpha < 1$ has an $N(1 + \log J)^2/J$ particle-variance upper bound and, when $J \geq N$, the same leading particle-MSE order, with constants depending on α . The particle bound increases with the effective memory $\sum_{r=0}^{N-1} \alpha^r$.

The population bound in Theorem S5 decreases to zero as $\alpha \uparrow 1$. At $\alpha = 1$, the terminal L^2 estimate gives Equation (S334) directly. Substitution into Equation (S346) would instead give $N^3(1 + \log J)^2/J + N^4/J^2$. The neutral-model example shows that the $(1 - \alpha)^{-2}$ dependence and the N^3/J endpoint bound can be loose.

A particle MSE bound for $\text{osc}(f) \leq 1$ is multiplied by $\text{osc}(f)^2$ when it is applied to an unnormalized functional. Thus a generic path-space bound of order N^2/J for an oscillation-one functional becomes N^4/J if it is applied once to a full score whose oscillation is of order N . Lemma S13 bounds each h_s before summing lags, so Theorem S5 needs no further rescaling. The resulting N^4/J generic upper bound is separate from the quadratic asymptotic-variance lower bound of Poyiadjis et al. (2011).

Remark 5 (Choosing α in the upper bound). *We tune the upper bound. For $0 \leq \alpha < 1$,*

$$\sum_{r=0}^{N-1} \alpha^r \leq \min\{N, (1 - \alpha)^{-1}\} \quad (\text{S347})$$

and Theorem S5 give

$$\begin{aligned} & \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ & \lesssim pG'(\theta)^2 \left[\left(\frac{N(1 + \log J)^2}{J} + \frac{N^2}{J^2} \right) \min\{N^2, (1 - \alpha)^{-2}\} \right. \\ & \quad \left. + N^2 \left\{ \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} \frac{(1 - \alpha)(3 - 2\varrho)}{3 - 2\varrho - \alpha(2\varrho - 1)} \right\}^2 \right]. \end{aligned} \quad (\text{S348})$$

For fixed $\frac{1}{2} < \varrho < 1$, before the effective memory reaches the horizon, balancing these two terms is equivalent to

$$\frac{(1 + \log J)^2}{NJ} + \frac{1}{J^2} \asymp \left\{ \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} \frac{3 - 2\varrho}{3 - 2\varrho - \alpha(2\varrho - 1)} \right\}^2 (1 - \alpha)^4. \quad (\text{S349})$$

Suppressing these fixed model constants gives the following rate-attaining choices for $N \geq 2$:

regime	$1 - \alpha$	MSE upper bound/ $\{pG'(\theta)^2\}$
$J(1 + \log J)^2 \lesssim N$	$\asymp J^{-1/2}$	N^2/J
$N \lesssim J(1 + \log J)^2 \lesssim N^3$	$\asymp \{(1 + \log J)^2/(NJ)\}^{1/4}$	$N^{3/2}(1 + \log J)/J^{1/2}$
$J(1 + \log J)^2 \gtrsim N^3$	0	N^3/J

(S350)

The first boundary is where the two particle coefficients have the same order. In the middle regime, compare the balanced bound with the independent terminal bound:

$$\min \left\{ \frac{N^{3/2}(1 + \log J)}{\sqrt{J}}, \frac{N^3}{J} \right\}.$$

They agree when $J(1 + \log J)^2$ has order N^3 . The choice $\alpha = 1$ attains the last rate by Equation (S334). To check minimization of the finite bound in Equation (S331), for $N \geq 2$,

$$\frac{\partial}{\partial \alpha} \left\{ \left(\frac{N(1 + \log J)^2}{J} + \frac{N^2}{J^2} \right) \left(\sum_{r=0}^{N-1} \alpha^r \right)^2 \right\} \Big|_{\alpha=1} = \left(\frac{N(1 + \log J)^2}{J} + \frac{N^2}{J^2} \right) N^2(N - 1) > 0, \quad (\text{S351})$$

The squared population term has derivative zero there, so decreasing α slightly from one reduces this finite upper bound. This need not change the rate or reduce the actual MSE. The first-row choice $J^{-1/2}$ is a rate for $1 - \alpha$; the corresponding MSE upper bound is N^2/J .

For fixed $\varrho \in (\frac{1}{2}, 1)$, its factors affect the tuning constants. At $\varrho = \frac{1}{2}$, the population term is zero and $\alpha = 0$ minimizes the finite particle upper bound. For $N = 1$ all choices of α give the same estimator. Finally, if the signed sum in Equation (S317) vanishes, Remark 1 gives, before horizon saturation and with fixed model constants,

$$1 - \alpha \asymp \left(\frac{(1 + \log J)^2}{NJ} + \frac{1}{J^2} \right)^{1/6},$$

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \lesssim p G'(\theta)^2 N^2 \left(\frac{(1 + \log J)^2}{N J} + \frac{1}{J^2} \right)^{2/3}. \quad (\text{S352})$$

Indeed, the particle contribution is of order $\{N(1 + \log J)^2/J + N^2/J^2\}(1 - \alpha)^{-2}$ and the squared population contribution is then at most order $N^2(1 - \alpha)^4$. This cancellation is an additional condition for the last rate.

Remark 6 (Geometric discounting and lag truncation). For $0 \leq k < N$, the truncations defined in Lemma S13 satisfy

$$\begin{aligned} \nabla_{\theta} \hat{s}_k^1(\theta) &= \sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N) \left(\sum_{s=N-k}^N h_s \right), \\ \nabla_{\theta} \ell(\theta) - \nabla_{\theta} s_k^1(\theta) &= \sum_{s=1}^{N-k-1} \sum_{u=s+k+1}^N \Delta_{u, u-s}. \end{aligned} \quad (\text{S353})$$

Equation (S312) with $q = k$ gives, for $\frac{1}{2} < \varrho < 1$,

$$\left\| \nabla_{\theta} \ell(\theta) - \nabla_{\theta} s_k^1(\theta) \right\|_2 \leq 2(N - k - 1) \sqrt{p} G'(\theta) \frac{1 - \sqrt{2(1 - \varrho)}}{1 + \sqrt{2(1 - \varrho)}} \left(\frac{2\varrho - 1}{3 - 2\varrho} \right)^k. \quad (\text{S354})$$

At $\varrho = \frac{1}{2}$ the population error is zero for every k ; at $k = N - 1$ it is zero for every ϱ . The finite-window particle calculation in Equations (S281) and (S282) gives

$$\begin{aligned} \text{Var}\{\nabla_{\theta} \hat{s}_k^1(\theta)\} &\lesssim \frac{N p G'(\theta)^2 (k + 1)^2 (1 + \log J)^2}{J}, \\ \left\| \mathbb{E} \nabla_{\theta} \hat{s}_k^1(\theta) - \nabla_{\theta} s_k^1(\theta) \right\|_2 &\lesssim \frac{N \sqrt{p} G'(\theta) (k + 1)}{J}. \end{aligned} \quad (\text{S355})$$

Equation (S285) writes DMOP- α as their geometric average. Its effective memory is

$$\sum_{k=0}^{N-1} \beta_k (k + 1) = \sum_{r=0}^{N-1} \alpha^r \quad (\text{S356})$$

Thus every fixed $\alpha < 1$ has an $N(1 + \log J)^2/J$ particle-variance upper bound, including the DMOP-0 case.

S8 Figures for Bayesian Inference

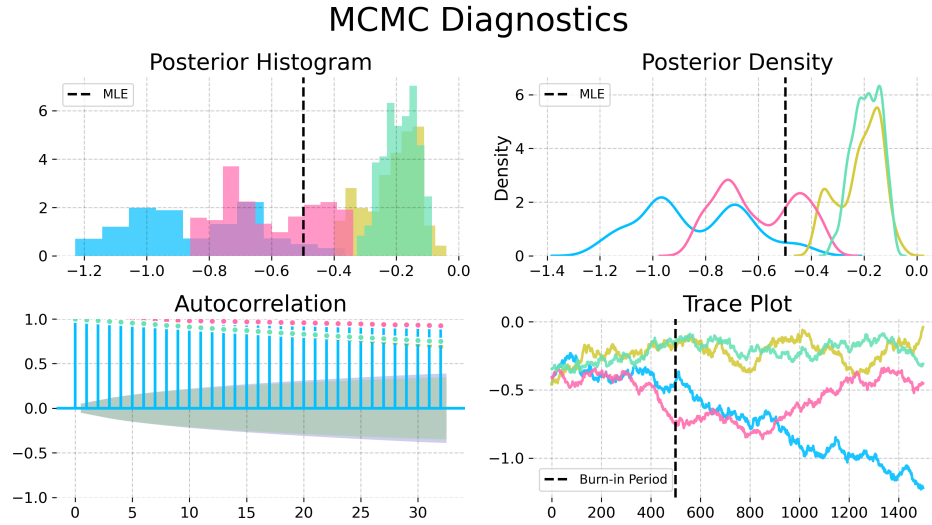


Figure S-1: Convergence diagnostics for the Metropolis-Hastings variant particle MCMC with the random walk proposal and an informative empirical prior from IF2, on the trend parameter in the Dhaka cholera model of King et al. (2008). The poor convergence shown here, using an MCMC algorithm that does not have access to derivatives, contrasts with Fig. 6 in the main text.

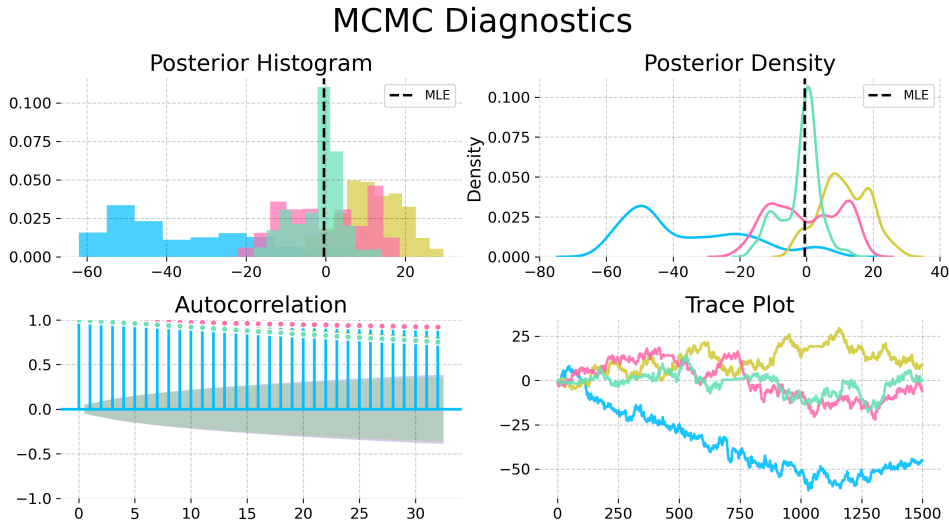


Figure S-2: Convergence diagnostics for a No-U-Turn Sampler (NUTS) with uniform priors on a compact set, on the trend parameter in the Dhaka cholera model of King et al. (2008). The NUTS sampler explores more of the posterior than the Metropolis-Hastings sampler, but fails to converge quickly. This poor convergence, compared with Fig. 6 in the main text, demonstrates the value of the empirical prior for obtaining successful convergence.

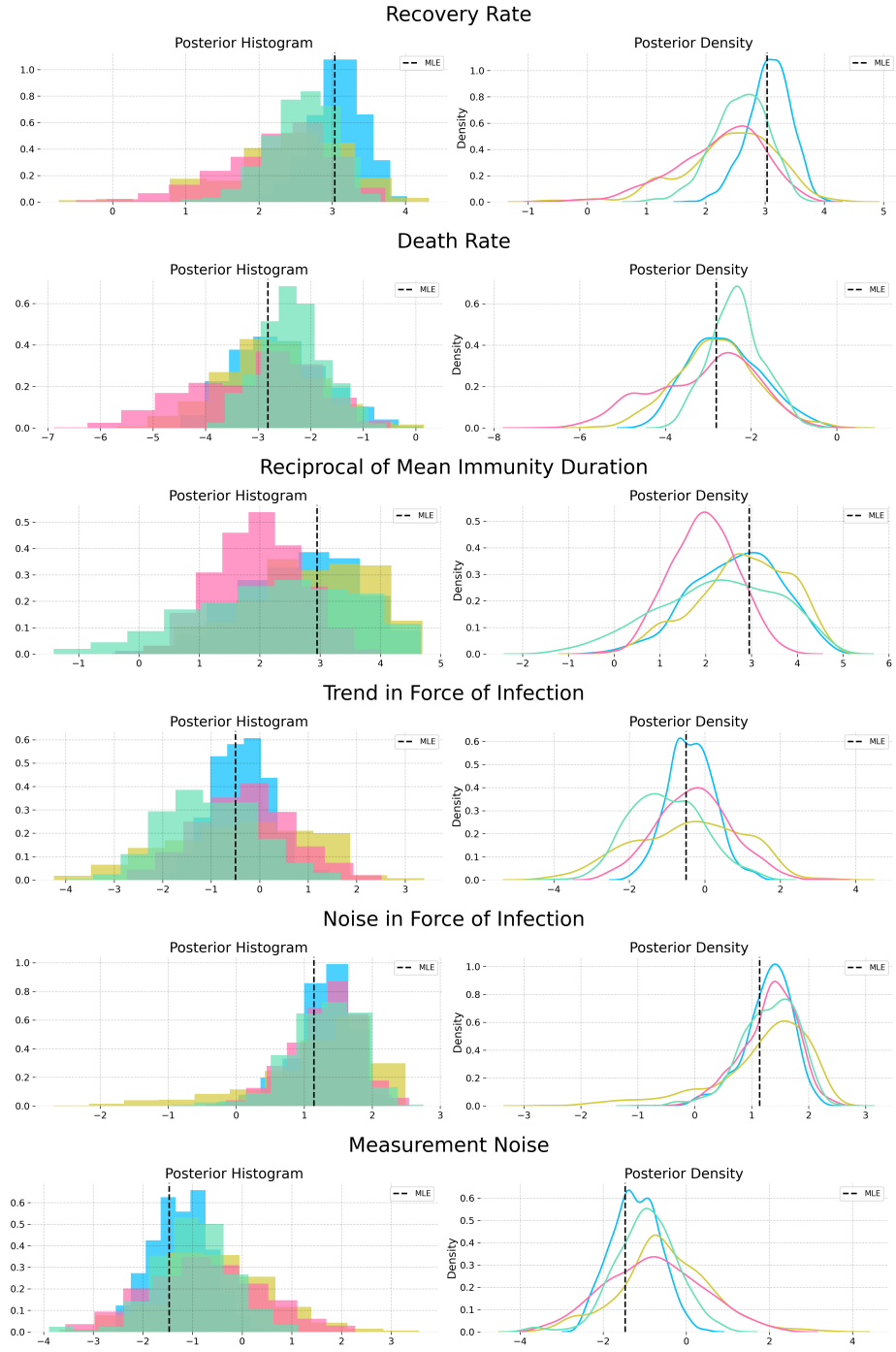


Figure S-3: Posterior estimates from NUTS powered by MOP- α with the informative nonparametric empirical prior obtained from a kernel density estimator on the IF2 parameter cloud. The posterior estimates from each chain are largely in agreement.

S9 Algorithmic Parameters

We include the full set of algorithmic parameters used for the Dhaka results in Table S-1. The warm start iterations are IF2 iterations, which are followed by gradient descent (GD) iterations when using IFAD. The IF2 cooling rate is the fractional reduction in the parameter perturbations over 50 iterations, so that a cooling rate of 0.5 corresponds to approximately 1% cooling per iteration. The random walk standard deviation (RWSD) scales the noise added at each filtering step to each regular parameter. Initial value parameters (IVPs), that affect the value of the latent process only at time t_0 , are treated differently: they are perturbed only at time t_0 , with a larger RWSD. Parameters are transformed to a unit scale, for example by taking logarithms or a logistic transformation, so that a single RWSD value works for all parameters.

Hyperparameter	Standard IF2	IFAD-0	IFAD-0.97	IFAD-1
Number of Particles (J)	5,000	5,000	5,000	5,000
Warm-start Iterations	1,950	60	175	60
GD Iterations	—	410	400	425
IF2 Cooling Rate	0.93	0.5	0.5	0.5
Initial RWSD	0.02	0.02	0.02	0.02
Initial RWSD (IVPs)	0.16	0.16	0.16	0.16
Gradient Optimizer	—	Adam	Adam	Adam
Momentum (β_1)	—	0.0	0.9	0.9
LR Schedule	—	Cosine Decay	Cosine Decay	Cosine Decay
Initial LR (η_0)	—	0.1	0.1	0.1
Warm-up Steps	—	10	10	10
Eval. Particles / Reps	5,000 / 36	5,000 / 36	5,000 / 36	5,000 / 36

Table S-1: Algorithmic parameters for the global search benchmarking on the Dhaka cholera model.

Following some experimentation, IFAD was implemented by plugging the DMOP gradient estimate into the Adam optimizer (Kingma and Ba, 2015). The momentum parameter in Adam was turned off for IFAD-0, since this low-variance estimate did not require it. Decreasing the learning rate (LR) was found to be helpful, and we selected the commonly used cosine decay. Warm-up steps were taken with Adam to stabilize its initial performance and avoid noisy

S10 Application to a Daphnia Population Model

S10.1 Model and data

The SIRJPF2 model developed by Yang et al. (2025) describes interactions between two species of *Daphnia*, their algal food, and a fungal parasite. We use the version with unit-specific infected-adult mortality to compare likelihood maximization by IFAD and the marginalized panel iterated filter (MPIF) of Wheeler et al. (2025).

The data come from the mesocosm experiment of Searle et al. (2016). We use the treatment containing the native species *D. dentifera*, the invasive species *D. lumholtzi*, and the parasite *Metschnikowia bicuspidata*, with eight mesocosms, each treated as one unit in the panel. Each mesocosm was observed at ten times, $t_n = 5n + 2$ days for $n = 1, \dots, 10$. At each observation time, a one-liter sample was taken and susceptible and infected adult females of both species were counted. Juvenile counts are excluded from the likelihood.

We write $S_u^a(t)$, $I_u^a(t)$, and $J_u^a(t)$ for the densities of susceptible adult females, infected adult females, and juveniles, where $u = 1, \dots, 8$ indexes the mesocosm and $a \in \{n, i\}$ denotes the native or invasive species. The densities of food and parasite spores are $F_u(t)$ and $P_u(t)$. These variables form an eight-dimensional latent state in each mesocosm. Removal by death, consumption, or sampling is denoted by R in the model name and does not require an additional state variable. The two host species interact through their shared food and parasite populations. The processes in different mesocosms are independent conditional on the parameters.

As in Yang et al. (2025), the transition dynamics follow an Itô stochastic differential equation (SDE):

$$\begin{aligned} dS_u^a &= \{\lambda_J J_u^a - (\theta_S^a + p^a f_S^a P_u + \delta) S_u^a\} dt + \sigma_S^a S_u^a dB_{S,u}^a, \\ dI_u^a &= \{p^a f_S^a S_u^a P_u - (\theta_{I,u}^a + \delta) I_u^a\} dt + \sigma_I^a I_u^a dB_{I,u}^a, \\ dJ_u^a &= \{r^a f_S^a F_u S_u^a - (\theta_J^a + \delta + \lambda_J) J_u^a\} dt + \sigma_J^a J_u^a dB_{J,u}^a, \\ dP_u &= \left\{ \sum_a [\beta \theta_{I,u}^a I_u^a - f_S^a (S_u^a + \xi I_u^a) P_u] - (\theta_P + \delta) P_u \right\} dt + \sigma_P P_u dB_{P,u}, \\ dF_u &= \left\{ \mu - \delta F_u - \sum_a f_S^a F_u (S_u^a + \xi I_u^a + J_u^a) \right\} dt + \sigma_F F_u dB_{F,u}, \end{aligned}$$

with independent standard Brownian motions B . The time argument is suppressed in the equations. Except for infected-adult mortality, estimated parameters are shared across mesocosms. The parameters r^a and f_S^a are the reproductive rate and host filtration rate, p^a controls infection following spore ingestion, and ξ is the filtration rate of infected adults relative to susceptible adults. The mortality or loss rates of susceptible adults, infected adults, juveniles, and spores are θ_S^a , $\theta_{I,u}^a$, θ_J^a , and θ_P , respectively; the σ parameters are process-noise coefficients. The relative filtration rate of juveniles is fixed at one.

We use the units of the public Python implementation:¹ *Daphnia* densities are individuals per liter, F is in 10^6 algal cells per liter, and P is in 10^3 spores per liter, with parameters expressed in the corresponding units. The sampling loss rate is fixed at $\delta = 0.013$ per day, the food replenishment rate at $\mu = 0.37$, the juvenile maturation rate at $\lambda_J = 0.1$ per day, and the spore yield per

¹[pypomp/Daphnia-tutorial](https://github.com/pypomp/Daphnia-tutorial), advanced tutorial, revision f318330.

infected-host death at $\beta = 30$, corresponding to 3×10^4 spores. The initial states are $S_u^n = 2.333$, $S_u^i = 0.667$, and $F_u = 16.667$, with the remaining states set to zero. Simulation begins at $t_0 = 1$ day, and parasites are introduced on day 4 at a density of 25×10^3 spores per liter. We numerically solve the SDE using the Euler-Maruyama method with a time step of 0.25 days, following the initial conditions, inoculation, and numerical boundary rules of the public implementation.

The measurement model for the adult counts is

$$Y_{S,u,n}^a \sim \text{NBinomial}(S_u^a(t_n), \tau_S^a), \quad Y_{I,u,n}^a \sim \text{NBinomial}(I_u^a(t_n), \tau_I^a),$$

where $\text{NBinomial}(m, \tau)$ has mean m and variance $m + m^2/\tau$ (Yang et al., 2025). The four counts at each observation time are independent conditional on the latent state. Since the sampled volume is one liter, the state densities and observed counts are on the same scale. The numerical implementation sets a lower bound of 10^{-10} on the negative binomial mean and size parameters. As in the public Python code, a state variable that crosses a numerical boundary is reset to zero, and that particle’s log weight for the mesocosm at the next observation is set to -150 . The boundary thresholds and the treatment of the inoculation time are specified in the code.

We fix $\sigma_S^n = \sigma_S^i = 0$ and estimate $\theta_{I,u}^n$ and $\theta_{I,u}^i$ separately for each mesocosm. The remaining 22 estimated parameters are shared across mesocosms, giving $22 + 2 \times 8 = 38$ estimated parameters. This is the model with unit-specific infected-adult mortality in Table S3 of the archived revision (Yang et al., 2026).

S10.2 Experimental settings

We tested IFAD with discount factors $\alpha = 0$, $\alpha = 0.97$, and $\alpha = 1$ against MPIF using Pypomp 1.0.5 (Abkemeier et al., 2025). For each method, we performed 50 searches from the same starting parameter vectors, generated by independent lognormal perturbations of the initial parameter vector in the public tutorial, with a standard deviation of 0.45 on the log scale. All positive estimated parameters were optimized on the log scale. Each IFAD search used an MPIF warm start of 200 iterations followed by 200 iterations of the Adam optimizer (Kingma and Ba, 2015), using DMOP- α gradient estimates. We ran MPIF alone for 680 iterations, chosen to approximate the computing time of IFAD; measured times are reported in Table S-3.

We used 1,000 particles per mesocosm during optimization. The initial random walk standard deviation (RWSD) for MPIF was 0.05 on the log scale, with geometric cooling by a factor of 0.7 over 50 iterations. Adam used $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$, with all eight mesocosms contributing to each update. The learning-rate schedule used ten steps of linear warmup multiplied by cosine decay with a final factor of 0.1. A summary of the algorithmic parameters is provided in Table S-2.

The learning-rate scale was selected using four additional starting points, separate from the 50 used for the comparison. We compared the six combinations of learning-rate scales 0.001, 0.005, and 0.01 with $\alpha \in \{0.9, 0.97\}$. The combination with the highest median final log-likelihood, evaluated by independent particle filtering, was 0.01 with $\alpha = 0.97$. We used the same learning-rate schedule for $\alpha = 0$ and $\alpha = 1$.

All reported log-likelihoods were evaluated independently of optimization using ordinary particle filters, with 2,000 particles per mesocosm and ten replicates. Likelihood estimates were averaged within each mesocosm before taking logarithms, and the eight mesocosm log-likelihoods were then summed. Each method’s best parameter vector was then held fixed and re-evaluated with ten

Hyperparameter	MPIF	IFAD-0	IFAD-0.97	IFAD-1
Particles per mesocosm (J)	1,000	1,000	1,000	1,000
MPIF Iterations	680	200	200	200
Adam Iterations	—	200	200	200
MPIF Cooling Factor	0.7	0.7	0.7	0.7
Initial RWSD (log scale)	0.05	0.05	0.05	0.05
Momentum (β_1)	—	0.9	0.9	0.9
Learning-rate Scale	—	0.01	0.01	0.01
Warm-up Steps	—	10	10	10
Eval. Particles / Reps	2,000 / 10	2,000 / 10	2,000 / 10	2,000 / 10

Table S-2: Algorithmic parameters for the Daphnia experiment. Particle numbers are per mesocosm; the cooling factor is the multiplier over 50 iterations.

fresh replicates at the same particle count, using the same model and Pypomp version on a CPU. The same evaluation method was used for the final results and for intermediate saved parameter vectors. The progress plot uses 25, 27, 26, and 22 saved parameter vectors per search for IFAD-0, IFAD-0.97, IFAD-1, and MPIF, respectively, including the final evaluations. Time is assigned in proportion to iteration number within each stage, using the measured total optimization time for that stage. Each set of 50 searches ran on one Tesla V100 16 GB GPU, with two searches per batch. Optimization times are summed over the 25 batches and include initial compilation but exclude the additional particle filter evaluations.

S10.3 Results

Our findings are summarized in Table S-3 and Figures S-4 and S-5. IFAD-0.97 and IFAD-1 attained maximum log-likelihoods of -858.48 and -858.27 , respectively, each with a Monte Carlo standard error (MCSE) of approximately 0.17. The maximum for MPIF was -860.06 , with MCSE 0.16. IFAD-0 attained -860.36 and had a median log-likelihood close to that of MPIF.

	Maximum	Re-evaluated	Median	MPIF time (s)	DMOP time (s)
IFAD-0.97	-858.48 (0.17)	-858.23 (0.16)	-861.42	535	1,309
IFAD-1	-858.27 (0.17)	-858.00 (0.14)	-861.46	520	1,313
IFAD-0	-860.36 (0.24)	-860.30 (0.21)	-864.12	503	1,314
MPIF	-860.06 (0.16)	-860.26 (0.14)	-864.27	1,591	-

Table S-3: Log-likelihoods from 50 searches per method, with MCSEs in parentheses. Re-evaluated uses fresh particle filters at each method’s original maximizing parameter vector.

Comparing searches from the same starting points, IFAD-0.97 and IFAD-1 each attained a higher final log-likelihood than MPIF in 49 of the 50 searches. The median paired differences were 2.86 and 2.79 log-likelihood units, respectively. The distributions of the final log-likelihoods for IFAD-0.97 and IFAD-1 were similar, which may be due to the short time series in each mesocosm.

Yang et al. (2026) report a maximum log-likelihood of -863.03 in the revised Table S3. Our MPIF searches attained a higher value, and IFAD-0.97 and IFAD-1 improved on this further.

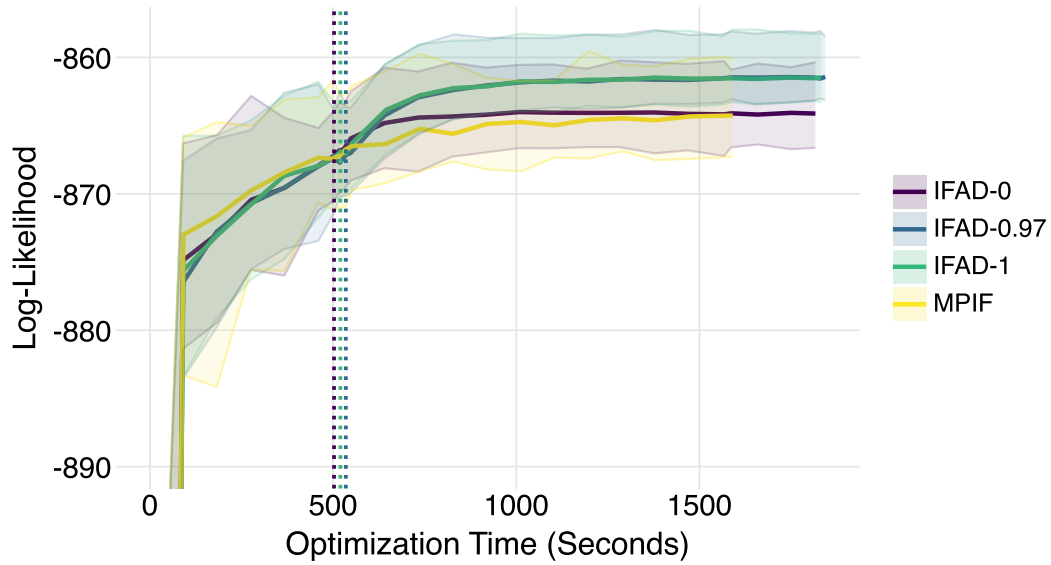


Figure S-4: Optimization progress of IFAD and MPIF for the Daphnia model. The solid lines show the median log-likelihood across 50 searches, and the shaded regions cover the area between the 10th and 100th percentiles. The vertical dotted lines indicate where the IFAD searches switch from MPIF to Adam optimization. The plot shows log-likelihoods above -890 ; all searches contribute to the percentiles.

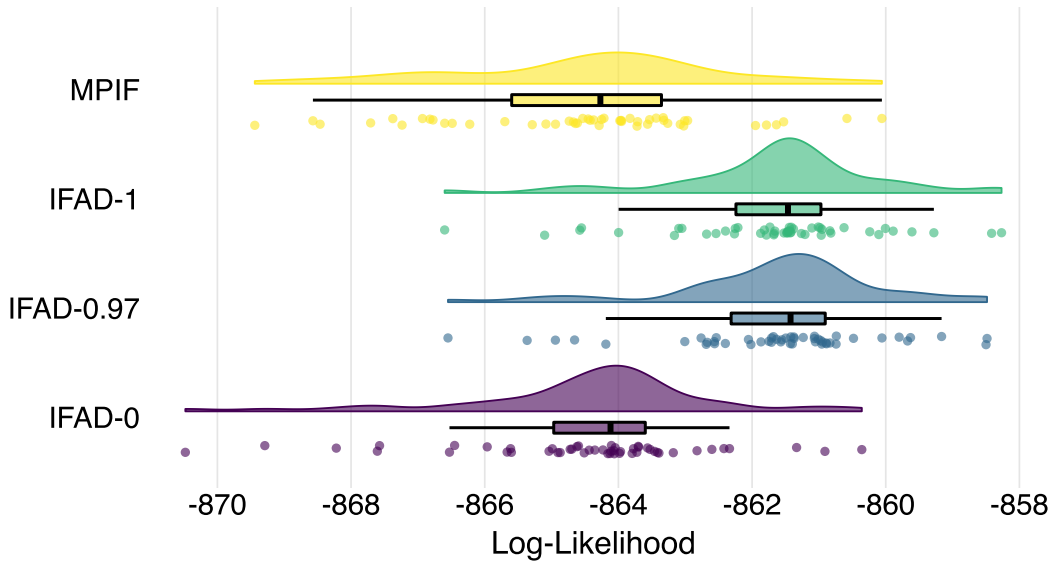


Figure S-5: Raincloud plot of the searches for the Daphnia model. The half-violin plots show the distributions of all 50 final log-likelihood estimates for each method, the boxplots show the medians and quartiles, and each point represents one search. IFAD-0.97 and IFAD-1 have similar distributions, with generally higher log-likelihoods than MPIF and IFAD-0.

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